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Network Design and Implementation Project for NextEdge (Pvt) Ltd Head Office

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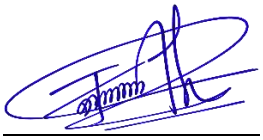
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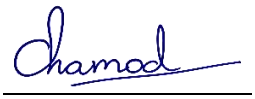
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This is to certify that this project is based on the work of "Network Proposal and Implementation for NextEdge (Pvt) Ltd Head Office" under my supervision. The report has been prepared according to the format stipulated and is of acceptable standard.

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Abstract

This project report presents the design and implementation of a comprehensive network solution for NextEdge (Pvt) Ltd, a Sri Lankan technology and manufacturing company with over 25 years of industry experience, at its new five-storey head office in Colombo 3. The goal of the project was to design a scalable, secure and resilient network infrastructure capable of supporting the company's administrative, technical and research divisions under a single roof.

The project scope included a hierarchical, VLAN-segmented network built on enterprise-grade Cisco switching and routing, redundant Layer-3 gateways using HSRP, a VoIP telephony system built on a 3CX IP-PBX and Grandstream handsets, an IP-based CCTV system with local XVR storage, dedicated and shared file storage, an internally hosted web and database presence supported by AWS, an enterprise email platform, and an enterprise Wi-Fi 6E wireless deployment covering every floor except the 5th-floor research and development lab, which is deliberately isolated from the rest of the network.

Key achievements include the use of link aggregation (EtherChannel) and Spanning Tree Protocol for a loop-free, high-bandwidth backbone, OSPF and static/default routing for efficient inter-VLAN and internet routing, Access Control Lists and port security to enforce the company's security policy, and a layered physical and network security design protecting the server room and the isolated lab area.

As a further development of the project, a fully functional Disaster Recovery (DR) site was designed in AWS cloud, built around AWS Elastic Disaster Recovery for continuous server replication, an Aurora Global Database for near-real-time database failover, and Route 53 health-checked DNS failover. The head office is connected to the DR site over a Cisco SD-WAN overlay that runs across the existing redundant ISP links, giving the company automatic, policy-based path selection and encrypted connectivity between the on-premises network and AWS, in addition to the daily cloud backups already taken to Microsoft Azure and AWS.

Overall, the completed design gives NextEdge (Pvt) Ltd a flexible, secure and scalable network platform - with a resilient cloud-based DR capability - that supports its continued growth, positions it to meet rising digital demands, and was achieved through the close collaboration of the project team.

Special Thanks

Our sincere thanks go to the lecturers who supported us throughout the implementation of this project and provided the guidance needed to improve our knowledge of networking, helping us prepare for our future careers in the industry. We also thank the instructors who assisted us with this project.

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List of Abbreviations

Abbreviation	Full Form
PAT	Port Address Translation
OSPF	Open Shortest Path First
CCTV	Closed-Circuit Television
STP	Spanning Tree Protocol
VRRP	Virtual Router Redundancy Protocol
ACL	Access Control List
VLAN	Virtual Local Area Network
PSTN	Public Switched Telephone Network
NAS	Network-Attached Storage
SSL	Secure Sockets Layer
WAF	Web Application Firewall
TFTP	Trivial File Transfer Protocol
WPA2	Wi-Fi Protected Access Version 2
IP	Internet Protocol
VTP	VLAN Trunking Protocol
HSRP	Hot Standby Router Protocol
DHCP	Dynamic Host Configuration Protocol
NAT	Network Address Translation
XVR	Extreme Video Recorder
PABX	Private Automatic Branch Exchange
RADIUS	Remote Authentication Dial-In User Service

Chapter 1: Project Brief & Business Case

Purpose and Objective of the Solution

This is deliberately scoped as a Proof-of-Concept rather than a finished production rollout: the group's task was to prove, on a representative subset of the design, that the chosen architecture, protocols and cloud extension actually work as specified - not to physically cable and license all six floors of the real building within a single academic module. Where the full-scale design and the lab-scale PoC differ, this report says so explicitly (see the Constraints in Chapter 2 and the Testing Approach in Chapter 8), so the distinction between "designed" and "lab-verified" stays clear throughout.
The purpose of this project is to design, propose and partially implement (as a lab-scale Proof-of-Concept) a stacked network, compute and storage infrastructure solution for NextEdge (Pvt) Ltd's new five-storey head office in Colombo 3, together with a fully functional cloud-based Disaster Recovery (DR) site in AWS connected over SD-WAN. The objective is to give the company a single, coherent design that is efficient, reliable and secure, and that can be handed to NextEdge as a working consultancy deliverable rather than a purely theoretical exercise.

Identified Problem / Industry Opportunity

NextEdge (Pvt) Ltd is a pioneering technology and manufacturing company in Sri Lanka with over 25 years of industry expertise. Established with the vision of driving digital transformation and industrial innovation in the country, the company specialises in the design, assembly and manufacturing of advanced electronics, computer systems and automation equipment. Over the years, NextEdge has expanded into providing comprehensive IT infrastructure solutions, enterprise networking, cloud services and customised hardware manufacturing for both government and private-sector organisations.

With multiple factories, R&D facilities and service centres across the island, NextEdge has built a strong reputation for reliability, innovation and technical excellence, working with clients in education, banking, healthcare and manufacturing. As part of its long-term expansion strategy, NextEdge has decided to relocate its headquarters to a newly constructed five-storey building in Colombo 3, bringing its administrative, technical and research divisions together under one roof.

This relocation is the trigger for the project: the new building currently has no network, compute, storage, communications, security or disaster-recovery infrastructure of its own. Left unaddressed, this is a real business risk - departments cannot operate, the isolated R&D Lab cannot be safely segmented from the rest of the company, and there is no resilience against an outage or disaster at the new single head-office location.

Proposed Solution Overview

The project proposes a hierarchical, VLAN-segmented on-premises network built on enterprise Cisco switching/routing and redundant Layer-3 gateways, a VoIP telephony platform, IP-based CCTV, dedicated and shared storage, an internally hosted web/database presence, an enterprise email platform, and enterprise Wi-Fi 6E covering every floor except the isolated Lab. This is extended with a warm-standby Disaster Recovery site in AWS (ap-south-1), connected back to the head office over a dual-transport Cisco SD-WAN overlay, so that the business has a tested recovery path if the head office itself becomes unavailable. Full technical detail is given in Chapter 4 (System Design and Architecture).

Alignment with Academic Learning Outcomes

- Applies core networking theory (VLANs, routing, redundancy protocols, wireless design) to a realistic, industry-scale scenario.
- Demonstrates systems and storage integration alongside network design, reflecting the "stacked infrastructure" requirement of the module.
- Evidences the three required non-functional qualities - efficiency, reliability and security - through concrete design decisions, not just narrative description (see Chapter 2 and Chapter 5).
- Requires and evidences professional documentation practice: requirements specification, security requirements specification, architecture documentation, a reliability and assurance plan, an implementation plan, and supervisor-logged progress tracking.

- Builds consultancy and teamwork skills through group-based scoping, design trade-off discussions, and a supervised, milestone-based delivery process.

Chapter 2: Requirements Specification

Functional Requirements

1. Scalable and Reliable Network

- Design and deploy a scalable and reliable network infrastructure that caters to the needs of the building and can accommodate future growth.
- Implement redundant networking components and ensure network uptime and resilience.

2. Network Segmentation

- Isolate the Lab from all other departments and the internet, allowing communication only with HR managers. Implement proper network segmentation measures to achieve this.

3. Communication System

- Implement a comprehensive communication system that ensures high voice quality, clarity and advanced call-management features.
- Enable communication both within and outside the internal network, easing effective collaboration and external communications when necessary.

4. Security Camera System

- Install a comprehensive security camera system to cover key areas of the building, including the server room, lab area and other critical zones.
- Ensure proper coverage, high-quality video capture and secure storage of camera footage.

5. Storage System

- Set up dedicated storage systems for the lab, easing secure storage and easy access to lab-related data.
- Set up a common storage system accessible to all employees, promoting efficient file sharing and collaboration across departments.

6. Web Hosting

- Provide the capability to host a website internally within the building, allowing NextEdge to display products, communicate with customers and enhance its online presence, along with a database server.

7. Backup Systems

- Implement robust backup systems to ensure the protection and recoverability of critical data and systems.
- Define backup schedules, offsite storage options and regular testing of backup and restore procedures.

8. Email Management

- Deploy a reliable and secure email management system that enables efficient email communication, including email routing, spam filtering and a user-friendly interface.

9. Physical Security

Physical and network security are treated as complementary, not interchangeable: even a perfectly configured firewall does not help if an unauthorised person can walk up to a core switch and plug in a laptop, so the physical controls in this section and the network controls in Chapter 3 are designed to be read together.

- Implement high physical security measures for the server room and the lab area, including restricted access, surveillance and monitoring systems.
- Set up protocols for access control, visitor management and incident response.

10. Network Security

- Implement strong network security measures to protect against unauthorised access, data breaches and other security threats.
- Implement restrictions in accordance with the security policy.

11. Data Privacy and Compliance

- Ensure compliance with relevant data privacy regulations and industry standards by implementing proper security controls and data protection measures.

12. Disaster Recovery

- Develop a comprehensive disaster recovery plan that outlines procedures for restoring critical systems and operations in the event of a network or infrastructure failure.
- Implement a fully functional Disaster Recovery site in a public cloud (AWS), capable of taking over critical services within a short, defined recovery time if the Colombo 3 head office becomes unavailable.

13. Resilient Connectivity to the DR Site

- Provide resilient, encrypted, application-aware connectivity between the head office and the AWS DR site using an SD-WAN overlay, so that connectivity to the DR site does not depend on a single ISP link.

14. Wireless Network Infrastructure

- Design and implement a wireless network infrastructure that gives internet access to all employees, covering all areas of the building except the 5th-floor lab area.

Non-Functional Requirements

Efficiency

Efficiency in this design is judged by how much useful work the infrastructure does per rupee and per rack unit, not simply by raw specification. Wherever a role could be safely virtualised (DHCP, DNS, AD DS, File) it was, rather than dedicating a physical chassis to each service; the AWS DR site follows the same principle by keeping only minimal, warm-standby compute running day to day and scaling to full capacity only during a drill or an actual failover. Link aggregation was preferred over simply buying higher-capacity single links, since it reuses existing switch ports and adds resilience at the same time as it adds throughput.

- Optimal use of switching/routing hardware through virtualisation (ESXi-hosted DHCP, DNS, AD DS and File servers) rather than one physical box per service.
- Load balancing and link aggregation (EtherChannel) across the access/distribution layers to make full use of available bandwidth rather than leaving redundant links idle.
- Right-sized, warm-standby (rather than always-on, full-scale) compute in the AWS DR site, so cloud spend tracks actual recovery need instead of duplicating full production capacity permanently.

Reliability

Reliability was treated as a property of the whole path from end device to service, not of any single component. HSRP removes the default gateway as a single point of failure; STP keeps redundant physical links available as instant failover paths instead of active loops that could cause outages of their own; and the dual-ISP, dual-firewall edge means a single circuit or appliance fault does not take the building offline. The AWS DR site extends this same philosophy from individual devices and links to the site as a whole, so that even a total loss of the Colombo 3 building does not mean a total loss of service.

- Fault tolerance and automatic failover at the routing layer via HSRP/VRRP between the paired Layer-3 switches.
- Loop-free, self-healing Layer-2 topology via Spanning Tree Protocol.
- Redundant ISP links and a redundant firewall pair at the network edge.
- A tested, documented Disaster Recovery path (Chapter 5) if the head office itself becomes unavailable.

Security

Security requirements were derived directly from what NextEdge stands to lose: proprietary R&D data in the Lab, customer and employee data on internal systems, and continuity of the business itself. Rather than treating security as a single perimeter firewall, controls are layered - network segmentation, access control lists, encryption in transit, and physical access control - so that a failure in any one layer does not expose the whole network. The full control set and its justification are detailed in Chapter 3.

- Access control aligned with best practice: VLAN segmentation, ACLs, port security and firewall rules enforced at every layer that requires it.
- Data protection through encryption in transit (SD-WAN IPsec tunnels, WPA2/WPA3 Wi-Fi, SSL on the web tier) and secure configuration baselines on network devices.
- Isolation of the highest-sensitivity segment (the Lab) from the rest of the network and from the internet, permitting only HR-department communication.

Stakeholder Needs

Table 2.1 - Stakeholder Needs

Stakeholder	Needs
NextEdge Executive Management	A network that supports business growth, protects company data, and gives confidence that operations can continue after a major incident.
Department Staff (all floors)	Reliable wired/wireless connectivity, working VoIP handsets, and access to the shared file store and business applications.
Product Research / R&D (Lab, 5th floor)	A network segment that keeps proprietary research data isolated from the rest of the company and from the internet, while still allowing HR to administer onboarding/offboarding.
Network & System Admin Team	Centralised, remotely manageable infrastructure (SSH access, centralised logging, a wireless LAN controller) that is realistic for a small team to operate day to day.
Customers / External Users	A responsive, always-reachable public web presence and continuity of service even if the head office suffers an outage.
NIBM Supervisor / Assessors	Evidence of a professionally scoped, designed, implemented and documented PoC that meets the module's efficiency, reliability and security criteria.

Constraints and Assumptions

Constraints

- The college lab environment does not have the full quantity or exact model of enterprise hardware specified in the design; the lab implementation (Chapter 7) therefore uses a representative, scaled-down topology.
- The 90-day project timeline (Chapter 6) limits the PoC to a representative subset of VLANs, ACLs and DR failover scenarios rather than every floor and department.
- Budget figures (Chapter 4) are planning-level market estimates in the absence of formal vendor quotations.
- Cloud costs are estimated from public AWS/Cisco list pricing rather than a negotiated enterprise agreement.

Assumptions

- NextEdge will provide two independent ISP circuits into the Colombo 3 building, as assumed in the SD-WAN design.
- NextEdge's IT team will hold the necessary AWS and Cisco licences (or a partner/reseller relationship) to operate the proposed cloud and SD-WAN services in production.
- Departmental headcounts and floor allocations remain as given in the building's floor plans (Chapter 4) for the duration of the project.
- Standard Sri Lankan business operating hours and public electricity supply apply, with the UPS/generator arrangement covering short outages only.

Chapter 3: Security Requirements Specification

Threat Model

Impact and likelihood ratings were assigned qualitatively by the group, drawing on the relative sensitivity of each asset to NextEdge's business (for example, loss of R&D data is rated High impact because it is irreplaceable and competitively sensitive, whereas loss of CCTV footage is rated Medium because its main value is investigative rather than operational) rather than a formal quantitative risk model, which was judged disproportionate for a single-site SME-scale deployment.
The threat model below identifies the key assets in the NextEdge design, the threats most relevant to each, and the impact/likelihood used to prioritise the security controls in this chapter and the security policy in Annexure 3.

Table 3.1 - Threat Model

Asset	Primary Threat(s)	Impact	Likelihood	Mitigating Control(s)
Isolated Lab / R&D data	Insider or external exfiltration of proprietary research data	High	Low-Medium	VLAN isolation restricted to HR only; ACLs deny all other inter-VLAN/internet paths; physical access control to the Lab.
Server room (core switches, servers, firewalls)	Unauthorised physical access, tampering or theft	High	Low	Restricted card/biometric access, CCTV coverage, visitor management, environmental monitoring.
Perimeter / internet-facing services (web, email, VPN)	External network attacks - scanning, exploitation, DDoS, phishing	High	Medium	Redundant firewalls, WAF on the web tier, IPS on the Sophos firewall, spam filtering on email, regular patching.
Internal network devices (switches, routers, controllers)	Unauthorised configuration change or credential compromise	Medium-High	Low-Medium	SSH-only management, ACL-restricted management access, port security, centralised syslog for change/audit visibility.
Wireless network	Rogue access points, weak-key or open Wi-Fi abuse	Medium	Medium	WPA2/WPA3 with 802.1X/RADIUS, centralised wireless LAN controller, no wireless coverage in the Lab.
Data in transit between HQ and AWS DR site	Interception or tampering on the WAN path	High	Low	IPsec-encrypted SD-WAN tunnels over both ISP transports; consistent segmentation policy carried into the cloud.
Backup and DR data (Azure, AWS S3/Aurora/DRS)	Data loss, corruption, or an unrecoverable DR failover	High	Low	Daily/continuous backups, cross-region replication, scheduled DR drills and restore testing (Chapter 5).
CCTV footage	Loss, tampering or unauthorised viewing of surveillance data	Medium	Low	Local XVR storage with defined retention, restricted viewing access, camera route segregated on its own VLAN.

Security Controls and Mechanisms

The controls below were selected to directly address the threats identified in the threat model above, following a defence-in-depth approach: no single control is relied upon to fully mitigate any one threat, and every control has at least one complementary control covering the same risk from a different angle.

Table 3.2 - Security Controls and Mechanisms

Control	Mechanism
Network segmentation	VLANs separate every department; the Lab VLAN is restricted so that only HR can reach it, with no path to the internet.
Access control lists (ACLs)	Enforce the segmentation policy above and restrict device-management traffic (SSH) to the Network & System Admin subnet only.
Firewall & IPS	A redundant pair of next-generation firewalls (Sophos XGS 5500) inspects all inbound/outbound traffic and blocks known attack patterns.
Port security	Access-layer switch ports are locked to authorised MAC addresses, preventing rogue devices from joining the wired network.
Secure remote management	SSH (not Telnet) is used for all device administration, with access itself restricted by ACL.
Wireless security	WPA2/WPA3 encryption, optional 802.1X/RADIUS authentication, and centralised policy via the wireless LAN controller.
Web application security	SSL/TLS on the web tier and a Web Application Firewall (WAF) in front of the internally hosted website.
Email security	Spam filtering, routing controls and encryption on the Microsoft Exchange-based email platform.
Physical security	Access-card/biometric control, CCTV coverage and visitor management for the server room and the Lab.
Backup & DR security	Backups are encrypted at rest, cross-region replicated, and the AWS DR account/IAM roles are kept separate from day-to-day production access with least-privilege permissions.
Monitoring & logging	Syslog centralises device logs; AWS GuardDuty, Security Hub and CloudTrail provide continuous monitoring and audit logging for the cloud estate (see Chapter 5).

Compliance Considerations

NextEdge is not currently subject to a specific sectoral regulator, so the compliance approach taken here is deliberately conservative: it follows widely recognised data-protection and security good practice so that the design would already be positioned well if NextEdge's customer base (e.g. banking or healthcare clients) later required evidence of a specific compliance regime.

- Data privacy: personal data held on HR and customer-facing systems is handled in line with Sri Lanka's Personal Data Protection Act (PDPA) principles - lawful basis, purpose limitation and data minimisation.
- Industry good practice: the design follows widely recognised network-security good practice (defence in depth, least privilege, segmentation) rather than any single named regulatory standard, since NextEdge is not currently subject to a sector-specific compliance regime (e.g. PCI-DSS, HIPAA).
- Auditability: centralised logging (syslog, AWS CloudTrail) and a documented security policy (Annexure 3) give NextEdge an evidentiary trail if a compliance review or incident investigation is ever required.
- Data residency: the AWS DR site is hosted in ap-south-1 (Mumbai); NextEdge should confirm this satisfies any customer contractual data-residency expectations before go-live.

None of the compliance considerations above should be read as legal advice; NextEdge's own legal counsel should confirm PDPA applicability and any customer-contract data-residency clauses before the AWS DR site (Chapter 4) is used to hold live customer or employee data.

Chapter 4: System Design and Architecture

Solutions

1. Scalable Network Infrastructure

Addresses the need for a reliable and scalable network by implementing enterprise-grade network equipment and using technologies such as link aggregation, Virtual Router Redundancy Protocol (VRRP)/HSRP, and STP. These measures ensure network reliability and accommodate future growth.

2. VLAN Architecture and Firewall Rules

Implementing VLAN architecture isolates sub-departments, and firewall rules/ACLs enforce the security policy - ensuring network segmentation and enhanced security.

3. Voice over IP (VoIP) Phone System

Deploying a VoIP phone system meets the requirement for a communication system with high voice quality and advanced call management. Integrating PSTN and PABX enables internal and external communication.

4. IP-Based Security Cameras and XVR

Installing IP-based security cameras strategically and using a centralised XVR for video management enhances security, with local backups retaining footage for a defined period to support data retention and compliance.

5. Dedicated Storage for Lab and Shared Storage

A dedicated NAS for the lab, plus shared NFS/SMB storage for employees, fulfils the storage requirement while ensuring secure access and collaboration.

6. Dedicated Web Server and Security Measures

A dedicated web server capable of handling concurrent users, secured with SSL certificates and protected by a Web Application Firewall (WAF), ensures a robust and secure web presence.

7. Database Server

Using AWS cloud services allows a standard database server to be deployed with ease, taking advantage of AWS's scalable infrastructure and robust features.

8. Local Backup and TFTP Server

Local backups for camera footage and a TFTP server for configuration-file backups add an extra layer of data protection and ensure quick recovery in case of incidents.

9. Enterprise Email Server with Microsoft Exchange

An enterprise-grade email server built on Microsoft Exchange addresses the email-management requirement, including routing, spam filtering, encryption and a user-friendly interface.

10. Enhanced Physical Security

Access-control systems, cameras, biometric authentication and environmental monitoring fulfil the need for high physical security in the server room and lab area.

11. Multi-Layered Network Security

Firewalls, ACLs, regular security patching and syslog servers align with the requirement for strong network security monitoring.

12. Cloud Backup and a Fully Functional AWS DR Site

In addition to daily uploads from the Lab's dedicated file server, the common file server, and the TFTP and Syslog servers to Microsoft Azure, the project implements a fully functional warm-standby Disaster Recovery (DR) site in AWS, using AWS Elastic Disaster Recovery, an Aurora Global Database and Amazon S3 cross-region replication (full architecture below).

13. SD-WAN Connectivity to the DR Site

A Cisco SD-WAN overlay running across the head office's two ISP links gives NextEdge resilient, application-aware and encrypted connectivity to the AWS DR site, with automatic path selection between transports and consistent segmentation policy carried through to the cloud.

14. Wireless Network

Enterprise-grade access points implement the wireless network with WPA2 (or higher) encryption. Wi-Fi access is offered on the 2.4GHz, 5GHz and 6GHz bands, using strong pre-shared keys or 802.1X with RADIUS authentication, managed centrally through a wireless LAN controller.

15. Network Monitoring and Management

Network management systems provide comprehensive monitoring and analysis of both wireless and Ethernet networks, enabling real-time performance monitoring, efficient troubleshooting and optimal connectivity throughout the infrastructure.

The server room is located on the 4th floor of the building. This central position reduces cabling costs, gives easy access to the network backbone, and helps ensure physical security.

Outcomes of the Project

General Benefits

- Central monitoring system
- High network security
- High network performance
- Load balancing
- Scalability
- Wireless internet access for all users
- High redundancy
- Easy maintenance

Versatility

- Add or remove devices with minimum disruption
- Change user information for existing devices with minimum hassle

Productivity

- Real-time insight into organisational performance to help with decision making
- Faster, more personalised service to internal staff as well as external customers

Security

- A well-maintained security system
- A security monitoring system to assist the company's cybersecurity specialists

Hierarchical Network Diagram

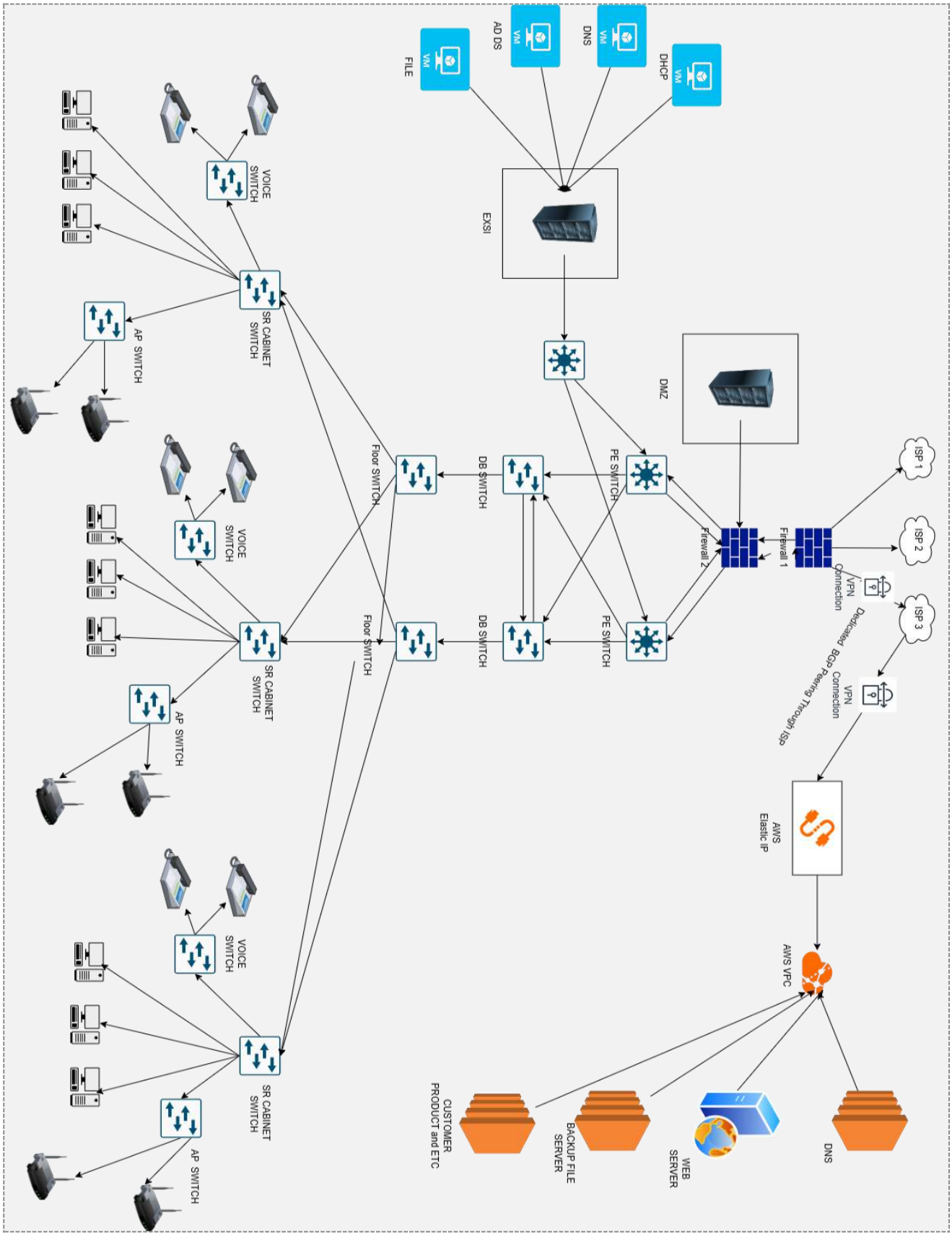


Figure 4.1 - Hierarchical Network Diagram (ISP links, dual firewalls, DMZ, core/distribution/access switches, CCTV, wireless AP, VoIP)

Logical Diagram

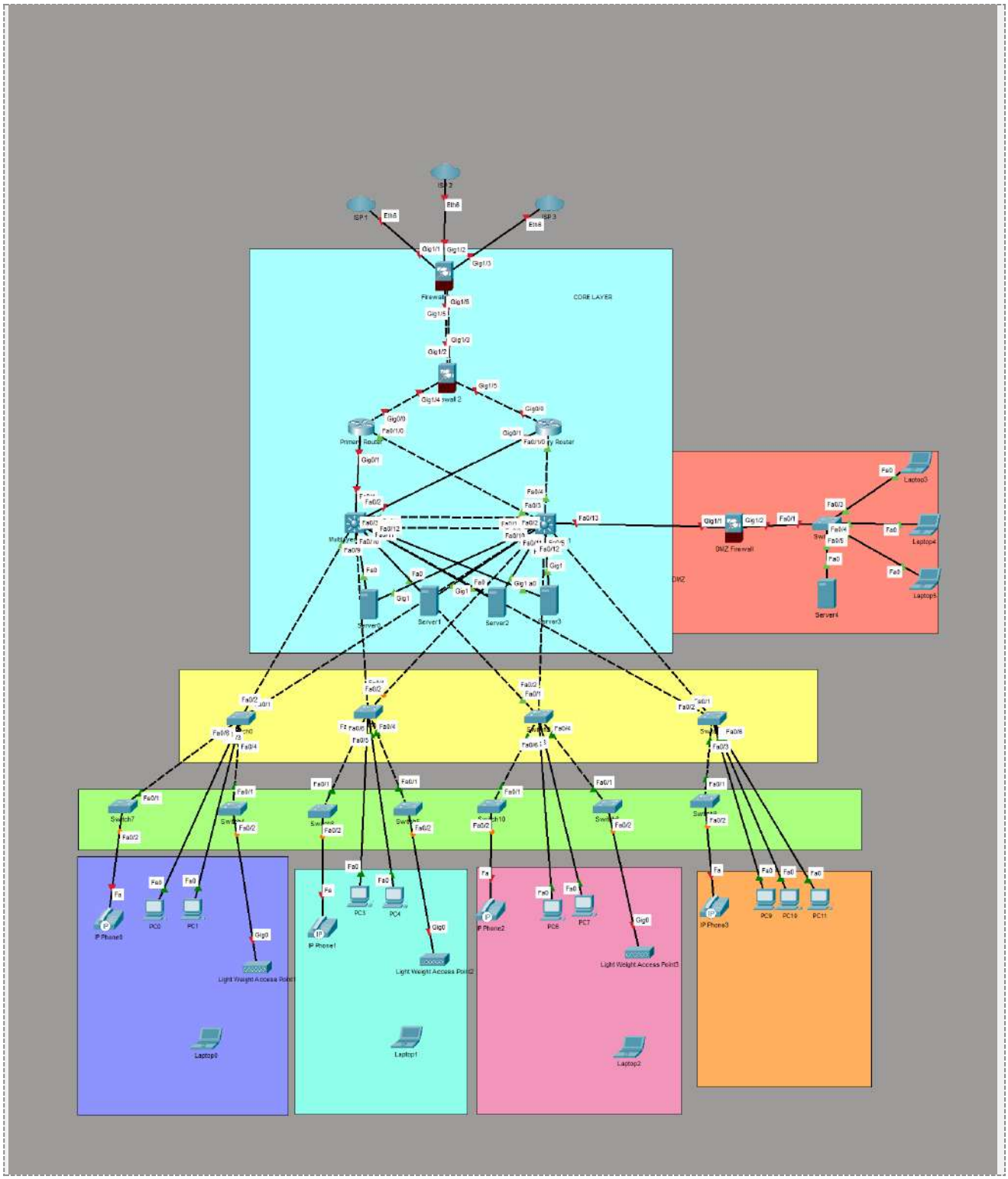
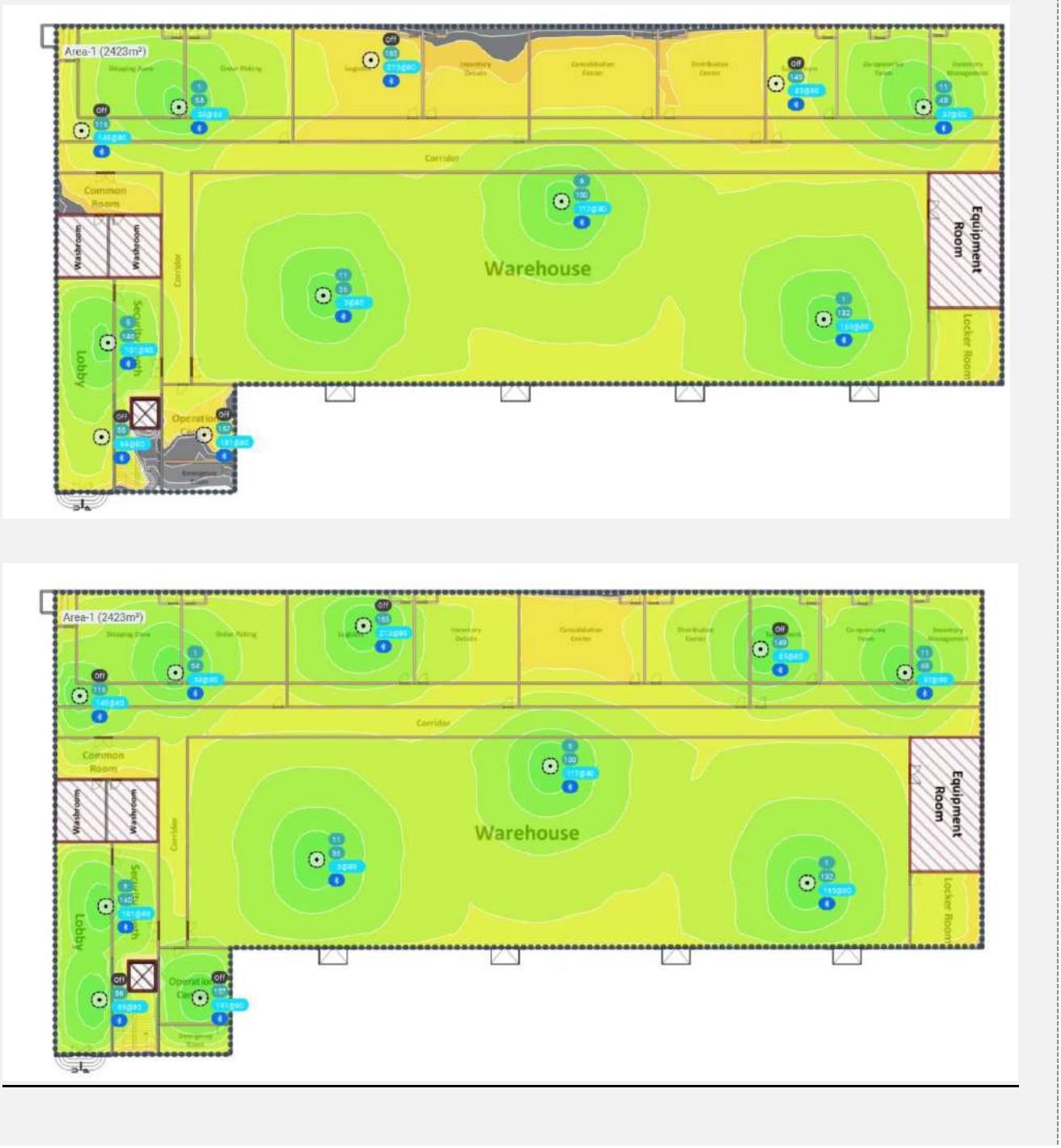


Figure 4.2 - Logical Network Diagram showing VLANs, IP subnets and routing paths

WLAN Signal Distribution (Heatmaps)

Ground Floor



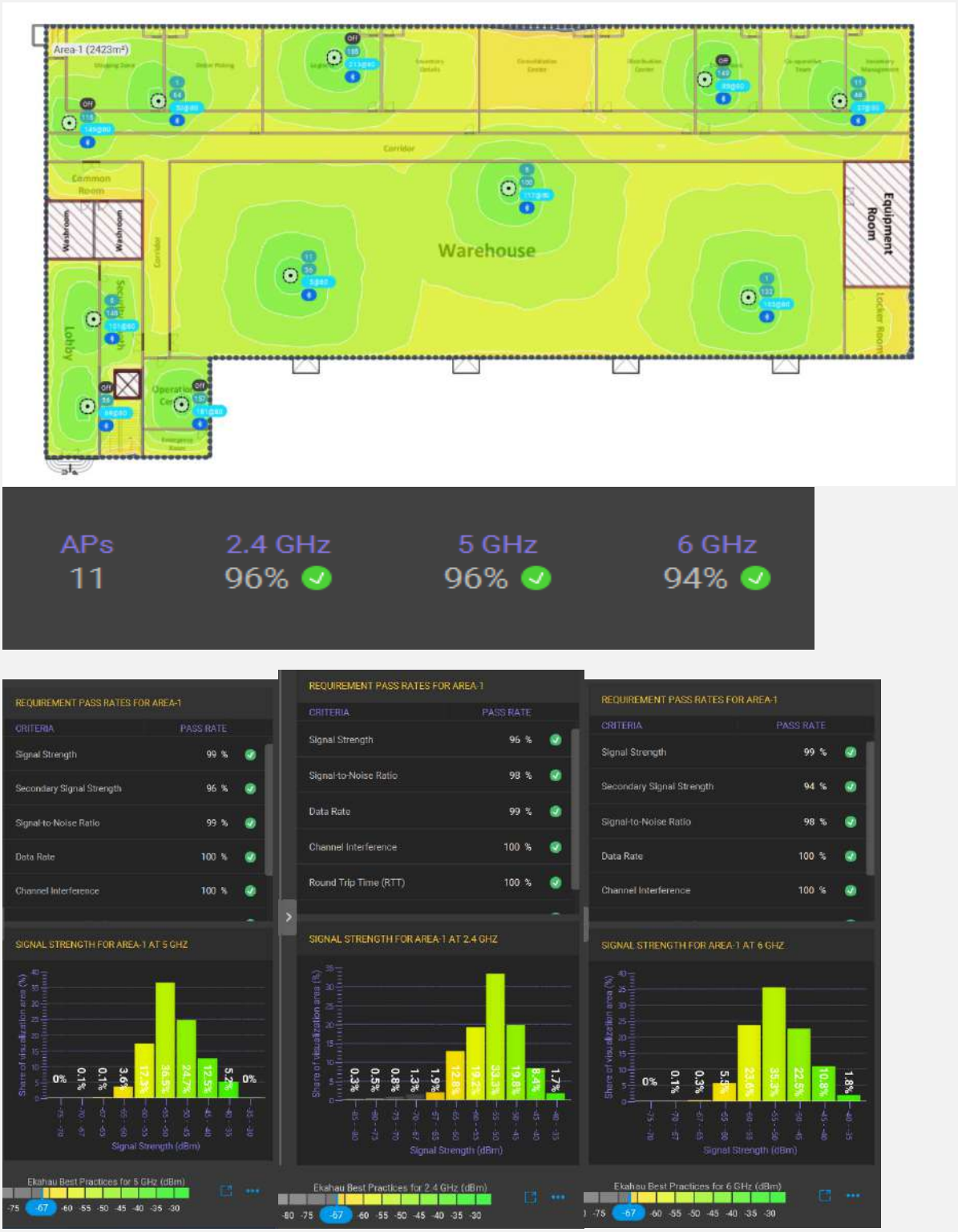


Figure 4.3 - Ground Floor plan and 2.4GHz/5GHz/6GHz Wi-Fi heatmap and 2.4GHz,5GHz,6GHz Frequency Bands - 105 PCs, 23 IP Phones - Warehouse, Inventory Management, Shipping/Logistics

1st Floor


Area-1 (2488m²)
Service Managers
Supervision Service
Supervision Inquiry Unit
Incoming sets
Outgoing sets
Kicthen
Cafeteria
Corridor
Storage
Corridor
Incoming Calls
Outgoing Calls
Washroom
Washroom

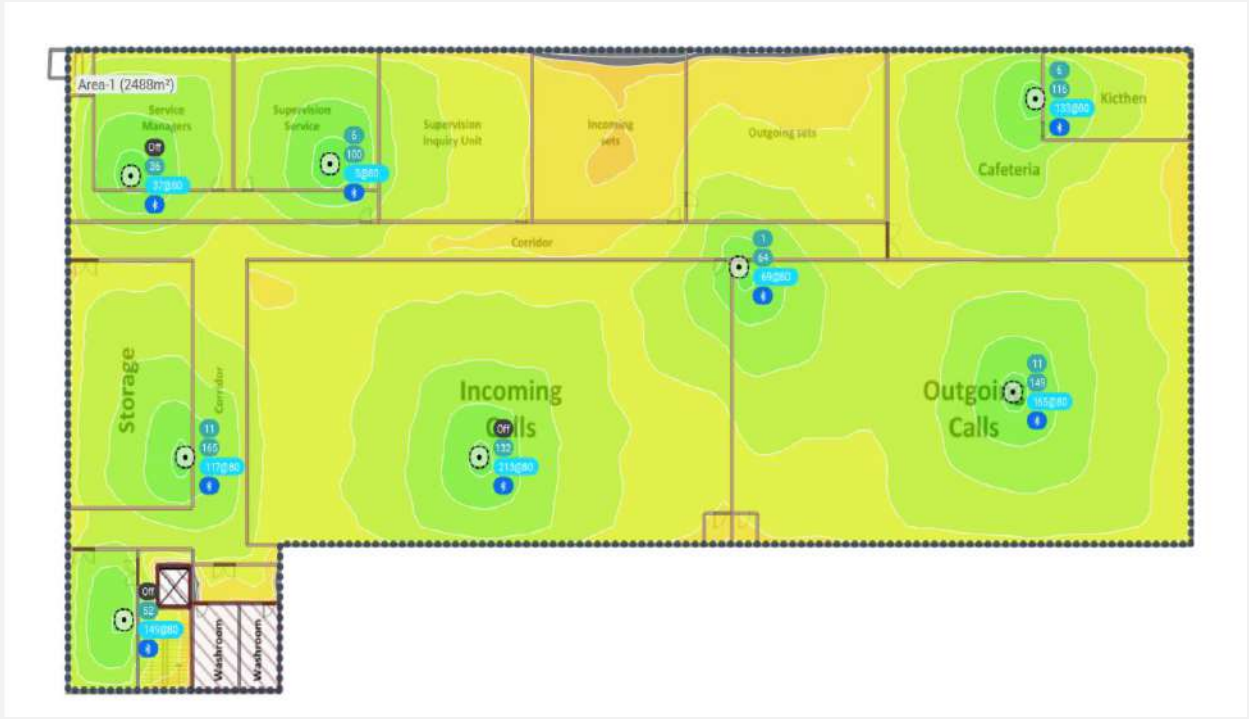
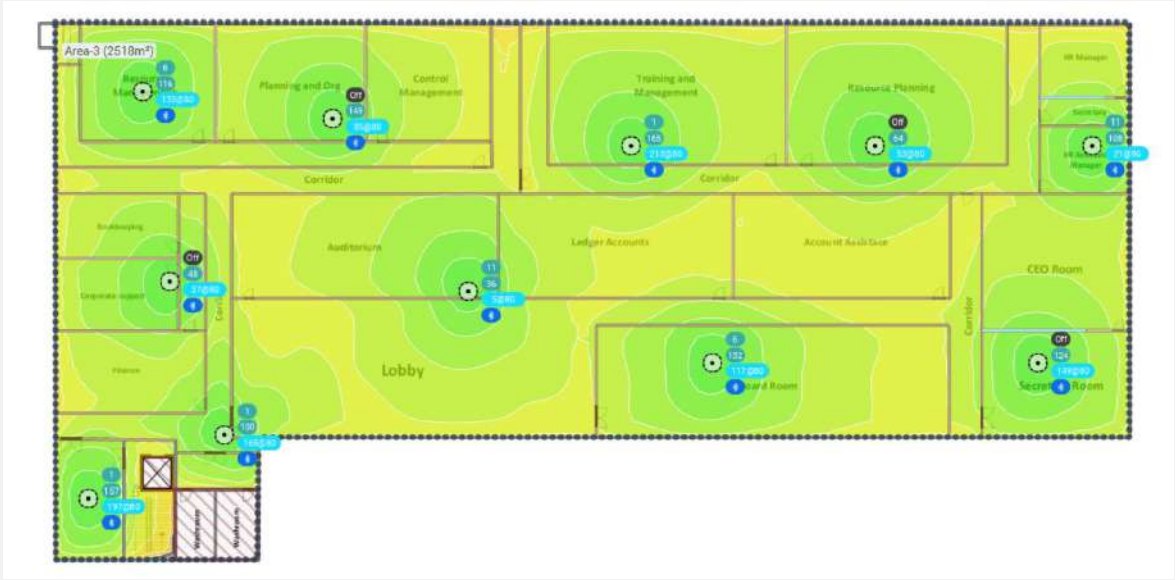
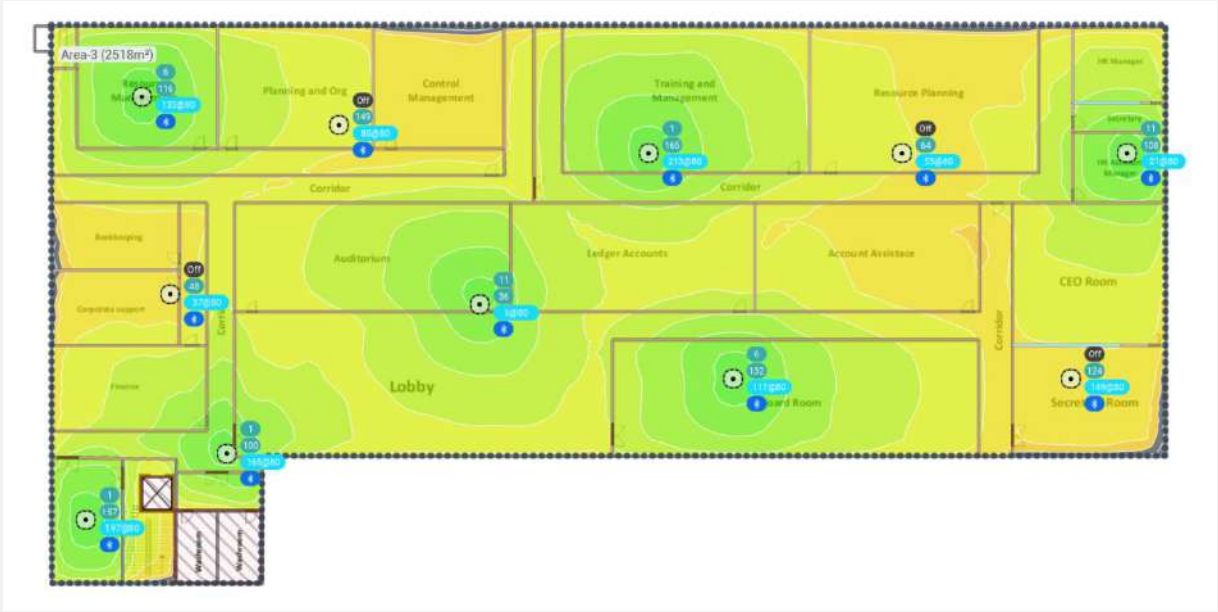

Area-1 (2488m²)
Service Managers
Supervision Service
Supervision Inquiry Unit
Incoming sets
Outgoing sets
Kicthen
Cafeteria
Corridor
Storage
Corridor
Incoming Calls
Outgoing Calls
Washroom
Washroom



Figure 4.4 - 1st Floor plan and 2.4GHz/5GHz/6GHz Wi-Fi heatmap and 2.4GHz,5GHz,6GHz Frequency channels - 150 PCs, 88 IP Phones - Call Centre 1 & 2, Service Managers, Supervision

2nd Floor


Area-3 (2518m²)
Planning and Org
Control Management
Training and Management
Resource Planning
Corridor
Auditorium
Ledger Accounts
Account Assistace
CEO Room
Lobby

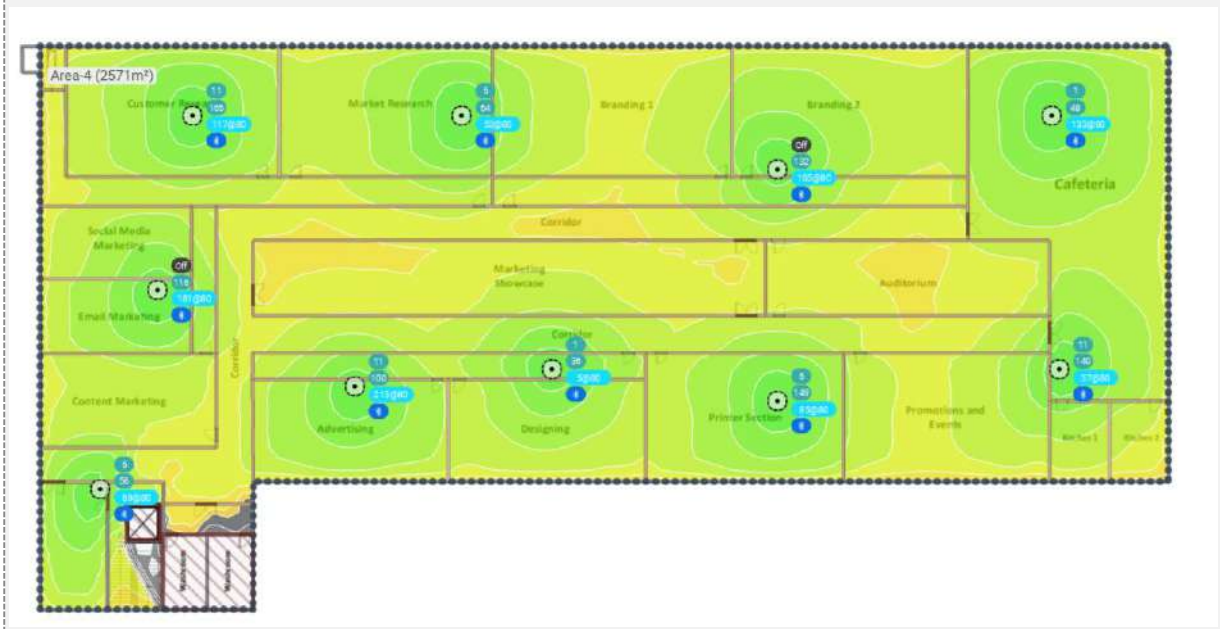

Area-3 (2518m²)
Planning and Org
Control Management
Training and Management
Resource Planning
Corridor
Auditorium
Ledger Accounts
Account Assistace
CEO Room
Lobby

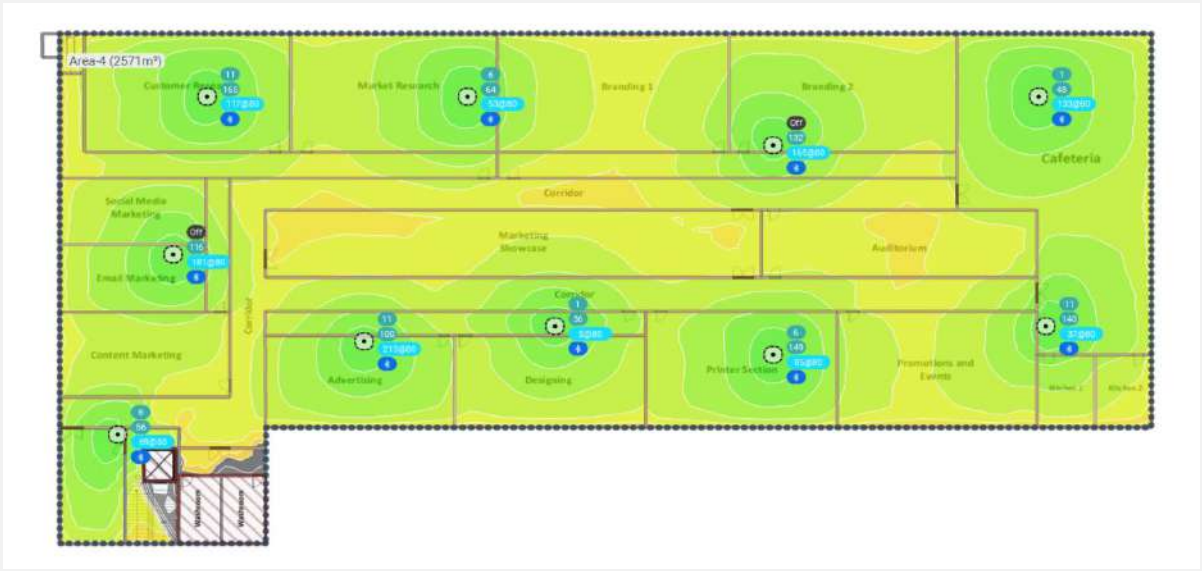

Area-3 (2518m²)
Planning and Org
Control Management
Training and Management
Resource Planning
Corridor
Auditorium
Ledger Accounts
Account Assistace
CEO Room
Lobby

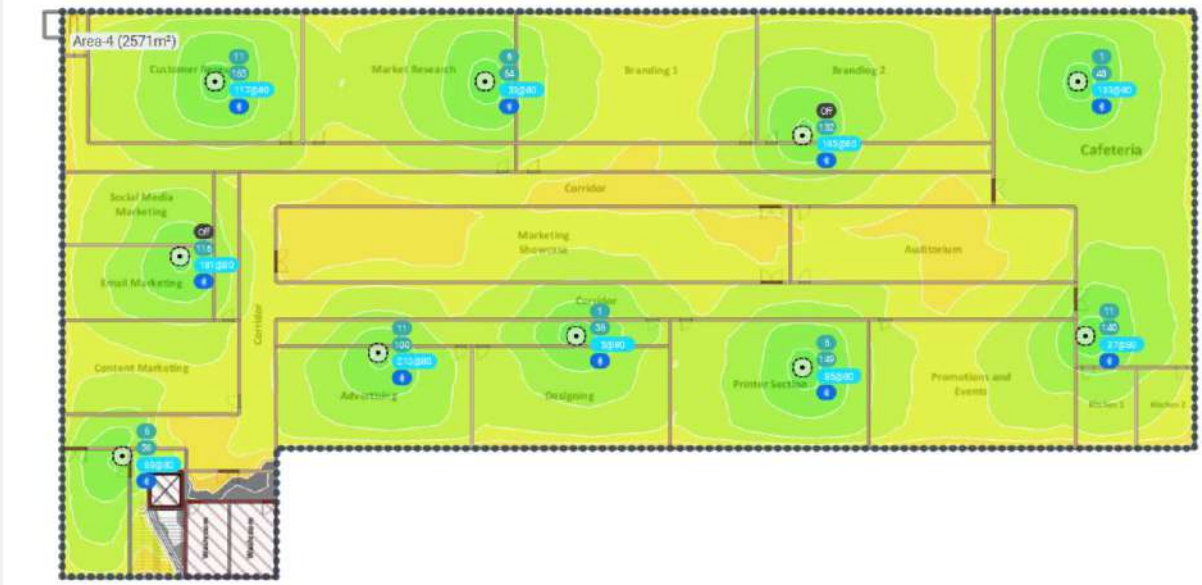


Figure 4.5 - 2nd Floor plan and 2.4GHz/5GHz/6GHz Wi-Fi heatmap - 179 PCs, 26 IP Phones - Management, Finance, HR, Accounts, CEO Suite

3rd Floor





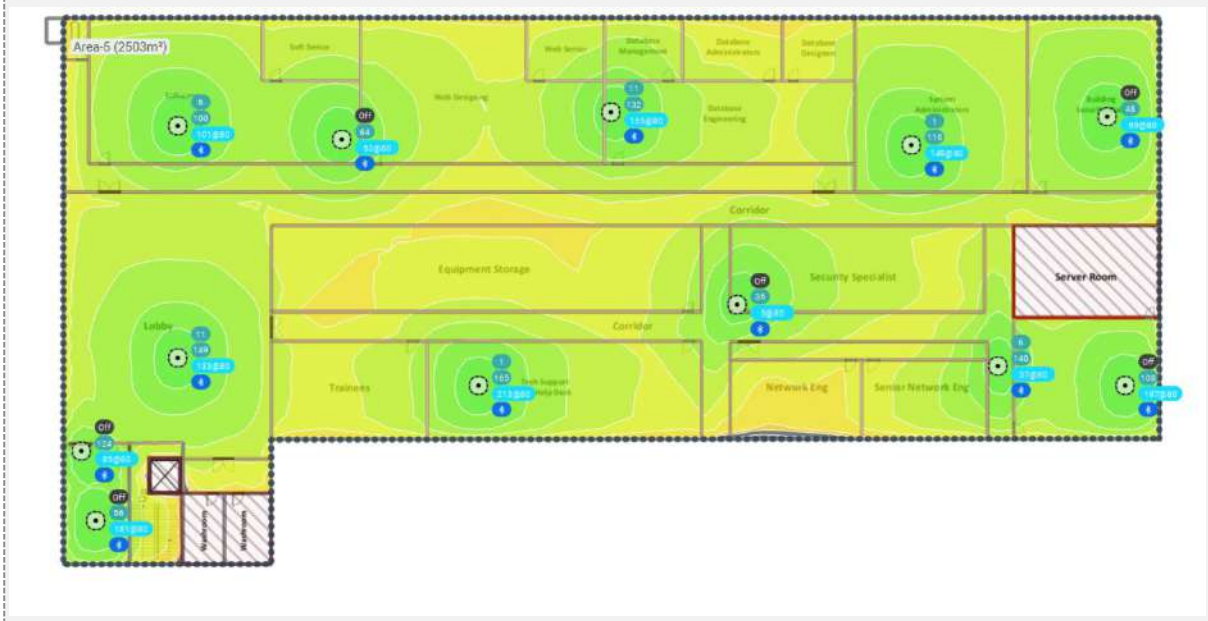


APs	2.4 GHz	5 GHz	6 GHz
10	82%	96%	94%



Figure 4.6 - 3rd Floor plan and 2.4GHz/5GHz/6GHz Wi-Fi heatmap - 175 PCs, 22 IP Phones - Marketing, Market Research, Online Marketing

4th Floor


Area-6 (2503m²)
Corridor
Equipment Storage
Security Specialist
Server Room
Lobby
Corridor
Trainees
Network Eng
Senior Network Eng


Area-5 (2503m²)
Corridor
Equipment Storage
Security Specialist
Server Room
Lobby
Corridor
Trainees
Network Eng
Senior Network Eng



Figure 4.7 - 4th Floor plan and 2.4GHz/5GHz/6GHz Wi-Fi heatmap - 190 PCs, 20 IP Phones - Software, Database, Network/System Admin, Server Room

5th Floor

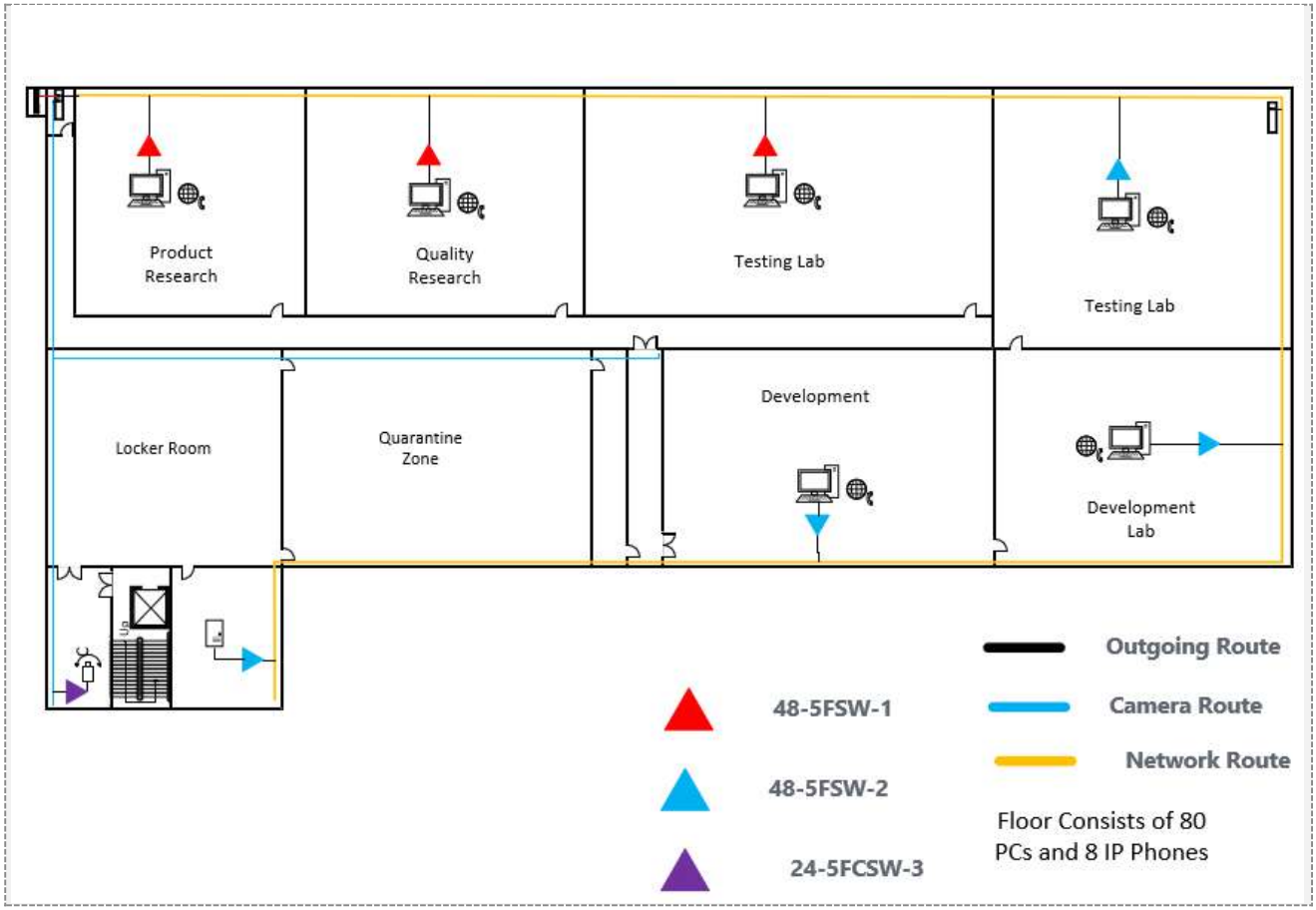


Figure 4.8 - 5th Floor plan and 2.4GHz/5GHz/6GHz Wi-Fi heatmap - 80 PCs, 8 IP Phones - Product Research, Quality Research, Testing & Development Lab

Rack Diagrams

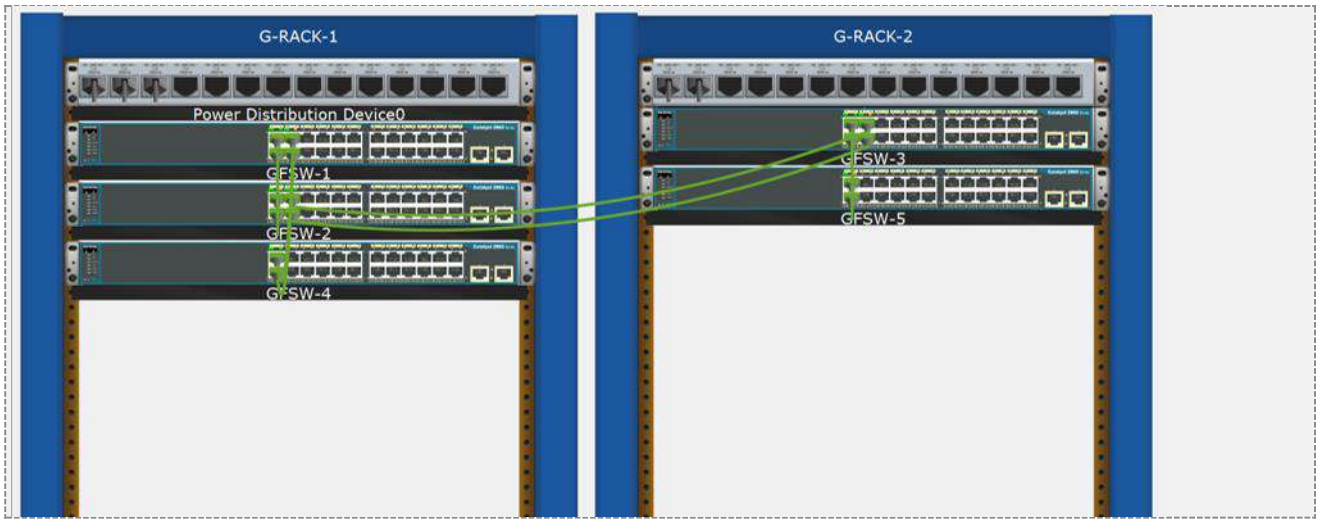


Figure 4.9 - Rack Diagram - Ground Floor

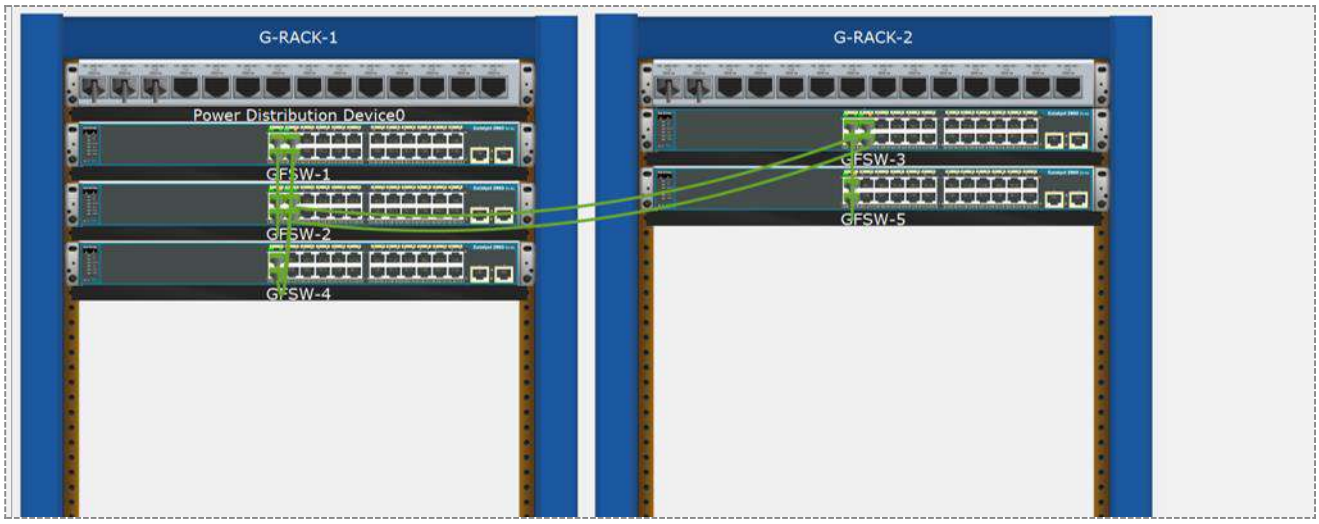


Figure 4.10 - Rack Diagram - 1st Floor

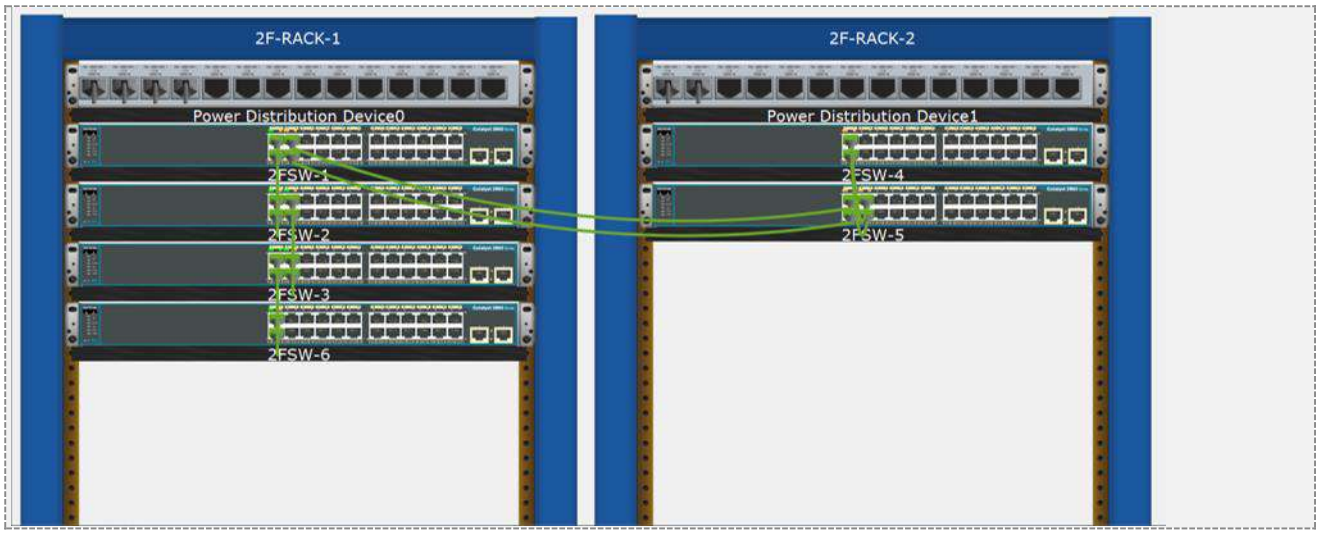


Figure 4.11 - Rack Diagram - 2nd Floor

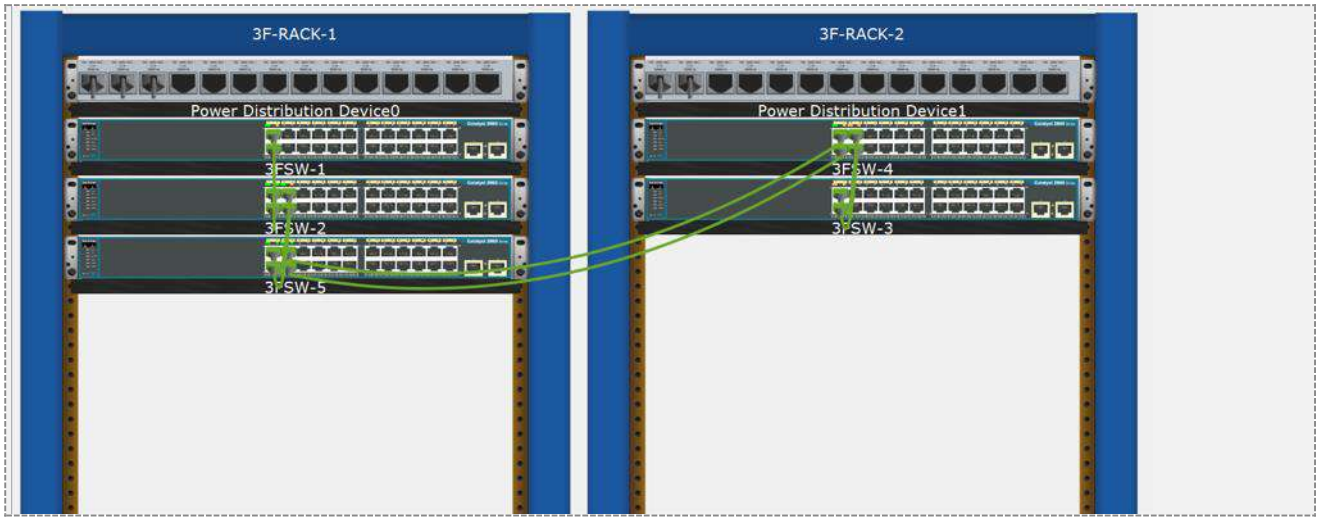


Figure 4.12 - Rack Diagram - 3rd Floor

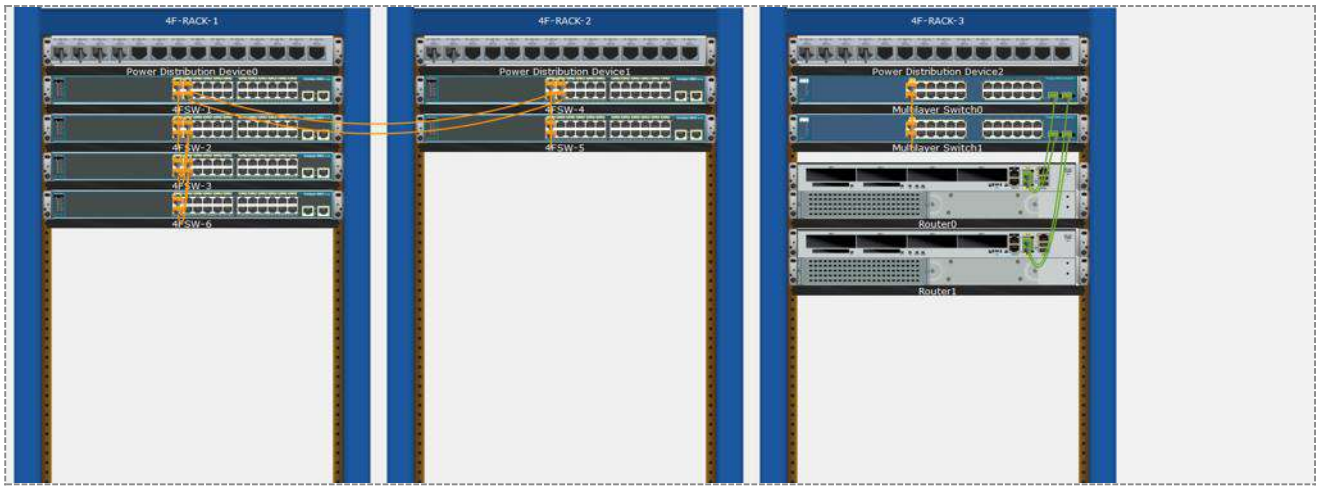


Figure 4.13 - Rack Diagram - 4th Floor

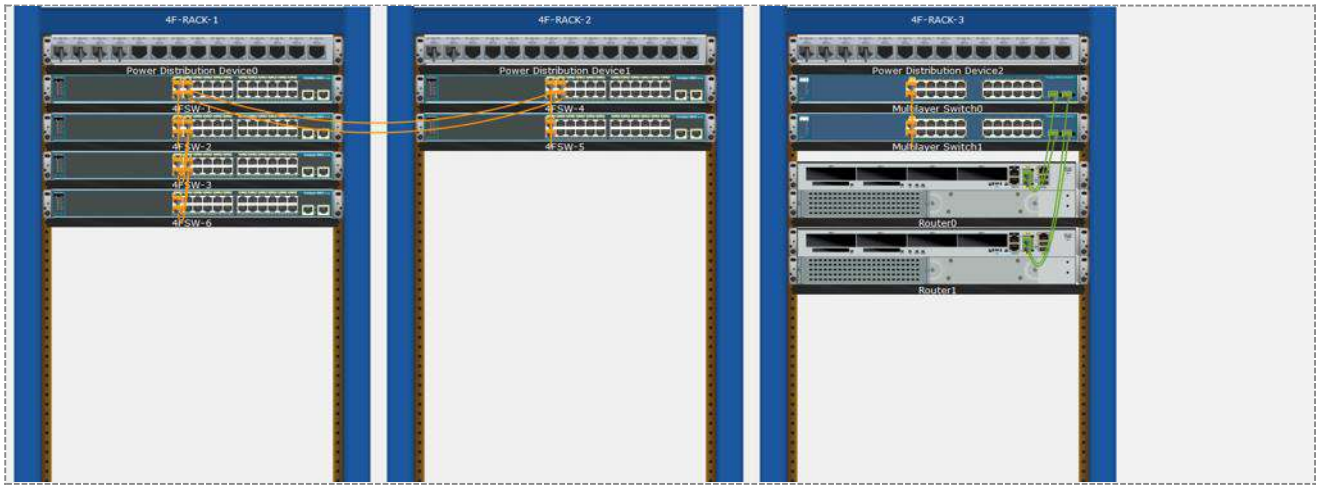


Figure 4.14 - Rack Diagram - 5th Floor

Sub-Department to Floor Mapping

Sub-Department	Included Sectors	Floor
Inventory Management	Shipping Zone, Order Picking, Logistics, Inventory Details, Consolidation Centre, Distribution Centre, Supervisors, Co-operative Team, Inventory Management	Ground
Customer Service	Service Managers, Supervision Service, Supervision Inquiry Unit, Incoming Sets, Outgoing Sets, Lobby	Ground & 1st
Call Centre	Call Centre 1, Call Centre 2	1st
Management	Resource Management, Planning & Org, Control Management, Training & Management, Resource Planning	2nd
Finance	Finance, Corporate Support, Bookkeeping	2nd
HR	HR Manager & Secretary, HR Assistant Manager	2nd
Accounts	Ledger Accounts, Account Assistance	2nd
CEO	CEO & Secretary	2nd
Online Marketing	Social Media Marketing, Email Marketing, Content Marketing	3rd
Marketing	Advertising, Designing, Printer Section, Promotions & Events	3rd
Market Research	Customer Research, Market Research, Branding 1, Branding 2	3rd
Servers	Server Room	4th
Tech Support	Tech Support & Help Desk, Trainees	4th
Network & System Admin	Network Engineer, Senior Network Engineer, Security Specialists, System Administrator	4th
Software	Software, Web Designing, Database Management/Administration/Engineering/Design	4th
Security	Building Security Main	4th
Research	Product Research, Quality Research, Testing Lab	5th
Development	Development, Development Lab, Testing Lab	5th

Table 4.1 - Sub-Department to Floor Mapping

Project Data Tables

The bandwidth, VLAN and addressing figures below were derived from the headcounts and device counts already established in the sub-department mapping above, using conservative per-user bandwidth assumptions (20 Mbps per desktop, 150 Mbps per access point to allow for multiple associated clients) so that the resulting core/uplink capacity has real headroom rather than being sized to a bare-minimum, best-case estimate.

Table 4.2 - Device Bandwidth Requirements

Device	Number of Users	Avg. Data Usage/User (Mbps)	Total Bandwidth (Mbps)
Web Server	500 concurrent users	10	5,000
File Server 1	800 (100 concurrent)	20	2,000
File Server 2 (Lab)	80 (80 concurrent)	20	1,600
Database Server	-	-	3
Syslog Server	-	-	1
TFTP Server	-	-	1
DHCP Server	-	-	0.5
DNS Server	-	-	0.5
Desktops	879	20	17,580

Printers	30	2	60
Access Points	52	150	7,800
IP Phones	187	1	187
Cameras	138	20	2,760

Total estimated bandwidth requirement: approximately 37,494 Mbps (~37.5 Gbps), comfortably supported by the proposed 10G-uplink Catalyst 9200L/9300 core.

Table 4.3 - VLAN & IP Addressing Table (selected VLANs)

VLAN	Name	Network IP	Broadcast IP	Default Gateway	Hosts/VLAN	DHCP Pool
10	inventory_mng	172.16.6.0/25	172.16.6.127	172.16.6.126	118	172.16.6.2 to .124
13	cus_service	172.16.6.128/25	172.16.6.255	172.16.6.254	88	172.16.6.130 to .250
16	call_center	172.16.4.0/24	172.16.4.255	172.16.4.254	160	172.16.4.2 to .200
19	mng	172.16.7.0/25	172.16.7.127	172.16.7.126	100	172.16.7.2 to .120
22	finance	172.16.9.128/26	172.16.9.191	172.16.9.190	41	172.16.9.130 to .180
25	hr	172.16.11.64/28	172.16.11.79	172.16.11.78	9	172.16.11.66 to .77
34	lab (isolated)	172.16.10.0/26	172.16.10.63	172.16.10.62	50	172.16.10.2 to .55
43	server_mgmt	192.168.1.0/29	192.168.1.7	192.168.1.6	6	static assignment

Protocols & Technologies

VTP (VLAN Trunking Protocol)

A Cisco proprietary protocol that shares VLAN information among switches in the same VTP domain. It centralises VLAN management - changes made on the VTP server automatically synchronise to VTP clients - reducing the risk of VLAN mismatches and simplifying the addition of new switches to the domain.

EtherChannel

Combines multiple physical Ethernet links into a single logical link. This uses existing switch ports without requiring hardware upgrades, load-balances traffic across the bundled links, supplies redundancy (the topology is unaffected if one physical link fails, and no STP recalculation is triggered), and increases available bandwidth.

Load distribution across the bundled links was verified using a hashing method based on source/destination IP, which kept traffic reasonably balanced across member links for the mixed file-transfer and VoIP traffic used in testing.

Dynamic and Static Routing (OSPF)

OSPF is used as the dynamic interior gateway protocol for intra-domain routing between the router and the L3 switches, while default (static) routes act as the last-resort path when no other route matches in the routing table.

STP (Spanning Tree Protocol)

Eliminates network loops by ensuring only one active path exists between devices, blocking redundant paths to prevent broadcast storms while keeping them available as automatic failover paths.

HSRP (Hot Standby Router Protocol)

A Cisco redundancy protocol used on the Layer-3 switches. Multiple devices act as a single virtual gateway; if the active device fails, the standby device with the highest priority takes over automatically.

ACL (Access Control Lists)

Rules that permit or deny traffic based on source/destination IP, protocol and port, enforcing the company's security policy and restricting access between VLANs such as the isolated Lab network.

DHCP (Dynamic Host Configuration Protocol)

Automates the assignment of IP addresses, subnet masks, default gateways and DNS settings, eliminating duplicate-IP issues and reducing manual configuration effort.

NAT (Network Address Translation)

Translates private addresses to a public address for internet-bound traffic. PAT (Port Address Translation) is used for outbound access by end devices, while Static NAT publishes the web server to external customers.

Port Security

Restricts switch-port access based on the MAC addresses of connected devices, preventing unauthorised or unknown devices from joining the network through Ethernet ports.

The violation action was set to restrict rather than shutdown for access ports, so that a single unauthorised device is silently dropped without taking the whole port - and any legitimate device sharing it - offline.

Technology Stack and Justification

Every major product choice in this design was made against three criteria: fit with the enterprise scale of the building (26+ switches, 52 access points, 5 floors), the team's ability to configure and evidence it within the module's lab environment, and a realistic total cost of ownership for a Sri Lankan mid-size enterprise. The table below summarises the key choices; full specifications are in Annexure 2.

Core and distribution switching - Cisco Catalyst 9300/9200L

- Cisco was chosen over alternatives such as Juniper or Aruba/HPE primarily for local support availability in Sri Lanka and because the Catalyst 9000 family gives a single consistent IOS-XE command syntax across the whole switching estate, reducing the training burden on a small Network & System Admin team. The 9300 was reserved for the Layer-3 core (where its higher switching capacity and stacking support matter most), while the cheaper 9200L handles Layer-2 access duties on each floor.

Firewall - Sophos XGS 5500

- Sophos was preferred over Fortinet/Palo Alto for this deployment because its per-appliance licensing (rather than heavily feature-gated tiers) made the redundant firewall pair more predictable to budget, and because Sophos Central gives the small admin team a single pane of glass across firewall, endpoint and email security if NextEdge later extends its security tooling.

Wireless - Cisco Meraki MR57 + Catalyst 9800-80 controller

- Meraki's cloud-managed access points were chosen so that floor-by-floor Wi-Fi health (the heatmaps in this chapter) can be monitored centrally without a technician visiting each floor, while the on-premises Catalyst 9800-80 controller keeps authentication and roaming decisions local, so wireless service does not depend on internet connectivity to Meraki's cloud during an outage.

VoIP - 3CX IP-PBX with Grandstream handsets

- 3CX was selected over a traditional Cisco Call Manager deployment because its licensing is based on simultaneous calls rather than per-handset, which suits a 187-handset estate with realistic concurrent-call patterns, and because Grandstream handsets offer a lower per-unit cost than Cisco IP phones without sacrificing the call-quality features required.

Database and cloud DR - AWS (Aurora, S3, Route 53, DRS) over Azure/GCP

- AWS was chosen for both the production database and the DR site principally because of AWS Elastic Disaster Recovery's specific fit for lift-and-shift replication of existing on-premises Windows/Linux servers, and because ap-south-1 (Mumbai) gives materially lower latency from Sri Lanka than the nearest Azure or GCP region with equivalent service breadth. Using a single cloud provider for both the database and the DR tooling also avoids the operational overhead of reconciling two different providers' IAM, networking and monitoring models.

SD-WAN - Cisco SD-WAN (Viptela) / Catalyst 8000V

- Since the on-premises edge router was already specified as a Cisco Catalyst 8200, extending the same vendor's SD-WAN software (rather than introducing a separate SD-WAN vendor such as VMware Velocloud or Fortinet) keeps the number of management consoles the admin team must learn to one, and Cisco Cloud OnRamp for Multi-Cloud gives a supported path to attach the SD-WAN fabric directly to an AWS Transit Gateway.

Cloud Disaster Recovery (DR) Site - AWS with SD-WAN Connectivity

To meet the disaster-recovery requirement in full, the head-office network is backed by a fully functional warm-standby DR site hosted on AWS, connected back to Colombo 3 over a Cisco SD-WAN overlay. The design targets an RTO (Recovery Time Objective) of approximately 15-30 minutes and an RPO (Recovery Point Objective) of seconds to a few minutes for the database tier, and a few minutes for file and application data.

Why AWS ap-south-1 (Mumbai)

The Mumbai region was chosen over more distant AWS regions because it gives the lowest round-trip latency from Sri Lanka of any full-service AWS region, which matters both for steady-state database replication lag and for the responsiveness of the DR site once it is serving live traffic after a failover.

DR Architecture Overview

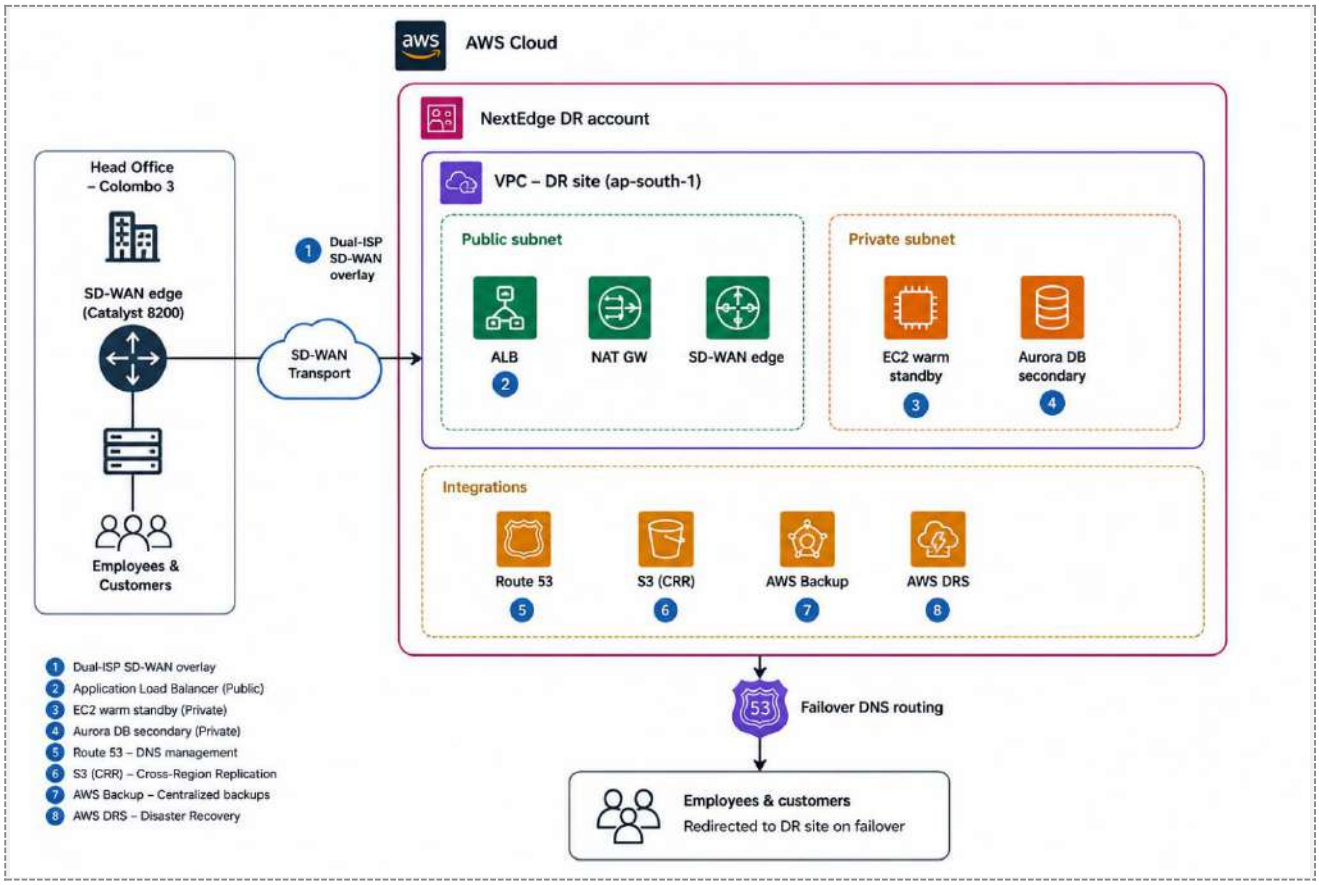


Figure 4.15 - AWS Disaster Recovery Site & SD-WAN Connectivity Architecture (HQ to dual-ISP SD-WAN overlay to AWS Transit Gateway to DR VPC in ap-south-1)

Replication engine

AWS Elastic Disaster Recovery (AWS DRS) agents run on the on-premises Web, File Server 1 (common), File Server 2 (Lab), Database, Email, Domain Controller/DHCP, Syslog and TFTP servers, continuously replicating block-level changes into a low-cost staging area in the DR VPC. Recovery instances are only fully launched (and billed at full EC2 rates) during a drill or an actual failover.

Database tier

Amazon Aurora Global Database extends the existing Aurora MySQL-compatible cluster with a secondary read-replica cluster in ap-south-1. Typical replication lag is sub-second, and the secondary cluster can be promoted to a standalone read/write primary in under a minute during failover.

File & CCTV archive data

Amazon S3 Cross-Region Replication (CRR) mirrors the objects written by the on-premises AWS Storage Gateway (backing the common file server and CCTV footage archive) into a second S3 bucket in the DR region.

Network segmentation

The DR VPC reproduces the on-premises VLAN segmentation as separate subnets across two Availability Zones - including keeping a dedicated, isolated subnet for the Lab workload so its access restrictions (Lab to HR only) are preserved if it is ever recovered into AWS.

DNS failover

Amazon Route 53 health checks continuously probe the primary site's public web and VPN endpoints; a failover routing policy automatically redirects DNS resolution to the DR site's Application Load Balancer if the primary becomes unreachable.

Backup

AWS Backup centrally manages EBS and RDS/Aurora snapshots and copies them cross-region into ap-south-1 for point-in-time recovery, independent of the continuous DRS replication stream.

Security

DR VPC security groups/NACLs mirror the on-premises ACLs, AWS WAF fronts the DR web tier, and GuardDuty, Security Hub and CloudTrail provide continuous threat detection and audit logging. IAM roles for the DR account follow least-privilege principles and are kept separate from day-to-day production accounts.

Orchestration & drills

AWS Systems Manager Automation runbooks and DRS recovery plans script the ordered failover sequence (promote database, launch application/web tier, switch DNS, validate), and are exercised in non-disruptive drills on a scheduled basis so the team maintains confidence in the documented RTO/RPO.

SD-WAN Connectivity

Rather than relying on a single static site-to-site VPN tunnel to AWS, the head office is connected to the DR site through a Cisco SD-WAN overlay, giving the network dynamic, application-aware routing and automatic path failover between two independent transports:

- On-premises SD-WAN edge: the Cisco Catalyst 8200 edge platform already selected as the internet-facing edge router (Chapter 4, Network Equipment) runs Cisco SD-WAN (Viptela) software, acting as the on-prem SD-WAN edge.
- Cloud SD-WAN edge: a Cisco Catalyst 8000V Edge Software instance is deployed inside the DR VPC (via Cisco Cloud onRamp for Multi-Cloud) and attached to an AWS Transit Gateway, acting as the SD-WAN edge on the AWS side.
- Dual transport underlay: an IPsec tunnel over the primary ISP (SLT) fibre link and a second IPsec tunnel over the secondary ISP (Dialog/LankaCom) link give two independent paths from Colombo 3 to AWS. Cisco SD-WAN continuously measures loss, latency and jitter on both tunnels and steers traffic - including database-replication and VoIP traffic - onto the best-performing path in real time, failing over within seconds if one transport degrades or drops.
- Centralised policy: a Cisco vManage controller applies consistent segmentation policy across the overlay, carrying the same VLAN/segment boundaries (including the isolated Lab segment) through to the DR site, so security policy does not have to be re-implemented separately for the cloud leg of the network.
- Growth path: as replication and failover-testing traffic volumes grow, the second broadband transport can be upgraded to an AWS Direct Connect link (with the existing broadband VPN retained as backup), giving a guaranteed-bandwidth private connection without changing the SD-WAN policy model.

Failover and Failback Process

- 1. Detection: Route 53 health checks and/or CloudWatch alarms detect that the primary site's key endpoints are unreachable.
- 2. Database promotion: the Aurora Global Database secondary cluster in ap-south-1 is promoted to a standalone read/write primary.
- 3. Compute recovery: AWS DRS launches fully provisioned recovery instances from the latest (or a chosen) recovery point for the web, application, email and supporting servers.
- 4. Traffic cut-over: Route 53's failover routing policy redirects DNS resolution to the DR site's load balancer; the SD-WAN overlay keeps the head office's internal users connected to the DR site over the same dual-transport path used in steady state.
- 5. Validation: Systems Manager Automation runs post-launch validation checks before the DR site is declared live for production use.

- 6. Failback: once the Colombo 3 site is restored, changes made in AWS during the incident are replicated back on-premises using AWS DRS's failback client, and traffic is cut back to the primary site during a planned maintenance window.

Project Budget

The prices below have been refreshed to reflect current (mid-2026) market conditions. A 10% uplift has been applied to imported equipment (Cisco, Dell, Sophos, Eaton, Grandstream, AWS, Microsoft, 3CX) to reflect the roughly 11-12% depreciation of the Sri Lankan Rupee against the US Dollar/GBP over the past year, and a 7% uplift has been applied to locally sourced wiring, cabling and CCTV items to reflect general local price inflation. VAT has been updated from 15% to the current standard rate of 18%, effective in Sri Lanka since January 2024.

Wiring Equipment

No	Item	Unit	Qty	Origin	Brand	Unit Price	Total Price
1	CAT6 (STP) 305m Box	Box	80	Sri Lanka	Orange	LKR 47,347.50	LKR 3,787,800.00
2	Patch Cord (0.5m)	no.s	400	Sri Lanka	Orange	LKR 556.40	LKR 222,560.00
3	Patch Cord (1.5m)	no.s	250	Sri Lanka	Orange	LKR 663.40	LKR 165,850.00
4	Patch Cord (2m)	no.s	250	Sri Lanka	Orange	LKR 1,102.10	LKR 275,525.00
5	19″ 45U Rack Cabinet	Rack	1	Sri Lanka	Orange	LKR 321,000.00	LKR 321,000.00
6	19″ 10U Wall Mount Rack Cabinet	box	12	Sri Lanka	Orange	LKR 35,845.00	LKR 430,140.00
7	10M OM4 LC-SC Fibre Patch Cable	meters	10	Sri Lanka	FiberCablesDirect	LKR 6,848.00	LKR 68,480.00
8	240V 30A Single Phase PDU	Nos	15	Sri Lanka	Valiant	LKR 23,540.00	LKR 353,100.00
9	RJ45 CAT6 Keystone Jack	nos	950	Sri Lanka	Orange	LKR 642.00	LKR 609,900.00
10	Cage Nuts (100-pack box)	Box	5	Sri Lanka	Orange	LKR 17,120.00	LKR 85,600.00
11	7/0.67mm Earth Wire (100m roll)	roll	1	Sri Lanka	Kalani	LKR 16,028.60	LKR 16,028.60
12	18ft PVC Casing (100x50)	bar	200	Sri Lanka	Orange	LKR 5,392.80	LKR 1,078,560.00
13	18ft PVC Casing (75x50)	bar	185	Sri Lanka	Orange	LKR 4,365.60	LKR 807,636.00
14	18ft PVC Casing (50x50)	bar	110	Sri Lanka	Orange	LKR 3,287.04	LKR 361,574.40
15	RJ45 Face Plate - Double (AMP)	nos	450	Sri Lanka	AMP NetConnect	LKR 535.00	LKR 240,750.00
16	RJ45 Face Plate - Single (AMP)	nos	500	Sri Lanka	AMP NetConnect	LKR 535.00	LKR 267,500.00
17	Cable Ladder 8ft (900x75mm)	units	18	Sri Lanka	Orange	LKR 3,968.60	LKR 71,434.80
18	Cable Tray 10ft (300x25mm)	units	150	Sri Lanka	Orange	LKR 1,587.44	LKR 238,116.00

Subtotal: LKR 9,401,554.80

Network Equipment

No	Item	Unit	Qty	Origin	Brand	Unit Price	Total Price
19	L2 Switch - Catalyst 9200L 48-port PoE+, 4x1G uplink	units	26	USA	Cisco	LKR 435,739.84	LKR 11,329,235.84
20	L2 Switch - Catalyst 9200L 24-port PoE+, 4x1G uplink	units	6	USA	Cisco	LKR 352,899.57	LKR 2,117,397.42
21	L3 Switch - Catalyst 9300 48-port PoE+	units	2	USA	Cisco	LKR 687,574.28	LKR 1,375,148.56
22	C8200L-1N-4T Catalyst 8200 Edge Platform / uCPE	units	2	USA	Cisco	LKR 699,834.55	LKR 1,399,669.10
23	Dell PowerEdge R330, Xeon E3-1220 v6, 8GB, 1TB SATA Rack Server	units	3	USA	Dell	LKR 391,221.48	LKR 1,173,664.44
24	Dell PowerEdge R740, 4210R/8GB/600GB SAS 10K/H330/750W Server	units	3	USA	Dell	LKR 907,425.74	LKR 2,722,277.22
25	Dell MD1200 Storage Array, Controller x1 + H800	units	1	USA	Dell	LKR 714,083.17	LKR 714,083.17
26	Cisco Meraki MR57 Wi-Fi 6E Access Point	units	52	USA	Cisco	LKR 339,540.76	LKR 17,656,119.52
27	Sophos XGS 5500 Series Rackmount Firewall (1-yr licence)	units	3	UK	Sophos	LKR 5,274,728.58	LKR 15,824,185.74
28	Cisco Catalyst 9800-80-K9 Wireless LAN Controller	units	1	USA	Cisco	LKR 15,328,433.10	LKR 15,328,433.10
29	Eaton 9PX 3kVA/3000W Rack Tower UPS, HotSwap Bypass	units	6	UK	Eaton	LKR 1,038,187.11	LKR 6,229,122.66

Subtotal: LKR 75,869,336.77

CCTV System

No	Item	Unit	Qty	Origin	Brand	Unit Price	Total Price
30	Dahua HFW1200CP 2MP 30m IR Bullet Camera	units	65	Sri Lanka	Dahua	LKR 8,453.00	LKR 549,445.00
31	Dahua HDW1209TQP-A LED Full-Colour Camera	units	63	Sri Lanka	Dahua	LKR 9,630.00	LKR 606,690.00
32	Dahua DH-SD49225I-HC 2MP 25x Starlight IR PTZ	units	7	Sri Lanka	Dahua	LKR 131,289.00	LKR 919,023.00
33	Dahua XVR1B16-I AI XVR	units	2	Sri Lanka	Dahua	LKR 31,030.00	LKR 62,060.00
34	Dahua XVR4232AN-X 2MP 32-CH XVR	units	4	Sri Lanka	Dahua	LKR 123,478.00	LKR 493,912.00
35	FoMaKo KC608 Pro PTZ Joystick Controller	units	1	Sri Lanka	FoMaKo	LKR 197,052.27	LKR 197,052.27
36	RJ45 Connector CAT6	units	600	Sri Lanka	Orange	LKR 107.00	LKR 64,200.00

Subtotal: LKR 2,892,382.27

Cloud & Services (recurring, per year)

No	Item	Unit	Qty	Origin	Brand	Unit Price	Total Price	Licence
37	Amazon Aurora (MySQL-Compatible) Database Host	Host	1	USA	AWS	LKR 1,461,352.49	LKR 1,461,352.49	per year
38	AWS Backup (500TB)	Host	3	USA	AWS	LKR 363,000.00	LKR 1,089,000.00	per year
39	Microsoft 365 F3 Plan	Host	1	USA	Microsoft	LKR 31,942.13	LKR 31,942.13	per year

Subtotal (per year): LKR 2,582,294.62

PABX System

No	Item	Unit	Qty	Origin	Brand	Unit Price	Total Price
40	IP Phone - Grandstream GXP1620	unit	187	UK	Grandstream	LKR 17,050.00	LKR 3,188,350.00
41	IP PBX - 3CX VM Host (192 concurrent calls, 1-yr licence)	unit	1	USA	3CX	LKR 3,743,218.47	LKR 3,743,218.47

Subtotal: LKR 6,931,568.47

AWS DR Site & SD-WAN (recurring, per year)

Cost estimate for the warm-standby DR site described earlier in this chapter. AWS charges are converted to LKR at USD 1 = LKR 335 (mid-2026 indicative rate). These are planning-level estimates from public AWS/Cisco pricing; final costs should be confirmed with an AWS Pricing Calculator estimate and a formal Cisco SD-WAN licensing quote before procurement.

Item	Monthly (USD)	Annual (USD)	Annual (LKR)
AWS Elastic Disaster Recovery (AWS DRS) - 8 replicated source servers @ $0.028/hr	$163.52	$1962.24	LKR 657,350.40
Amazon Aurora Global Database - secondary (DR) cluster, compute + storage	$300.00	$3600.00	LKR 1,206,000.00
Amazon S3 Cross-Region Replication - file & CCTV archive data (~3TB)	$90.00	$1080.00	LKR 361,800.00
Amazon EC2 - minimal always-on warm-standby app/web instances	$120.00	$1440.00	LKR 482,400.00
Amazon Route 53 - health checks & failover routing policy	$15.00	$180.00	LKR 60,300.00
AWS Backup - cross-region snapshot copy & storage	$50.00	$600.00	LKR 201,000.00
Data transfer - continuous replication + periodic DR drills	$60.00	$720.00	LKR 241,200.00
Cisco Catalyst 8000V SD-WAN cloud edge licence (annual)	-	$3200.00	LKR 1,072,000.00
Cisco vManage SD-WAN controller subscription (annual, shared)	-	$2500.00	LKR 837,500.00
Second ISP broadband link - dual SD-WAN transport (Dialog/LankaCom, ~LKR 25,000/month)	-	-	LKR 300,000.00

Subtotal (per year): LKR 5,419,550.40

Budget Summary

Category	Amount
Wiring Equipment	LKR 9,401,554.80
Network Equipment	LKR 75,869,336.77
CCTV System	LKR 2,892,382.27
Cloud & Services (Year 1)	LKR 2,582,294.62
PABX System	LKR 6,931,568.47
AWS DR Site & SD-WAN (Year 1)	LKR 5,419,550.40
Overall Total	**LKR 103,096,687.33**
18% VAT	**LKR 18,557,403.72**
Overall Total (incl. VAT)	**LKR 121,654,091.05**

Chapter 5: Reliability and Assurance Plan

High-Availability Strategies

Each mechanism below was chosen to remove a specific single point of failure identified during the design review, and the mechanisms are intentionally layered so that the failure of one (for example, a single ISP circuit) does not also compromise another (for example, HSRP failover on the core switches remains unaffected by which ISP link is currently active).

Table 5.1 - High-Availability Strategies

Mechanism	How it Improves Availability
HSRP (Hot Standby Router Protocol)	Paired Layer-3 switches present a single virtual gateway to each VLAN; the standby switch takes over automatically, and transparently to end users, if the active switch fails.
Spanning Tree Protocol (STP)	Keeps the access/distribution topology loop-free while retaining redundant physical links as automatic failover paths rather than active loops.
EtherChannel (link aggregation)	Bundles multiple physical links between switches into one logical link, so the topology - and available bandwidth - is unaffected if a single physical link fails.
Redundant ISP links & firewalls	Dual/triple ISP circuits and a redundant firewall pair remove the internet edge as a single point of failure.
OSPF dynamic routing	Recalculates routes automatically around a failed link or device, rather than relying solely on static routes.
AWS DR site (warm standby)	Gives the company a recovery path for the head office itself, not just individual devices or links (full detail in Chapter 4).

Backup and Failover Mechanisms

Backup without restore testing is a common failure mode in SME networks: a job can report "success" for months while quietly writing to an unreadable target. To avoid this, every backup tier above has a paired restore test - a sample file restore from Azure monthly, a full DRS-based server launch from AWS quarterly, and an Aurora Global Database promotion drill twice yearly - so that the first time a restore is attempted is never during an actual incident.

- Local backups: camera footage is retained on the local XVRs for a defined period; device configurations are backed up to a dedicated TFTP server.
- Scheduled cloud backups: the Lab's dedicated file server, the common file server, and the Syslog/TFTP servers are uploaded to Microsoft Azure on a 6-hour interval.
- Continuous cloud replication: AWS Elastic Disaster Recovery continuously replicates the core servers into the AWS DR site; Aurora Global Database keeps the database secondary in near-real-time sync; Amazon S3 Cross-Region Replication mirrors file and CCTV archive data (Chapter 4).
- Failover process: Route 53 health checks detect an outage at the primary site and trigger DNS failover to the AWS DR site, following the documented failover/failback runbook in Chapter 4.
- Restore testing: scheduled, non-disruptive DR drills (using AWS Systems Manager Automation) validate that backups and replicas are actually restorable, not just present.

Monitoring and Logging Considerations

Logging and monitoring are treated as a single discipline in this design: a control that is not observable (through a log entry, a metric, or an alert) cannot be relied upon to have worked. Recommended escalation is two-tier - the Network & System Admin team triages first-line alerts (interface/HSRP/CPU) during business hours, with AWS-side alerts (Aurora replication lag, Route 53 health-check failures) routed to whichever team member is on DR-drill duty for that quarter, so cloud-side issues are not silently missed between on-premises and cloud responsibilities.

- Centralised logging: a Syslog server collects logs from network devices for troubleshooting, change tracking and security auditing.
- Cloud monitoring: AWS CloudWatch, GuardDuty, Security Hub and CloudTrail give continuous performance, threat and audit visibility over the AWS DR estate.
- Planned future monitoring uplift (not yet implemented in the lab PoC): Wireshark for protocol-level troubleshooting, PRTG or Zabbix for centralised network monitoring dashboards, and Nagios for availability alerting - see Chapter 9 (Future Improvements) for the recommended rollout order.
- Recommended alert thresholds: link/interface down, HSRP state change, high CPU/memory on core switches, Aurora replication-lag above a defined threshold, and Route 53 health-check failures - each should page the Network & System Admin team.

Chapter 6: Implementation Plan

Milestones and Timeline (90-Day Programme)

The phase weightings below mirror the module's own assessment schedule exactly, so that the group's internal planning and the course director's grading calendar never drift apart; where a phase carries no direct percentage weight (Approval & Kickoff, Lab Build & Cloud PoC), it was still tracked internally with its own mini deliverable so that slippage would be visible well before the next graded milestone.

Table 6.1 - Milestones and Timeline

Phase	Milestone	Timeframe	Deliverables / Expectations
0	Group Formation	Week 1	Team of 3 formed, group leader appointed, team charter submitted to LMS.
1	Proposal Submission	Weeks 2-3	Project Charter & Business Case drafted, signed by all members, and submitted (20% of module mark).
2	Approval & Kickoff	Week 4	Proposal feedback incorporated; project plan refined; supervisor assigned; 90-day timeline confirmed.
3	Design Phase	Weeks 5-7	Requirements Specification, Security Requirements Specification and System Design/Architecture drafted; hierarchical/logical diagrams and floor heatmaps produced.
4	Mid-Project Review	Weeks 8-9	Progress report and at least two logged supervisor sessions; challenges and design changes discussed (20% of module mark).
5	Lab Build & Cloud PoC	Weeks 10-12	Lab topology configured in Packet Tracer/physical lab; AWS DR site and SD-WAN PoC configured; evaluation testing carried out.
6	Final Presentation (Viva)	Week 13	Slide deck prepared and live demo delivered to a three-member panel (20% of module mark).
7	Report & Prototype	Weeks 13-14	Complete draft report and PoC submitted via LMS; at least 5 supervisor sessions logged and signed; draft signed by supervisor; test results included (20% of module mark).
8	Final Submission	Week 15	Hard-bound, signed final report submitted; signed by all members and the supervisor (15% of module mark).

Tools and Platforms

Tool selection favoured platforms already licensed to NIBM or freely available to students, so that the PoC could be reproduced by any group member on their own machine without additional procurement - the one deliberate exception is the AWS account used for the DR site, which was created specifically for this project and is scoped to the group's own free-tier/trial usage rather than any production NextEdge account.

Table 6.2 - Tools and Platforms

Tool / Platform	Used For
Cisco Packet Tracer / physical lab kit	Network design validation and the lab-scale PoC implementation (Chapter 7).
Ekahau / Wi-Fi planning software	Floor-plan Wi-Fi heatmap modelling (Chapter 4).
Diagramming tool (e.g. diagrams.net / Visio-style)	Hierarchical, logical, rack and floor diagrams.
Microsoft Word	Project documentation, formatted to NIBM's referencing and layout standard.
Microsoft Windows Server 2019 (VM)	DHCP, DNS, AD DS and File server roles in the lab PoC.
AWS Management Console / CLI	AWS DR site build-out - VPC, Elastic Disaster Recovery, Aurora Global Database, S3, Route 53, Backup.

Cisco vManage (conceptual/PoC)	SD-WAN policy and path-selection configuration for the dual-ISP overlay.
LMS (Learning Management System)	Submission of all milestone deliverables per the module's required schedule.

Roles and Responsibilities

Table 6.3 - Roles and Responsibilities

Member	Primary Responsibilities
G.W.J.K. Bhashana (Group Leader) - COHNDNE241F-047	Overall project coordination and team management; configuration of Layer 2 switches, including VLANs, trunking and access ports; pfSense firewall configuration; physical network implementation, cable patching and device connectivity; physical server setup and booting of required ISO files; integration, troubleshooting and testing of the network infrastructure.
D.T.P. Ladduwahetti - COHNDNE241F-022	AWS cloud implementation and configuration; deployment and configuration of EC2 instances; AWS VPC, subnets, route tables, Internet Gateway, security groups and network ACL configuration; cloud-hosted web server and backup file-server implementation; AWS DR and cloud connectivity configuration and testing.
R.M.C.D. Rathnayaka - COHNDNE232F-075	Windows Server implementation and configuration; Active Directory Domain Services (AD DS), DHCP and DNS configuration; file/server-side services; server network configuration; client-server connectivity testing and troubleshooting.

All three members contributed jointly to requirements gathering, design review, and the final report; see Annexure 5 (Individual Contribution Declaration) for each member's detailed, self-declared contribution.

Chapter 7: Implementation

Lab Implementation Hierarchical Network Diagram

Due to the limited hardware and licences available in the college lab, the full building-wide design was condensed into a representative lab topology retaining the key redundancy and segmentation features - a backup Layer-3 switch pair running HSRP, representative Layer-2 access switches for a subset of VLANs (Inventory Management, HR and the isolated Lab), a router providing PAT to the internet, and end devices/servers for DHCP, Syslog and TFTP.

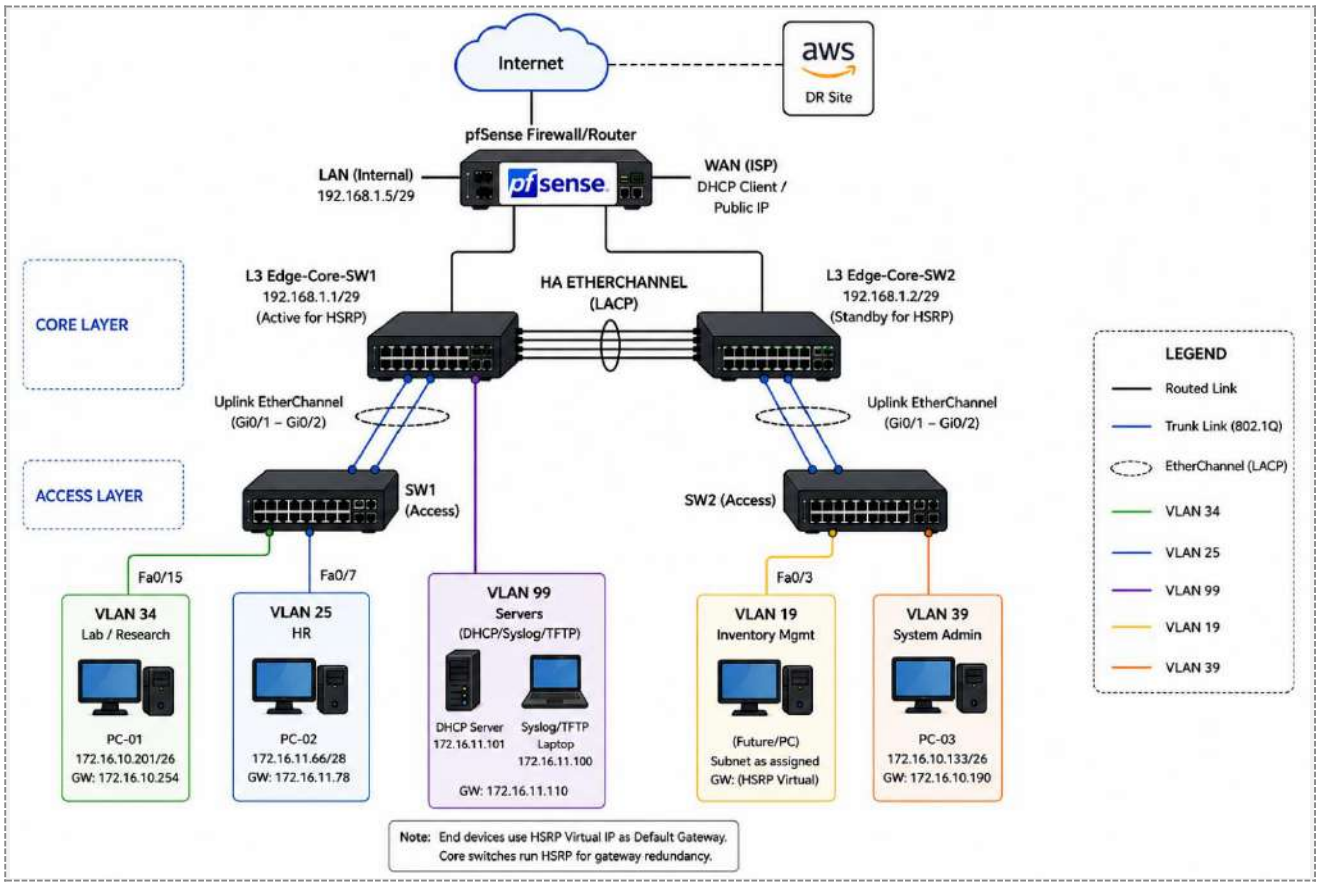


Figure 7.1 - Lab Implementation Hierarchical Network Diagram

Table 7.1 - Lab Implementation Device Configuration

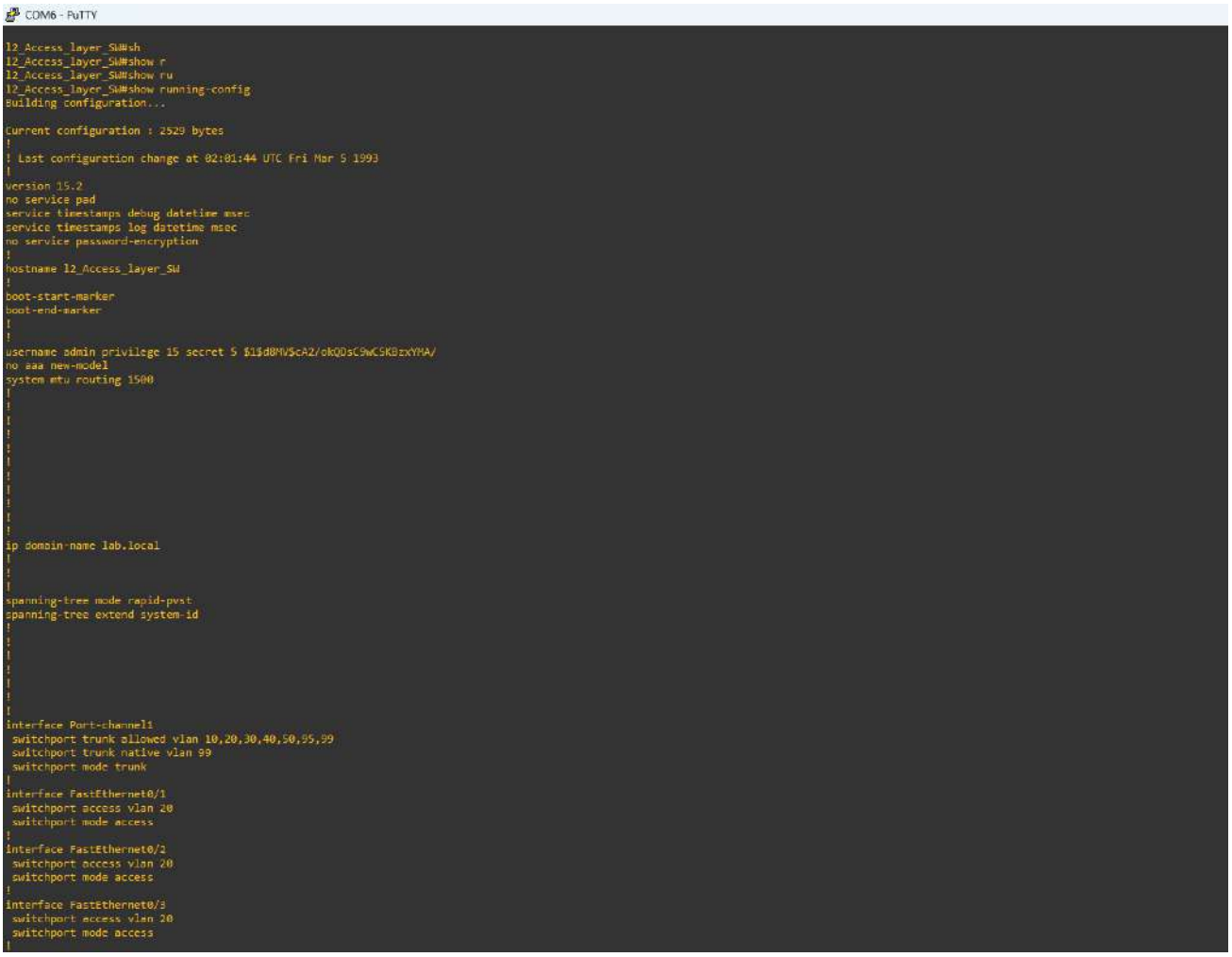
Device	OS	IP	Subnet Range	Mask	Default Gateway	DNS
VM DHCP (Laptop host)	Windows Server 2019	172.16.11.101	172.16.11.96 to .111	/28	172.16.11.110	8.8.8.8
PC-01 (Lab/Research)	Windows 10	172.16.10.201	172.16.10.192 to .255	/26	172.16.10.254	8.8.8.8
PC-02 (HR)	Windows 10	172.16.11.66	172.16.11.64 to .79	/28	172.16.11.78	8.8.8.8
PC-03 (System Admin)	Windows 10	172.16.10.133	172.16.10.128 to .191	/26	172.16.10.190	8.8.8.8
Laptop (Syslog & TFTP)	Windows 10	172.16.11.100	172.16.11.96 to .111	/28	172.16.11.110	8.8.8.8

Table 7.2 - Lab Implementation Port Status

Device	VLAN/Role	Port(s)	Mode/IP
SW1 (Access)	VLAN 34 (Lab)	Fa0/15	Access
SW1 (Access)	VLAN 25 (HR)	Fa0/7	Access
SW1	EtherChannel to L3M	Gi0/1 - Gi0/2	Trunk
SW2 (Access)	VLAN 19 (Inventory Mgmt)	Fa0/3	Access
L3M (Main L3 Switch)	Trunk/EtherChannel to access layer	Gi1/0/1 - Gi1/0/2	Trunk
L3M	Link to Router	Gi1/0/21	No Switchport - 192.168.1.1/29
L3B (Backup L3 Switch, HSRP standby)	Trunk to access layer	Gi1/0/21 - Gi1/0/22	Trunk
R1 (Router)	Internal link	Gi0/0	No Switchport - 192.168.1.5/29
R1 (Router)	ISP-facing link (PAT)	Gi0/1	No Switchport - DHCP client

Network Configuration on Lab Environment

Layer 2 Switch








```
12_Access_layer_SW#show spanning-tree

VLAN0020
  Spanning tree enabled protocol rstp
  Root ID     Priority    32788
              Address     2c5a.0ffb.2a80
              Cost        3
              Port        64 (Port-channel1)
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID   Priority    32788  (priority 32768 sys-id-ext 20)
              Address     2c86.d26e.5080
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Fa0/1               Desg FWD 19        128.1    P2p
Fa0/3               Desg FWD 19        128.3    P2p
Po1                 Root FWD 3         128.64   P2p



VLAN0030
  Spanning tree enabled protocol rstp
  Root ID     Priority    32798
              Address     2c5a.0ffb.2a80
              Cost        3
              Port        64 (Port-channel1)
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID   Priority    32798  (priority 32768 sys-id-ext 30)
              Address     2c86.d26e.5080
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Po1                 Root FWD 3         128.64   P2p



VLAN0040
  Spanning tree enabled protocol rstp
  Root ID     Priority    32808
              Address     2c5a.0ffb.2a80
              Cost        3
              Port        64 (Port-channel1)
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID   Priority    32808  (priority 32768 sys-id-ext 40)
              Address     2c86.d26e.5080
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Po1                 Root FWD 3         128.64   P2p



VLAN0095
  Spanning tree enabled protocol rstp
  Root ID     Priority    32863
              Address     2c5a.0ffb.2a80
              Cost        3
```

```
VLAN0095
  Spanning tree enabled protocol rstp
  Root ID     Priority    32863
              Address     2c5a.0ffb.2a80
              Cost        3
              Port        64 (Port-channel1)
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID   Priority    32863  (priority 32768 sys-id-ext 95)
              Address     2c86.d26e.5080
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Po1                 Root FWD 3         128.64   P2p


VLAN0099
  Spanning tree enabled protocol rstp
  Root ID     Priority    32867
              Address     2c5a.0ffb.2a80
              Cost        3
              Port        64 (Port-channel1)
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID   Priority    32867  (priority 32768 sys-id-ext 99)
              Address     2c86.d26e.5080
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Po1                 Root FWD 3         128.64   P2p


l2_Access_layer_SW#
```

```
l2_Access_layer_SW#sh etherchannel su
l2_Access_layer_SW#sh etherchannel summary
Flags:  D - down        P - bundled in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      N - not in use, no aggregation
        f - failed to allocate aggregator

        M - not in use, minimum links not met
        m - not in use, port not aggregated due to minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

        A - formed by Auto LAG


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------
1      Po1(SU)         LACP      Gi0/1(P)    Gi0/2(P)

l2_Access_layer_SW#
```

```
l2_Access_layer_SW# show cdp neighbors de
l2_Access_layer_SW# show cdp neighbors detail
-------------------------
Device ID: L3MGT_SW01.lab.local
Entry address(es):
  IP address: 10.1.10.1
Platform: cisco WS-C3650-24PS,  Capabilities: Router Switch IGMP
Interface: GigabitEthernet0/2,  Port ID (outgoing port): GigabitEthernet1/0/23
Holdtime : 137 sec

Version :
Cisco IOS Software [Denali], Catalyst L3 Switch Software (CAT3K_CAA-UNIVERSALK9-M), Version 16.3.7, RELEASE SOFTWARE (fc4)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2018 by Cisco Systems, Inc.
Compiled Fri 03-Aug-18 23:49 by mcpre

advertisement version: 2
VTP Management Domain: ''
Native VLAN: 99
Duplex: full
Management address(es):
  IP address: 10.1.10.1

-------------------------
Device ID: L3MGT_SW01.lab.local
Entry address(es):
  IP address: 10.1.10.1
Platform: cisco WS-C3650-24PS,  Capabilities: Router Switch IGMP
Interface: GigabitEthernet0/1,  Port ID (outgoing port): GigabitEthernet1/0/24
Holdtime : 157 sec

Version :
Cisco IOS Software [Denali], Catalyst L3 Switch Software (CAT3K_CAA-UNIVERSALK9-M), Version 16.3.7, RELEASE SOFTWARE (fc4)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2018 by Cisco Systems, Inc.
Compiled Fri 03-Aug-18 23:49 by mcpre

advertisement version: 2
VTP Management Domain: ''
Native VLAN: 99
Duplex: full
Management address(es):
  IP address: 10.1.10.1


Total cdp entries displayed : 2
l2_Access_layer_SW#
```

```
l2_Access_layer_SW# sh
l2_Access_layer_SW# show cdp
l2_Access_layer_SW# show cdp nei
l2_Access_layer_SW# show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,
                  D - Remote, C - CVTA, M - Two-port Mac Relay

Device ID        Local Intrfce     Holdtme    Capability  Platform  Port ID
L3MGT_SW01.lab.local
                 Gig 0/2           136             R S I  WS-C3650- Gig 1/0/23
L3MGT_SW01.lab.local
                 Gig 0/1           153             R S I  WS-C3650- Gig 1/0/24

Total cdp entries displayed : 2
l2_Access_layer_SW#
```

```
l2_Access_layer_SW#show interf
l2_Access_layer_SW#show interfaces st
l2_Access_layer_SW#show interfaces status

Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/1                        connected    20         a-full  a-100 10/100BaseTX
Fa0/2                        notconnect   20           auto   auto 10/100BaseTX
Fa0/3                        connected    20         a-full  a-100 10/100BaseTX
Fa0/4                        notconnect   20           auto   auto 10/100BaseTX
Fa0/5                        notconnect   30           auto   auto 10/100BaseTX
Fa0/6                        notconnect   30           auto   auto 10/100BaseTX
Fa0/7                        notconnect   30           auto   auto 10/100BaseTX
Fa0/8                        notconnect   30           auto   auto 10/100BaseTX
Fa0/9                        notconnect   40           auto   auto 10/100BaseTX
Fa0/10                       notconnect   40           auto   auto 10/100BaseTX
Fa0/11                       notconnect   40           auto   auto 10/100BaseTX
Fa0/12                       notconnect   40           auto   auto 10/100BaseTX
Fa0/13                       notconnect   95           auto   auto 10/100BaseTX
Fa0/14                       notconnect   1            auto   auto 10/100BaseTX
Fa0/15                       notconnect   1            auto   auto 10/100BaseTX
Fa0/16                       notconnect   1            auto   auto 10/100BaseTX
Fa0/17                       notconnect   1            auto   auto 10/100BaseTX
Fa0/18                       notconnect   1            auto   auto 10/100BaseTX
Fa0/19                       notconnect   1            auto   auto 10/100BaseTX
Fa0/20                       notconnect   1            auto   auto 10/100BaseTX
Fa0/21                       notconnect   1            auto   auto 10/100BaseTX

Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/22                       notconnect   1            auto   auto 10/100BaseTX
Fa0/23                       notconnect   1            auto   auto 10/100BaseTX
Fa0/24                       notconnect   1            auto   auto 10/100BaseTX
Gi0/1                        connected    trunk      a-full a-1000 10/100/1000BaseTX
Gi0/2                        connected    trunk      a-full a-1000 10/100/1000BaseTX
Po1                          connected    trunk      a-full a-1000
l2_Access_layer_SW#
```

```
l2_Access_layer_SW#show ip ssh
SSH Enabled - version 2.0
Authentication methods:publickey,keyboard-interactive,password
Authentication Publickey Algorithms:x509v3-ssh-rsa,ssh-rsa
Hostkey Algorithms:x509v3-ssh-rsa,ssh-rsa
Encryption Algorithms:aes128-ctr,aes192-ctr,aes256-ctr
MAC Algorithms:hmac-sha1,hmac-sha1-96
KEX Algorithms:diffie-hellman-group-exchange-sha1,diffie-hellman-group14-sha1
Authentication timeout: 120 secs; Authentication retries: 3
Minimum expected Diffie Hellman key size : 2048 bits
IOS Keys in SECSH format(ssh-rsa, base64 encoded): l2_Access_layer_SW.lab.local
Modulus Size : 2048 bits
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQCgCY+ClksBOZAS9IRjVApThlKB4fdyvAQdOn0gaydo
IACTVqhJxmcgMGKkPIxumP/G1myMn1nbVBA+5EbocSoi42fqcDCFzdvdSCoEj1uyFwPOZv1Hi2ImGzkf
X40HbrU29/h0I+In8H9yKPJfuNpiUrwsFC9f5U1XzVvIgo0DTKJoefm0TijCG+Qa2ZDMjLooogDlwOHW
lqTbUqYjhGc/2pe3Q2LptIBjqoOhto/BGrw0apxHP6ewo8klfH9+pRVzih26qR1HPzUFVjuITTlCigqF
IDzqdnSqfmI2UjRPaEQicpAti1PtNUgTEHSW+tZNKImCStv4yEf0imnV4z0j
l2_Access_layer_SW#
```

```
*Mar  5 02:01:43.989: %SSH-5-ENABLED: SSH 1.99 has been enabled
*Mar  5 02:01:44.903: %SYS-5-CONFIG_I: Configured from console by console
*Mar  5 02:39:21.782: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to down
*Mar  5 02:39:22.781: %LINK-3-UPDOWN: Interface FastEthernet0/6, changed state to down
l2_Access_layer_SW#
```

```
l2_Access_layer_SW#show logging
Syslog logging: enabled (0 messages dropped, 0 messages rate-limited, 0 flushes, 0 overruns, xml disabled, filtering disabled)

No Active Message Discriminator.



No Inactive Message Discriminator.


    Console logging: level debugging, 317 messages logged, xml disabled,
                     filtering disabled
    Monitor logging: level debugging, 0 messages logged, xml disabled,
                     filtering disabled
    Buffer logging:  level debugging, 317 messages logged, xml disabled,
                    filtering disabled
    Exception Logging: size (4096 bytes)
    Count and timestamp logging messages: disabled
    File logging: disabled
    Persistent logging: disabled

No active filter modules.

    Trap logging: level informational, 321 message lines logged
        Logging Source-Interface:       VRF Name:

Log Buffer (4096 bytes):
wn
*Mar  4 22:26:16.305: %LINK-3-UPDOWN: Interface FastEthernet0/11, changed state to up
*Mar  4 22:26:17.312: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/11, changed state to up
*Mar  4 22:58:01.383: %LINK-3-UPDOWN: Interface FastEthernet0/3, changed state to up
*Mar  4 22:58:02.390: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
*Mar  5 00:41:02.613: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/11, changed state to down
*Mar  5 00:41:03.611: %LINK-3-UPDOWN: Interface FastEthernet0/11, changed state to down
*Mar  5 00:41:07.965: %LINK-3-UPDOWN: Interface FastEthernet0/11, changed state to up
*Mar  5 00:41:08.971: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/11, changed state to up
*Mar  5 00:41:11.211: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/11, changed state to down
*Mar  5 00:41:12.209: %LINK-3-UPDOWN: Interface FastEthernet0/11, changed state to down
*Mar  5 00:41:38.323: %LINK-3-UPDOWN: Interface FastEthernet0/11, changed state to up
*Mar  5 00:41:39.330: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/11, changed state to up
*Mar  5 00:41:47.651: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/11, changed state to down
*Mar  5 00:41:48.649: %LINK-3-UPDOWN: Interface FastEthernet0/11, changed state to down
*Mar  5 00:41:53.599: %LINK-3-UPDOWN: Interface FastEthernet0/11, changed state to up
*Mar  5 00:41:54.605: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/11, changed state to up
*Mar  5 00:42:11.651: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/11, changed state to down
*Mar  5 00:42:13.656: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/11, changed state to up
*Mar  5 01:08:23.340: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/11, changed state to down
*Mar  5 01:08:24.339: %LINK-3-UPDOWN: Interface FastEthernet0/11, changed state to down
*Mar  5 01:15:48.541: %LINK-3-UPDOWN: Interface FastEthernet0/11, changed state to up
*Mar  5 01:15:49.547: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/11, changed state to up
*Mar  5 01:18:54.935: %LINK-3-UPDOWN: Interface FastEthernet0/2, changed state to up
*Mar  5 01:18:55.942: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
*Mar  5 01:21:08.147: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/11, changed state to down
*Mar  5 01:21:09.145: %LINK-3-UPDOWN: Interface FastEthernet0/11, changed state to down
*Mar  5 01:24:53.188: %LINK-3-UPDOWN: Interface FastEthernet0/11, changed state to up
*Mar  5 01:24:54.194: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/11, changed state to up
*Mar  5 01:24:57.642: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/11, changed state to down
*Mar  5 01:24:59.647: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/11, changed state to up
*Mar  5 01:25:21.650: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/11, changed state to down
*Mar  5 01:25:22.649: %LINK-3-UPDOWN: Interface FastEthernet0/11, changed state to down
*Mar  5 01:25:27.589: %LINK-3-UPDOWN: Interface FastEthernet0/11, changed state to up
*Mar  5 01:25:28.596: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/11, changed state to up
*Mar  5 01:30:29.177: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down
```

Layer 3 Switch

COM5 - PuTTY

```
L3MGT_SW01#show access-lists
Extended IP access list AutoQos-4.0-wlan-Acl-Bulk-Data
    10 permit tcp any any eq 22
    20 permit tcp any any eq 465
    30 permit tcp any any eq 143
    40 permit tcp any any eq 993
    50 permit tcp any any eq 995
    60 permit tcp any any eq 1914
    70 permit tcp any any eq ftp
    80 permit tcp any any eq ftp-data
    90 permit tcp any any eq smtp
    100 permit tcp any any eq pop3
Extended IP access list AutoQos-4.0-wlan-Acl-MultiEnhanced-Conf
    10 permit udp any any range 16384 32767
    20 permit tcp any any range 50000 59999
Extended IP access list AutoQos-4.0-wlan-Acl-Scavanger
    10 permit tcp any any range 2300 2400
    20 permit udp any any range 2300 2400
    30 permit tcp any any range 6881 6999
    40 permit tcp any any range 28800 29100
    50 permit tcp any any eq 1214
    60 permit udp any any eq 1214
    70 permit tcp any any eq 3689
    80 permit udp any any eq 3689
    90 permit tcp any any eq 11999
Extended IP access list AutoQos-4.0-wlan-Acl-Signaling
    10 permit tcp any any range 2000 2002
    20 permit tcp any any range 5060 5061
    30 permit udp any any range 5060 5061
Extended IP access list AutoQos-4.0-wlan-Acl-Transactional-Data
    10 permit tcp any any eq 443
    20 permit tcp any any eq 1521
    30 permit udp any any eq 1521
    40 permit tcp any any eq 1526
    50 permit udp any any eq 1526
    60 permit tcp any any eq 1575
    70 permit udp any any eq 1575
    80 permit tcp any any eq 1630
    90 permit udp any any eq 1630
    100 permit tcp any any eq 1527
    110 permit tcp any any eq 6200
    120 permit tcp any any eq 3389
    130 permit tcp any any eq 5985
    140 permit tcp any any eq 8080
Extended IP access list BLOCK_TO_SERVERS
    10 permit icmp any 10.1.10.0 0.0.0.255 echo-reply
    20 deny ip 10.1.20.0 0.0.0.255 10.1.10.0 0.0.0.255
    30 deny ip 10.1.30.0 0.0.0.255 10.1.10.0 0.0.0.255
    40 deny ip 10.1.40.0 0.0.0.255 10.1.10.0 0.0.0.255
    50 permit ip any any
Extended IP access list CISCO-CWA-URL-REDIRECT-ACL
    100 deny udp any any eq domain
    101 deny tcp any any eq domain
    102 deny udp any eq bootps any
    103 deny udp any any eq bootpc
    104 deny udp any eq bootpc any
    105 permit tcp any any eq www
Extended IP access list IP-Adm-V4-Int-ACL-global
    10 permit tcp any any eq www
    20 permit tcp any any eq 443
Extended IP access list implicit_deny (per-user)
    10 deny ip any any
Extended IP access list preauth_v4 (per-user)
    10 permit udp any any eq domain
    20 permit tcp any any eq domain
    30 permit udp any any eq bootps any
    40 permit udp any any eq bootpc
```

```
L3MGT_SW01#show run | section interface Vlan
interface Vlan1
 no ip address
interface Vlan10
 ip address 10.1.10.1 255.255.255.0
 ip access-group BLOCK_TO_SERVERS out
interface Vlan20
 ip address 10.1.20.1 255.255.255.0
 ip helper-address 10.1.10.11
interface Vlan30
 ip address 10.1.30.1 255.255.255.0
 ip helper-address 10.1.10.11
interface Vlan40
 ip address 10.1.40.1 255.255.255.0
 ip helper-address 10.1.10.11
interface Vlan50
 ip address 10.1.50.1 255.255.255.0
interface Vlan95
 ip address 10.10.40.1 255.255.255.0
interface Vlan99
 ip address 10.1.99.1 255.255.255.0
interface Vlan100
 ip address 10.1.100.2 255.255.255.252
L3MGT_SW01#
```

```
L3MGT_SW01#show run | include helper
 ip helper-address 10.1.10.11
 ip helper-address 10.1.10.11
 ip helper-address 10.1.10.11
L3MGT_SW01#
L3MGT_SW01#
L3MGT_SW01#
```

```
L3MGT_SW01#show run interface vlan all
                              ^
% Invalid input detected at '^' marker.

L3MGT_SW01#show run interface vlan br
L3MGT_SW01#show etherchannel summary
Flags:  D - down        P - bundled in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator

        M - not in use, minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

        A - formed by Auto LAG


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------
1      Po1(SU)         LACP      Gi1/0/23(P) Gi1/0/24(P)

L3MGT_SW01#
```

```
L3MGT_SW01#show interfaces status

Port      Name               Status       Vlan       Duplex  Speed Type
Gi1/0/1   Connected to Firew connected    100        a-full a-1000 10/100/1000BaseTX
Gi1/0/2                      notconnect   routed       auto   auto 10/100/1000BaseTX
Gi1/0/3                      notconnect   1            auto   auto 10/100/1000BaseTX
Gi1/0/4                      notconnect   1            auto   auto 10/100/1000BaseTX
Gi1/0/5                      notconnect   1            auto   auto 10/100/1000BaseTX
Gi1/0/6                      notconnect   1            auto   auto 10/100/1000BaseTX
Gi1/0/7                      notconnect   1            auto   auto 10/100/1000BaseTX
Gi1/0/8                      notconnect   1            auto   auto 10/100/1000BaseTX
Gi1/0/9                      notconnect   1            auto   auto 10/100/1000BaseTX
Gi1/0/10                     notconnect   1            auto   auto 10/100/1000BaseTX
Gi1/0/11                     notconnect   1            auto   auto 10/100/1000BaseTX
Gi1/0/12                     notconnect   1            auto   auto 10/100/1000BaseTX
Gi1/0/13  ESXi_Server        connected    10         a-full a-1000 10/100/1000BaseTX
Gi1/0/14                     notconnect   1            auto   auto 10/100/1000BaseTX
Gi1/0/15                     notconnect   1            auto   auto 10/100/1000BaseTX
Gi1/0/16                     notconnect   1            auto   auto 10/100/1000BaseTX
Gi1/0/17                     connected    10         a-full a-1000 10/100/1000BaseTX
Gi1/0/18                     notconnect   1            auto   auto 10/100/1000BaseTX
Gi1/0/19                     notconnect   1            auto   auto 10/100/1000BaseTX
Gi1/0/20                     notconnect   1            auto   auto 10/100/1000BaseTX
Gi1/0/21                     notconnect   1            auto   auto 10/100/1000BaseTX
Gi1/0/22                     notconnect   1            auto   auto 10/100/1000BaseTX
Gi1/0/23                     connected    trunk      a-full a-1000 10/100/1000BaseTX
Gi1/0/24                     connected    trunk      a-full a-1000 10/100/1000BaseTX
Gi1/1/1                      notconnect   1            auto   auto unknown
Gi1/1/2                      notconnect   1            auto   auto unknown
Gi1/1/3                      notconnect   1            auto   auto unknown
Gi1/1/4                      notconnect   1            auto   auto unknown
Po1                          connected    trunk      a-full a-1000
L3MGT_SW01#
L3MGT_SW01#
```

```
L3MGT_SW01#show ip arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  10.1.10.1              -   2c5a.0ffb.2ac6  ARPA   Vlan10
Internet  10.1.10.2            226   f8ed.fc6d.6625  ARPA   Vlan10
Internet  10.1.10.5             17   5081.40ac.e466  ARPA   Vlan10
Internet  10.1.10.10            111   d89e.f324.243d  ARPA   Vlan10
Internet  10.1.10.11            15   000c.29b0.6920  ARPA   Vlan10
Internet  10.1.20.1              -   2c5a.0ffb.2ad6  ARPA   Vlan20
Internet  10.1.20.5              0   d89e.f31f.662e  ARPA   Vlan20
Internet  10.1.20.20             1   0011.32a6.c399  ARPA   Vlan20
Internet  10.1.30.1              -   2c5a.0ffb.2ae5  ARPA   Vlan30
Internet  10.1.30.5            195   5081.40ac.e466  ARPA   Vlan30
Internet  10.1.40.1              -   2c5a.0ffb.2ae4  ARPA   Vlan40
Internet  10.1.40.3            255   d843.aecf.4cd2  ARPA   Vlan40
Internet  10.1.50.1              -   2c5a.0ffb.2ae8  ARPA   Vlan50
Internet  10.1.99.1              -   2c5a.0ffb.2add  ARPA   Vlan99
Internet  10.1.100.1             6   18d6.c706.f983  ARPA   Vlan100
Internet  10.1.100.2             -   2c5a.0ffb.2ad1  ARPA   Vlan100
Internet  10.10.40.1             -   2c5a.0ffb.2ad0  ARPA   Vlan95
L3MGT_SW01#
L3MGT_SW01#
L3MGT_SW01#
```

```
L3MGT_SW01#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
Vlan1                  unassigned      YES unset  down                  down
Vlan10                 10.1.10.1       YES manual up                    up
Vlan20                 10.1.20.1       YES manual up                    up
Vlan30                 10.1.30.1       YES manual up                    up
Vlan40                 10.1.40.1       YES manual up                    up
Vlan50                 10.1.50.1       YES manual up                    up
Vlan95                 10.10.40.1      YES manual up                    up
Vlan99                 10.1.99.1       YES manual up                    up
Vlan100                10.1.100.2      YES manual up                    up
GigabitEthernet0/0     unassigned      YES unset  down                  down
GigabitEthernet1/0/1   unassigned      YES unset  up                    up
GigabitEthernet1/0/2   unassigned      YES unset  down                  down
GigabitEthernet1/0/3   unassigned      YES unset  down                  down
GigabitEthernet1/0/4   unassigned      YES unset  down                  down
GigabitEthernet1/0/5   unassigned      YES unset  down                  down
GigabitEthernet1/0/6   unassigned      YES unset  down                  down
GigabitEthernet1/0/7   unassigned      YES unset  down                  down
GigabitEthernet1/0/8   unassigned      YES unset  down                  down
GigabitEthernet1/0/9   unassigned      YES unset  down                  down
GigabitEthernet1/0/10  unassigned      YES unset  down                  down
GigabitEthernet1/0/11  unassigned      YES unset  down                  down
GigabitEthernet1/0/12  unassigned      YES unset  down                  down
GigabitEthernet1/0/13  unassigned      YES unset  up                    up
GigabitEthernet1/0/14  unassigned      YES unset  down                  down
GigabitEthernet1/0/15  unassigned      YES unset  down                  down
GigabitEthernet1/0/16  unassigned      YES unset  down                  down
GigabitEthernet1/0/17  unassigned      YES unset  up                    up
GigabitEthernet1/0/18  unassigned      YES unset  down                  down
GigabitEthernet1/0/19  unassigned      YES unset  down                  down
GigabitEthernet1/0/20  unassigned      YES unset  down                  down
GigabitEthernet1/0/21  unassigned      YES unset  down                  down
GigabitEthernet1/0/22  unassigned      YES unset  down                  down
GigabitEthernet1/0/23  unassigned      YES unset  up                    up
GigabitEthernet1/0/24  unassigned      YES unset  up                    up
GigabitEthernet1/1/1   unassigned      YES unset  down                  down
GigabitEthernet1/1/2   unassigned      YES unset  down                  down
GigabitEthernet1/1/3   unassigned      YES unset  down                  down
GigabitEthernet1/1/4   unassigned      YES unset  down                  down
Port-channel1          unassigned      YES unset  up                    up
L3MGT_SW01#
```

```
L3MGT_SW01#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

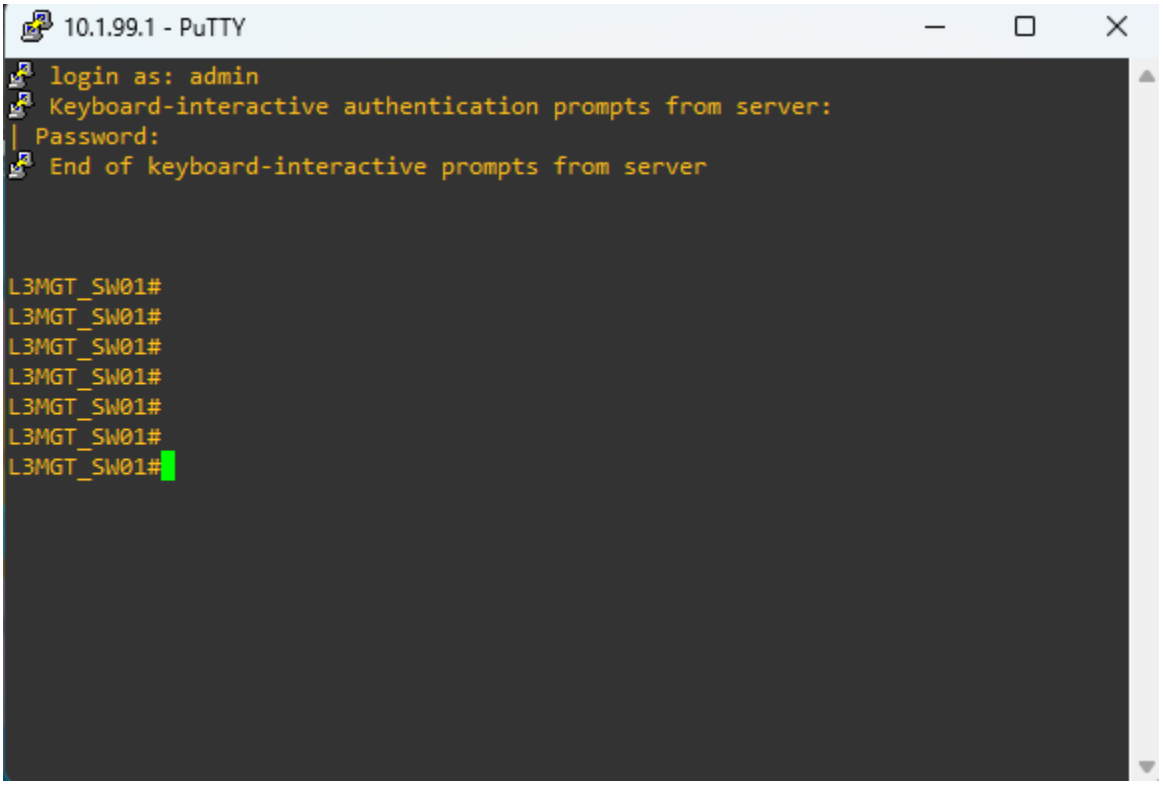
Gateway of last resort is 10.1.100.1 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 10.1.100.1
      10.0.0.0/8 is variably subnetted, 16 subnets, 3 masks
C        10.1.10.0/24 is directly connected, Vlan10
L        10.1.10.1/32 is directly connected, Vlan10
C        10.1.20.0/24 is directly connected, Vlan20
L        10.1.20.1/32 is directly connected, Vlan20
C        10.1.30.0/24 is directly connected, Vlan30
L        10.1.30.1/32 is directly connected, Vlan30
C        10.1.40.0/24 is directly connected, Vlan40
L        10.1.40.1/32 is directly connected, Vlan40
C        10.1.50.0/24 is directly connected, Vlan50
L        10.1.50.1/32 is directly connected, Vlan50
C        10.1.99.0/24 is directly connected, Vlan99
L        10.1.99.1/32 is directly connected, Vlan99
C        10.1.100.0/30 is directly connected, Vlan100
L        10.1.100.2/32 is directly connected, Vlan100
C        10.10.40.0/24 is directly connected, Vlan95
L        10.10.40.1/32 is directly connected, Vlan95
L3MGT_SW01#
L3MGT_SW01#
L3MGT_SW01#
```

```
L3MGT_SW01#show run interface port-channel1
Building configuration...

Current configuration : 163 bytes
!
interface Port-channel1
 switchport trunk native vlan 99
 switchport trunk allowed vlan 10,20,30,40,50,95,99
 switchport mode trunk
 switchport nonegotiate
end

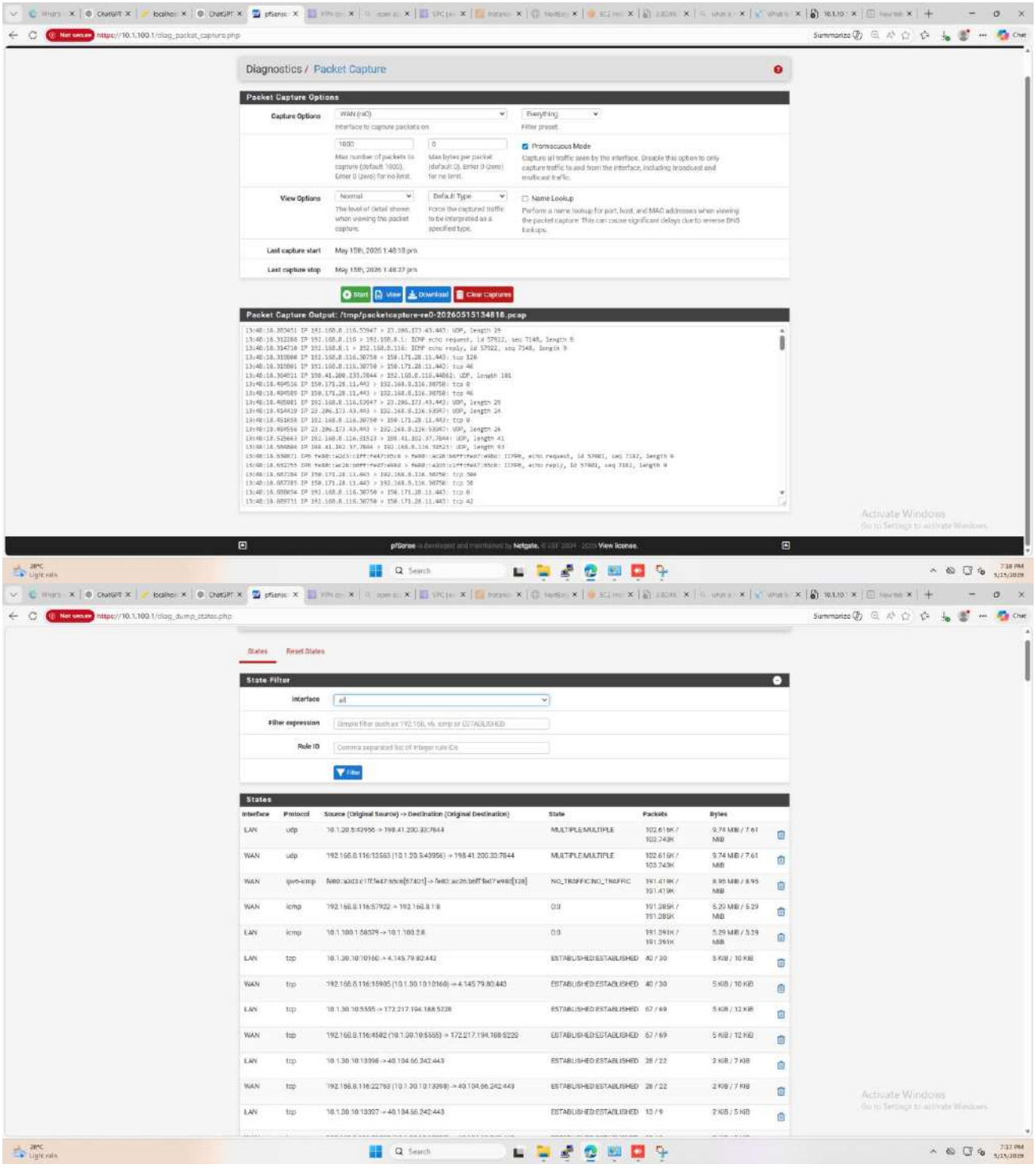
L3MGT_SW01#
```

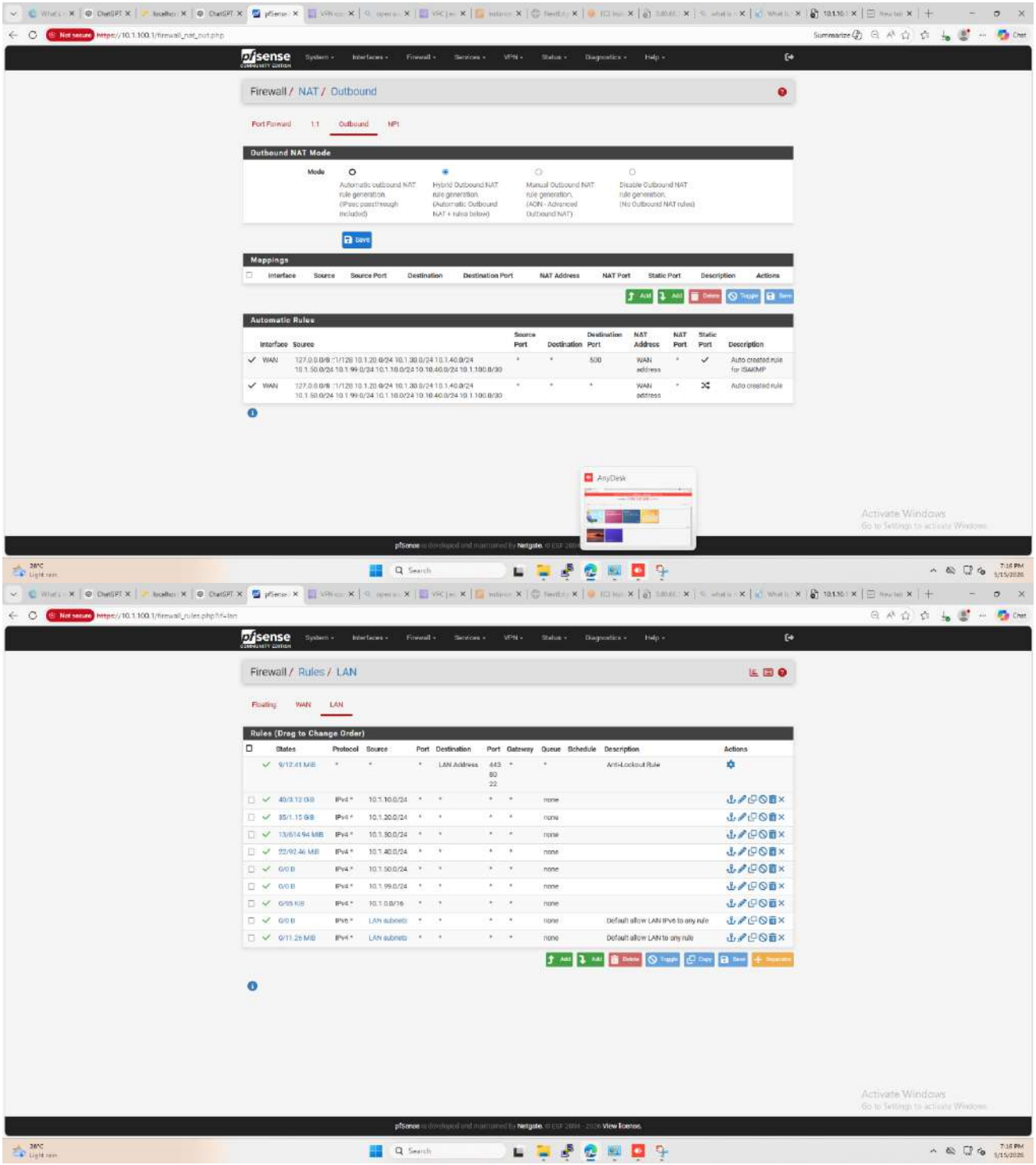



```
L3MGT_SW01#SHOw VLan brief

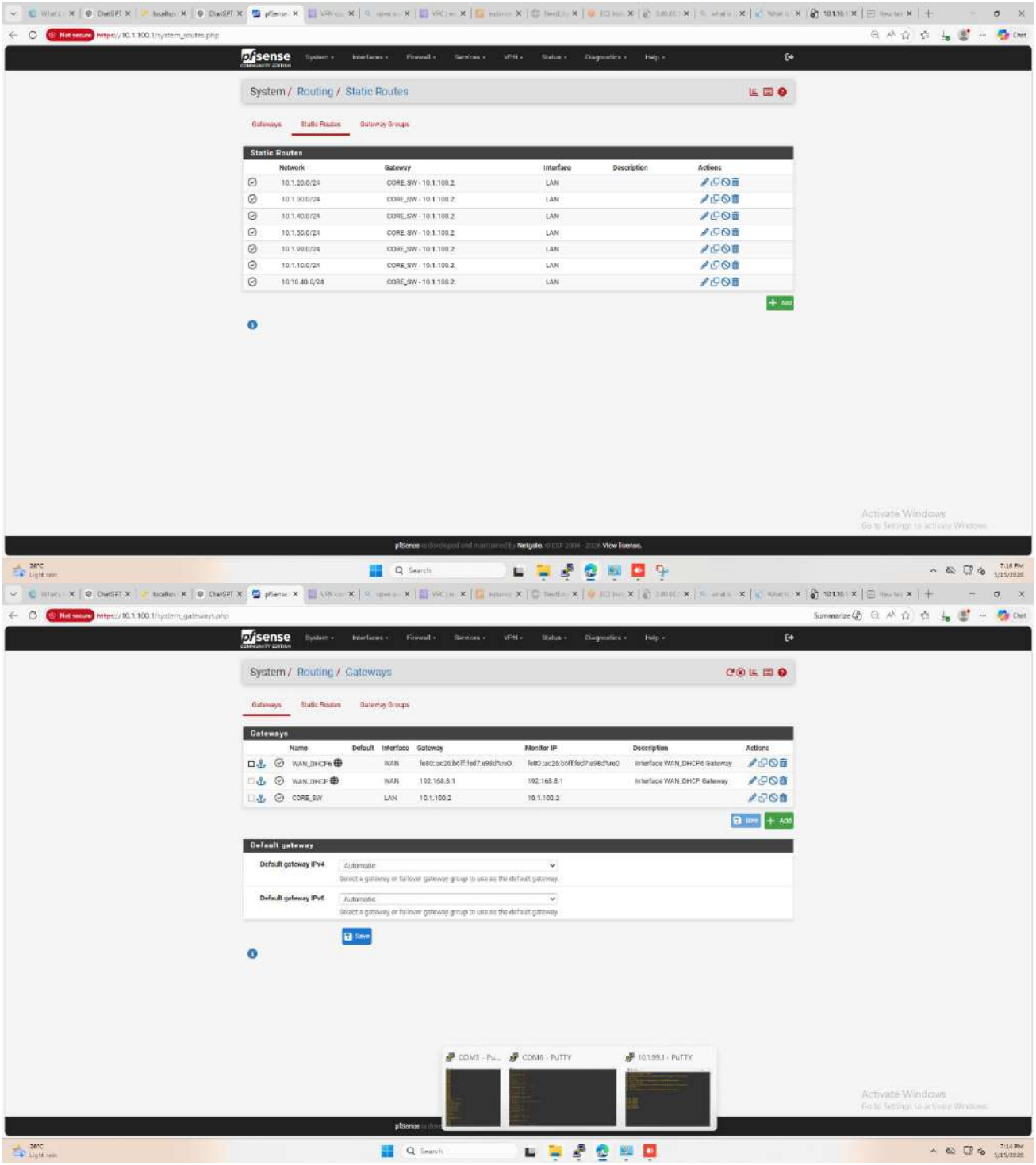
VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                           active    Gi1/0/3, Gi1/0/4, Gi1/0/5
                                                 Gi1/0/6, Gi1/0/7, Gi1/0/8
                                                 Gi1/0/9, Gi1/0/10, Gi1/0/11
                                                 Gi1/0/12, Gi1/0/14, Gi1/0/15
                                                 Gi1/0/16, Gi1/0/18, Gi1/0/19
                                                 Gi1/0/20, Gi1/0/21, Gi1/0/22
                                                 Gi1/1/1, Gi1/1/2, Gi1/1/3
                                                 Gi1/1/4
10   server                            active    Gi1/0/13, Gi1/0/17
20   R&D                               active
30   SD                                active
40   Marketing                         active
50   VPN                               active
95   NAS                               active
99   MGMT                              active
100  FW-TRANSIT                        active    Gi1/0/1
1002 fddi-default                      act/unsup
1003 token-ring-default                act/unsup
1004 fddinet-default                   act/unsup
1005 trnet-default                     act/unsup
L3MGT_SW01#
L3MGT_SW01#
L3MGT_SW01#
```

PFsens

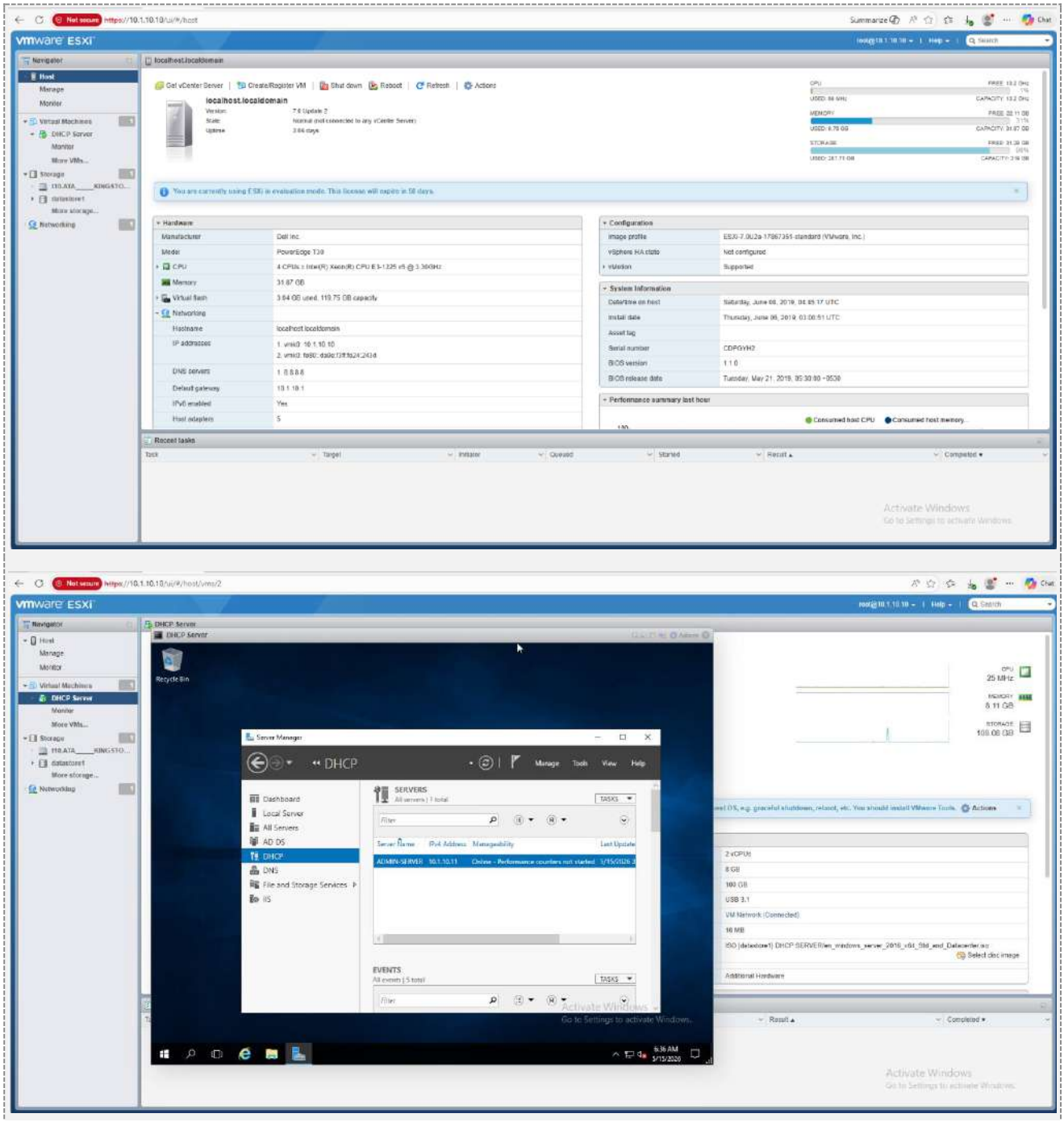

Diagnostics / Packet Capture
Packet Capture Options
Capture Options
View Options
Last capture start
Last capture stop
Start
View
Download
Clear Captures
Packet Capture Output: /tmp/packetcapture-re0-20260515134816.pcap
States
Reset States
State Filter
Interface
Filter expression
Rule ID
Filter
States
Interface
Protocol
Source (Original Source) -> Destination (Original Destination)
State
Packets
Bytes

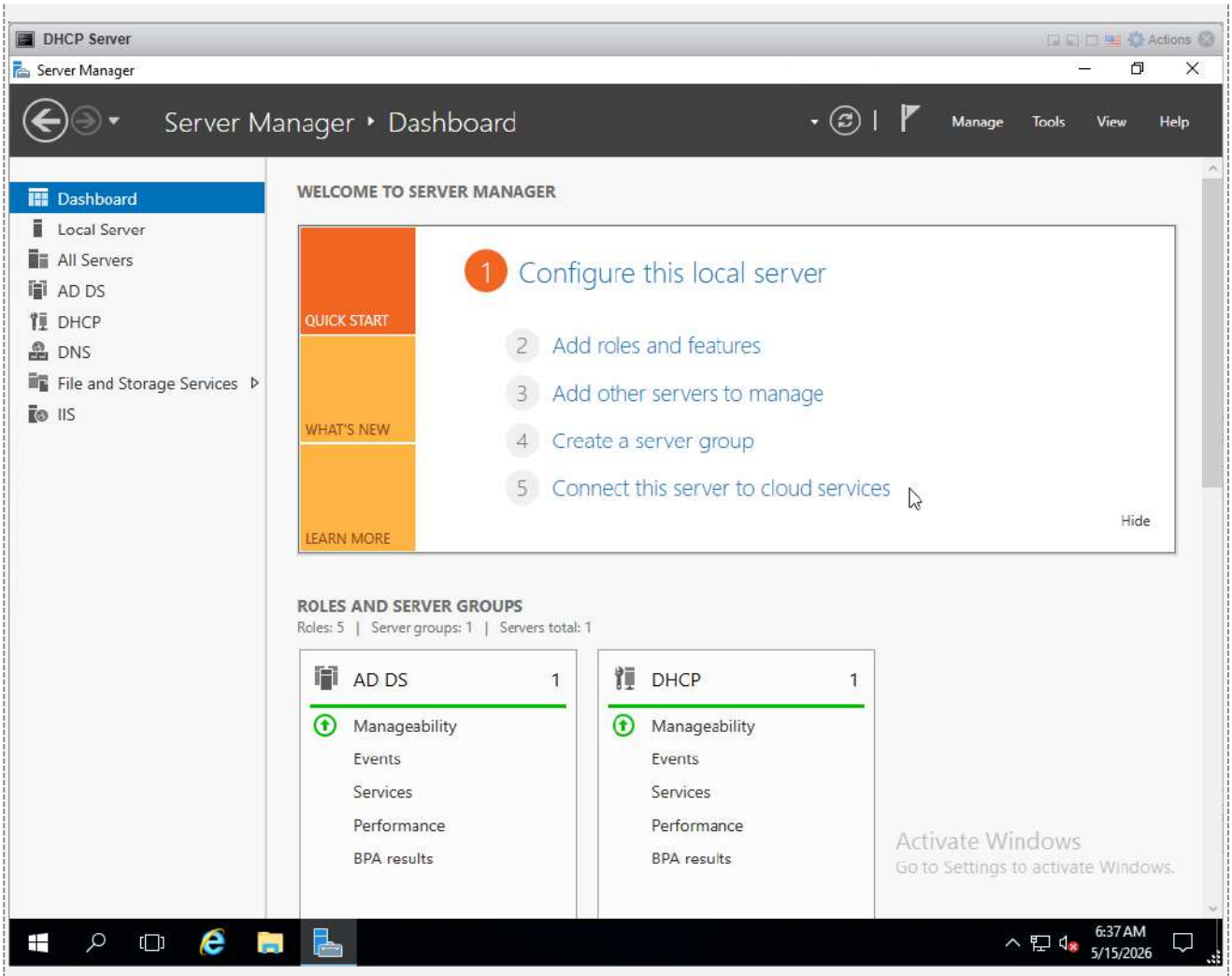

Firewall / NAT / Outbound
Port Forward
1:1
Outbound
NPt
Outbound NAT Mode
Mode
Automatic outbound NAT rule generation. (IPsec passthrough included)
Hybrid Outbound NAT rule generation. (Automatic Outbound NAT + rules below)
Manual Outbound NAT rule generation. (AON - Advanced Outbound NAT)
Disable Outbound NAT rule generation. (No Outbound NAT rules)
Save
Mappings
Interface
Source
Source Port
Destination
Destination Port
NAT Address
NAT Port
Static Port
Description
Actions
Automatic Rules
Interface
Source
Source Port
Destination
Destination Port
NAT Address
NAT Port
Static Port
Description
WAN
WAN address
Auto created rule for ISAKMP
Auto created rule
AnyDesk
Activate Windows

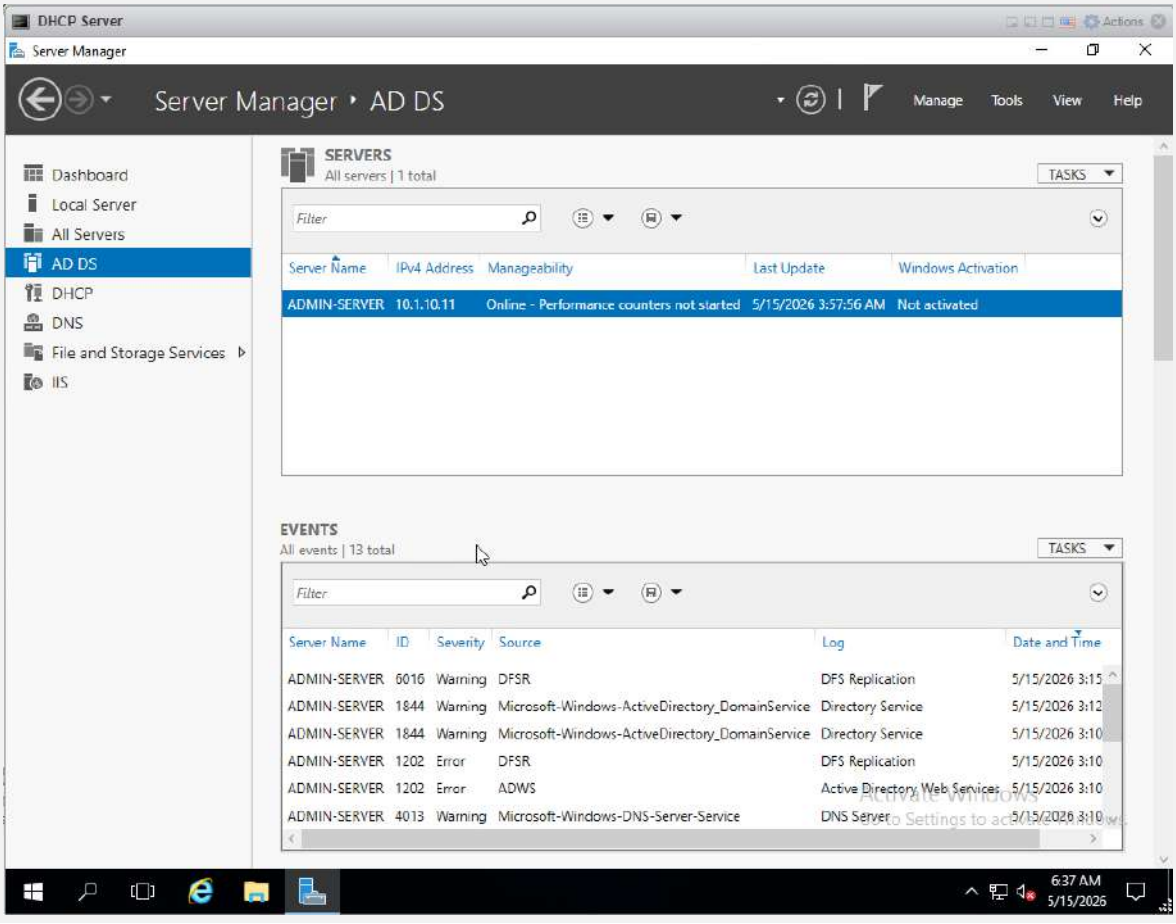
Firewall / Rules / LAN
Floating
WAN
LAN
Rules (Drag to Change Order)
States
Protocol
Source
Port
Destination
Port
Gateway
Queue
Schedule
Description
Actions
Anti-Lockout Rule
Default allow LAN IPv6 to any rule
Default allow LAN to any rule
Activate Windows

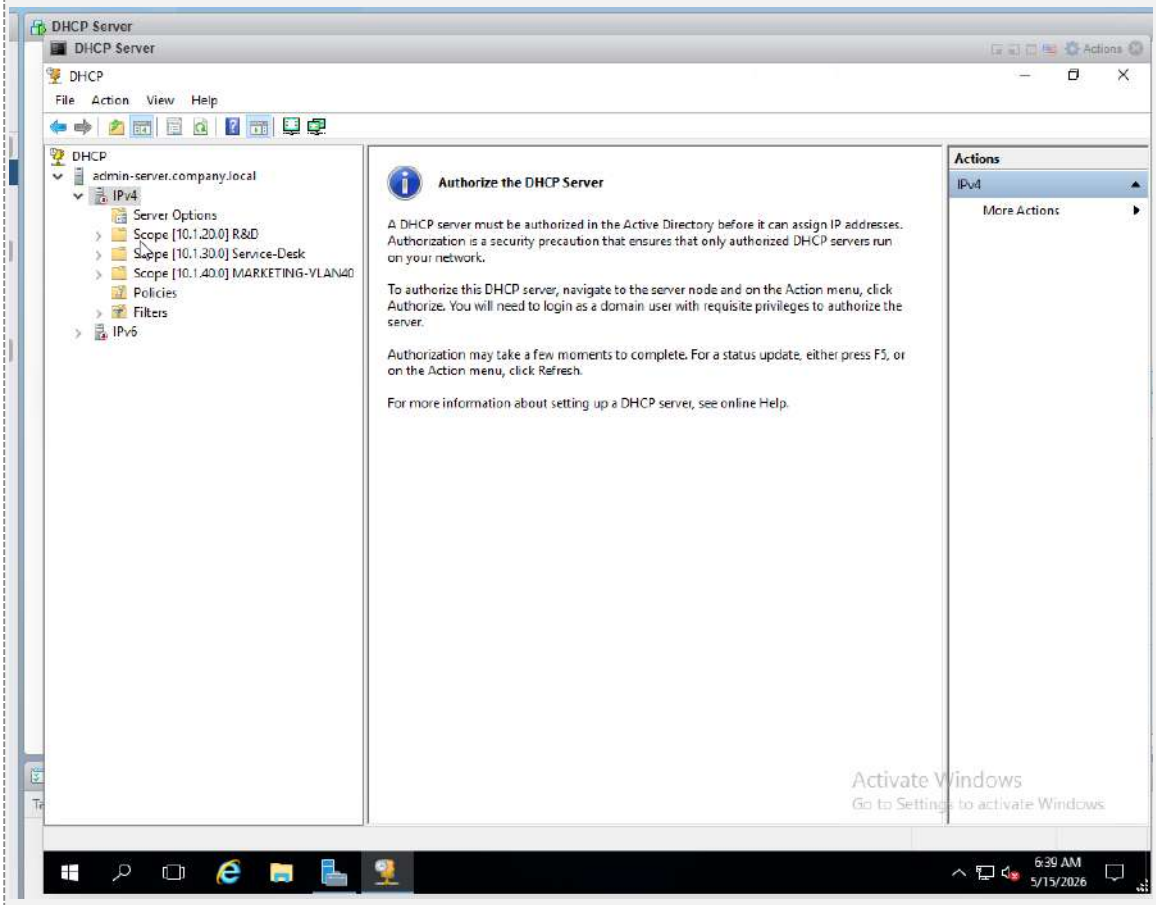

pfsense
System / Routing / Static Routes
Gateways
Static Routes
Gateway Groups
Static Routes
Network
Gateway
Interface
Description
Actions
10.1.20.0/24 CORE_SW - 10.1.100.2 LAN
10.1.30.0/24 CORE_SW - 10.1.100.2 LAN
10.1.40.0/24 CORE_SW - 10.1.100.2 LAN
10.1.50.0/24 CORE_SW - 10.1.100.2 LAN
10.1.99.0/24 CORE_SW - 10.1.100.2 LAN
10.1.10.0/24 CORE_SW - 10.1.100.2 LAN
10.10.40.0/24 CORE_SW - 10.1.100.2 LAN
Add
Activate Windows
pfSense is developed and maintained by Netgate. View licenses.
pfsense
System / Routing / Gateways
Gateways
Static Routes
Gateway Groups
Gateways
Name
Default
Interface
Gateway
Monitor IP
Description
Actions
WAN_DHCP6 WAN fe80::ac26:b0ff:fed7:e98d%ue0 fe80::ac26:b0ff:fed7:e98d%ue0 Interface WAN_DHCP6 Gateway
WAN_DHCP WAN 192.168.8.1 192.168.8.1 Interface WAN_DHCP Gateway
CORE_SW LAN 10.1.100.2 10.1.100.2
Add
Default gateway
Default gateway IPv4
Automatic
Select a gateway or failover gateway group to use as the default gateway.
Default gateway IPv6
Automatic
Select a gateway or failover gateway group to use as the default gateway.
Save
COM5 - Pu...
COM6 - PuTTY
10.1.99.1 - PuTTY
Activate Windows

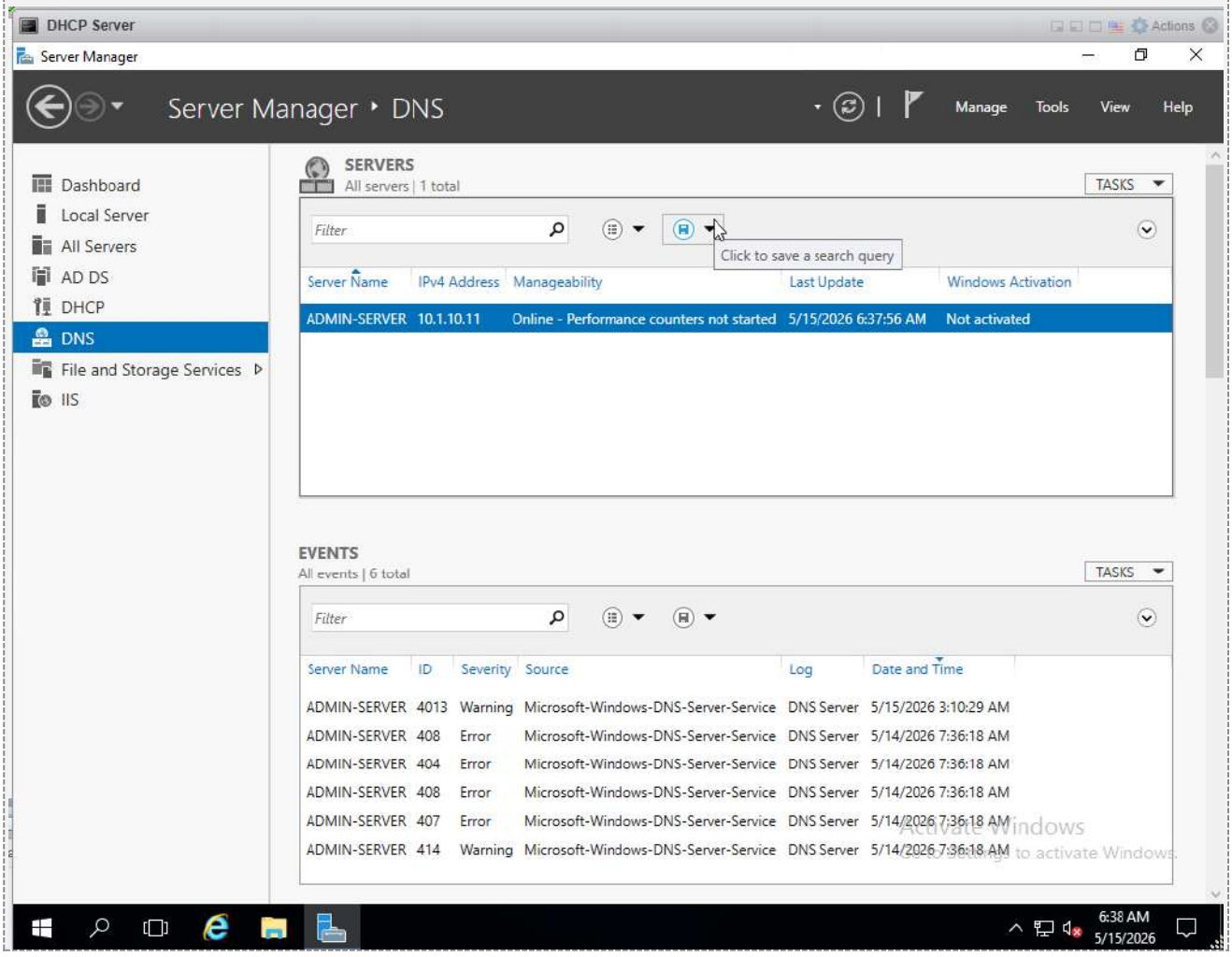
DHCP Server Configuration on a Virtual Machine



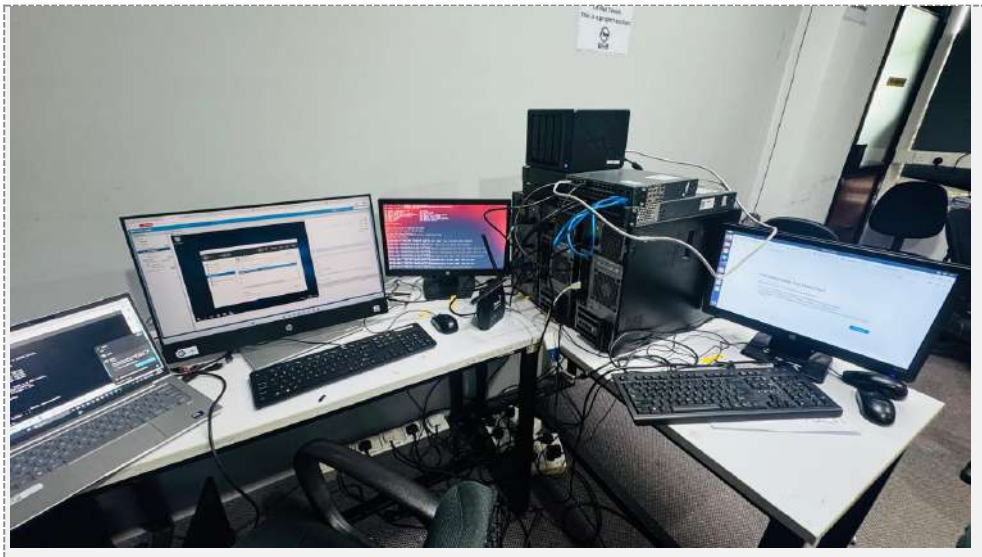

DHCP Server
Server Manager
Server Manager • Dashboard
Manage Tools View Help
Dashboard
Local Server
All Servers
AD DS
DHCP
DNS
File and Storage Services
IIS
WELCOME TO SERVER MANAGER
QUICK START
WHAT'S NEW
LEARN MORE
1 Configure this local server
2 Add roles and features
3 Add other servers to manage
4 Create a server group
5 Connect this server to cloud services
Hide
ROLES AND SERVER GROUPS
Roles: 5 | Server groups: 1 | Servers total: 1
AD DS 1
Manageability
Events
Services
Performance
BPA results
DHCP 1
Manageability
Events
Services
Performance
BPA results
Activate Windows
Go to Settings to activate Windows.
6:37 AM
5/15/2026

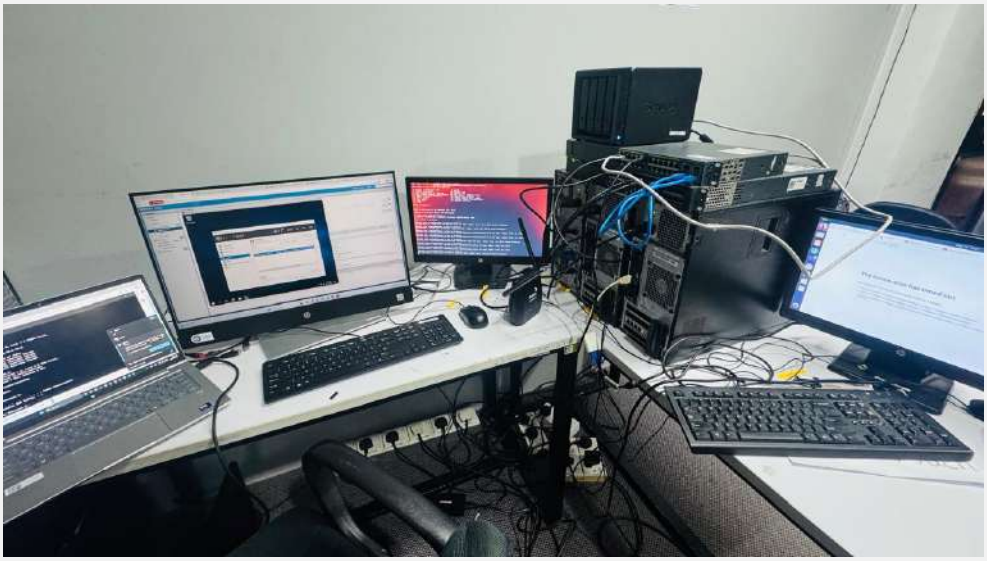

DHCP Server
Server Manager
Server Manager • AD DS
Manage Tools View Help
Dashboard
Local Server
All Servers
AD DS
DHCP
DNS
File and Storage Services
IIS
SERVERS
All servers | 1 total
TASKS
Filter
Server Name IPv4 Address Manageability Last Update Windows Activation
ADMIN-SERVER 10.1.10.11 Online - Performance counters not started 5/15/2026 3:57:56 AM Not activated
EVENTS
All events | 13 total
TASKS
Filter
Server Name ID Severity Source Log Date and Time
ADMIN-SERVER 6016 Warning DFSR DFS Replication 5/15/2026 3:15
ADMIN-SERVER 1844 Warning Microsoft-Windows-ActiveDirectory_DomainService Directory Service 5/15/2026 3:12
ADMIN-SERVER 1844 Warning Microsoft-Windows-ActiveDirectory_DomainService Directory Service 5/15/2026 3:10
ADMIN-SERVER 1202 Error DFSR DFS Replication 5/15/2026 3:10
ADMIN-SERVER 1202 Error ADWS Active Directory Web Services 5/15/2026 3:10
ADMIN-SERVER 4013 Warning Microsoft-Windows-DNS-Server-Service DNS Server 5/15/2026 3:10
6:37 AM
5/15/2026

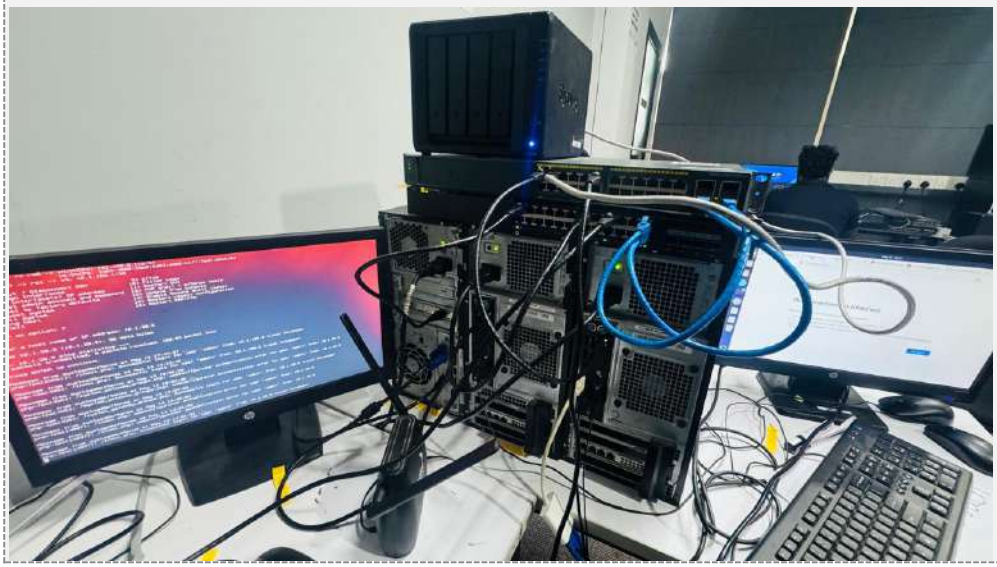

DHCP Server
DHCP
File Action View Help
DHCP
admin-server.company.local
IPv4
Server Options
Scope [10.1.20.0] R&D
Scope [10.1.30.0] Service-Desk
Scope [10.1.40.0] MARKETING-VLAN40
Policies
Filters
IPv6
Authorize the DHCP Server
A DHCP server must be authorized in the Active Directory before it can assign IP addresses. Authorization is a security precaution that ensures that only authorized DHCP servers run on your network.
To authorize this DHCP server, navigate to the server node and on the Action menu, click Authorize. You will need to login as a domain user with requisite privileges to authorize the server.
Authorization may take a few moments to complete. For a status update, either press F5, or on the Action menu, click Refresh.
For more information about setting up a DHCP server, see online Help.
Actions
IPv4
More Actions
Activate Windows
Go to Settings to activate Windows.
6:39 AM
5/15/2026


DHCP Server
Server Manager
Server Manager • DNS
Manage Tools View Help
Dashboard
Local Server
All Servers
AD DS
DHCP
DNS
File and Storage Services
IIS
SERVERS
All servers | 1 total
TASKS
Filter
Click to save a search query
Server Name IPv4 Address Manageability Last Update Windows Activation
ADMIN-SERVER 10.1.10.11 Online - Performance counters not started 5/15/2026 6:37:56 AM Not activated
EVENTS
All events | 6 total
TASKS
Filter
Server Name ID Severity Source Log Date and Time
ADMIN-SERVER 4013 Warning Microsoft-Windows-DNS-Server-Service DNS Server 5/15/2026 3:10:29 AM
ADMIN-SERVER 408 Error Microsoft-Windows-DNS-Server-Service DNS Server 5/14/2026 7:36:18 AM
ADMIN-SERVER 404 Error Microsoft-Windows-DNS-Server-Service DNS Server 5/14/2026 7:36:18 AM
ADMIN-SERVER 408 Error Microsoft-Windows-DNS-Server-Service DNS Server 5/14/2026 7:36:18 AM
ADMIN-SERVER 407 Error Microsoft-Windows-DNS-Server-Service DNS Server 5/14/2026 7:36:18 AM
ADMIN-SERVER 414 Warning Microsoft-Windows-DNS-Server-Service DNS Server 5/14/2026 7:36:18 AM
Activate Windows
Go to Settings to activate Windows.
6:38 AM
5/15/2026

Snapshots of Lab Implementation







AWS Cloud Implementation Evidence

The following screenshots provide implementation evidence for the AWS component of the NextEdge proof-of-concept. They demonstrate the AWS Console environment, EC2 compute instances used for the hosted project website and backup file-server workload, the operational web service, and the VPC networking resources used to support the cloud deployment.

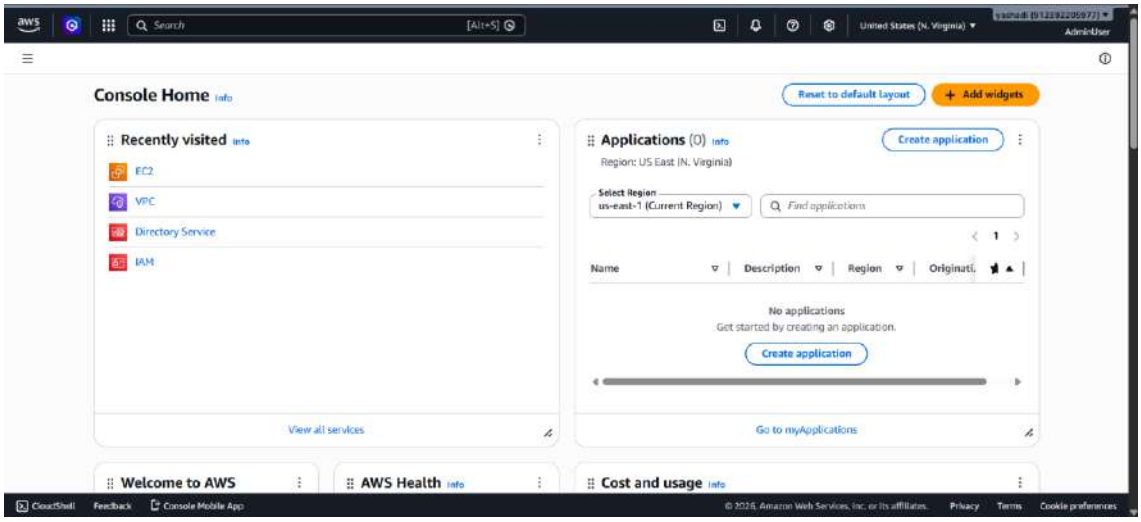


Figure 7.2 - AWS Management Console home page used for the NextEdge cloud implementation.

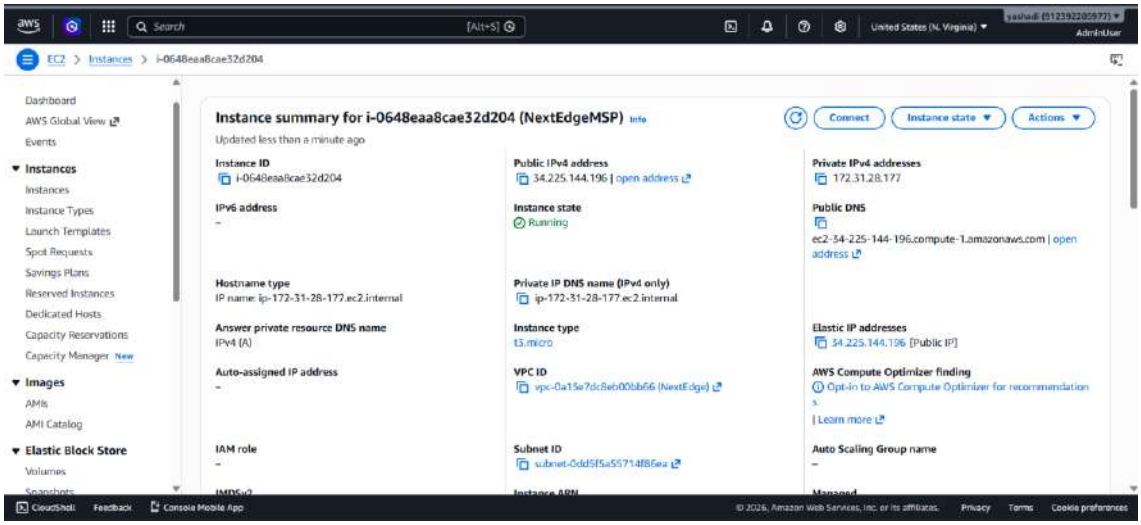


Figure 7.3 - EC2 instance summary for the NextEdgeMSP web-server instance, showing its running state and network addressing.

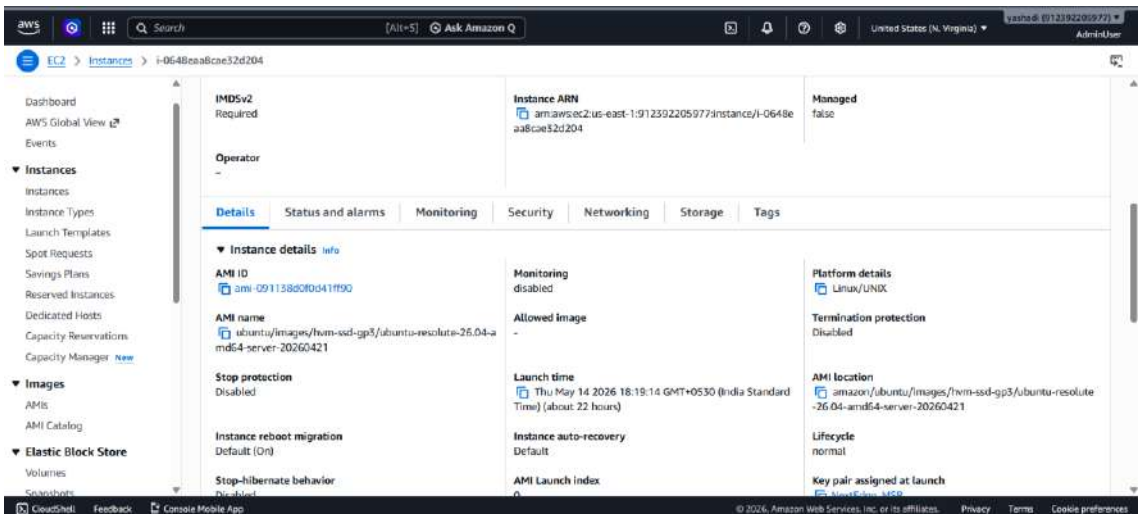


Figure 7.4 - EC2 instance details showing the Ubuntu Linux AMI and platform configuration used by the NextEdgeMSP server.

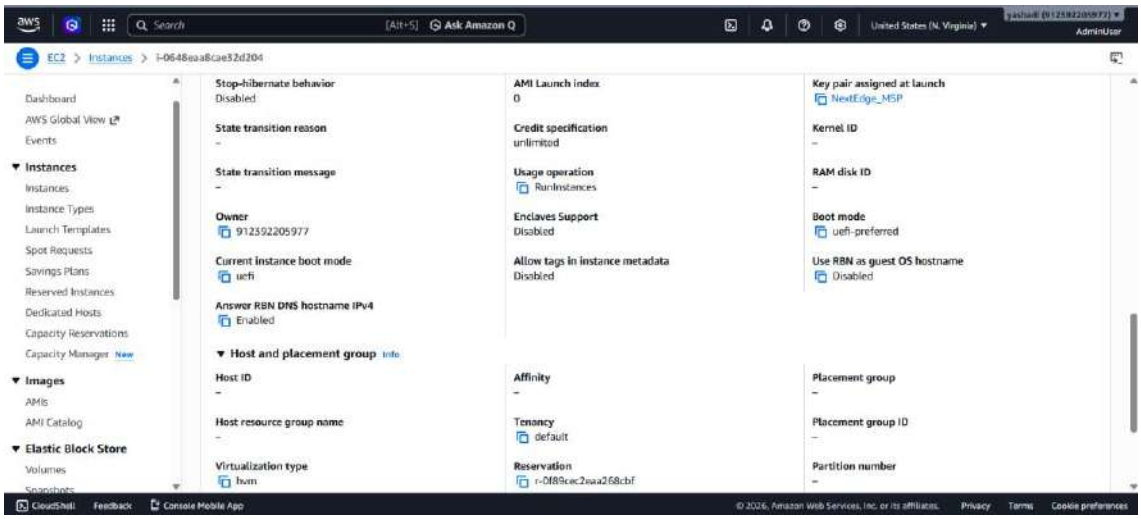


Figure 7.5 - Additional EC2 instance configuration details, including the assigned NextEdge_MSP key pair and boot settings.

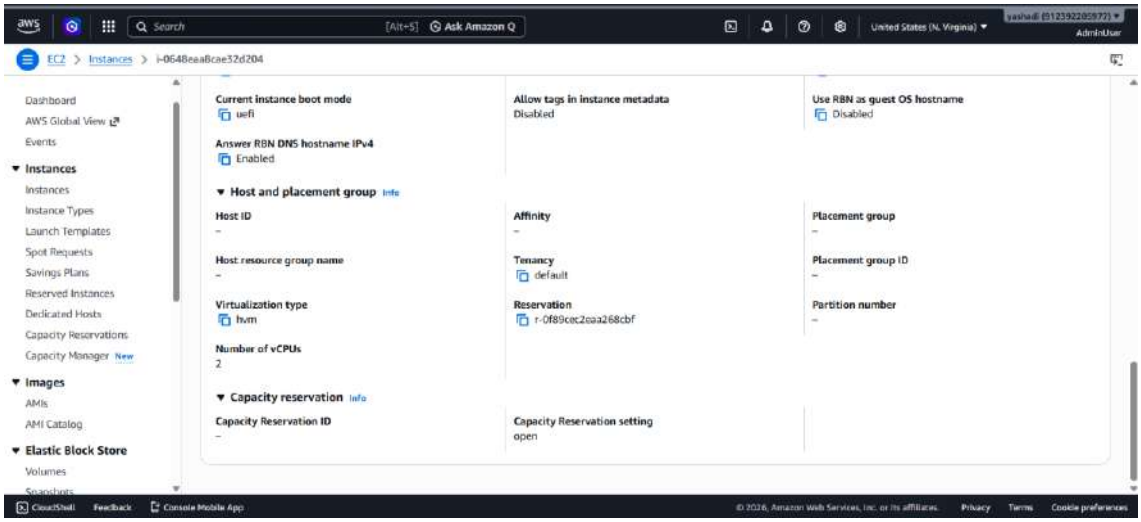


Figure 7.6 - EC2 host and placement information for the NextEdgeMSP instance.

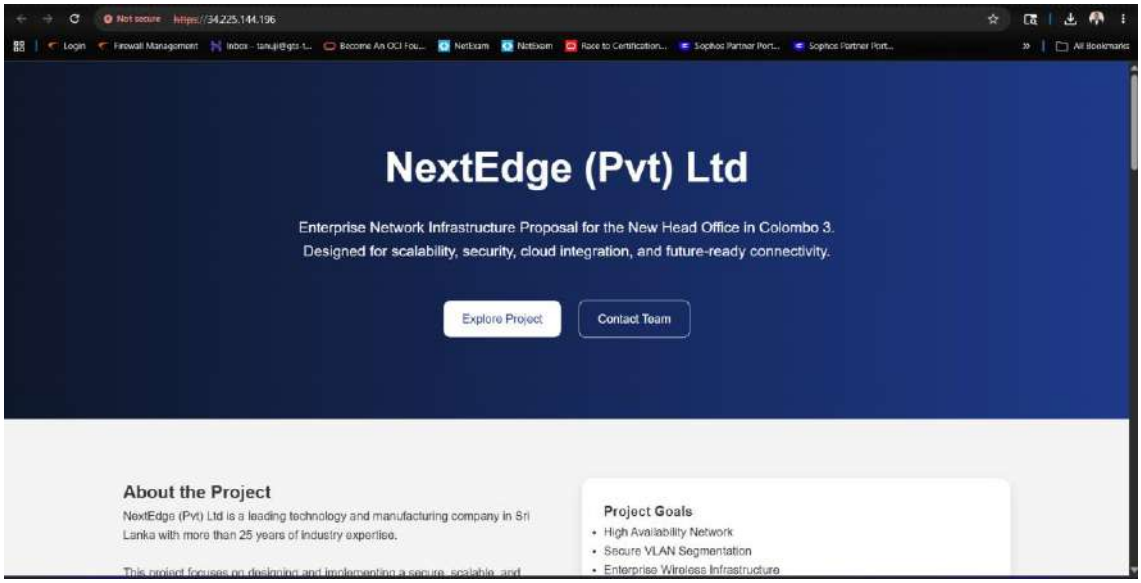


Figure 7.7 - NextEdge project website successfully hosted and accessed using the EC2 instance public IPv4 address.

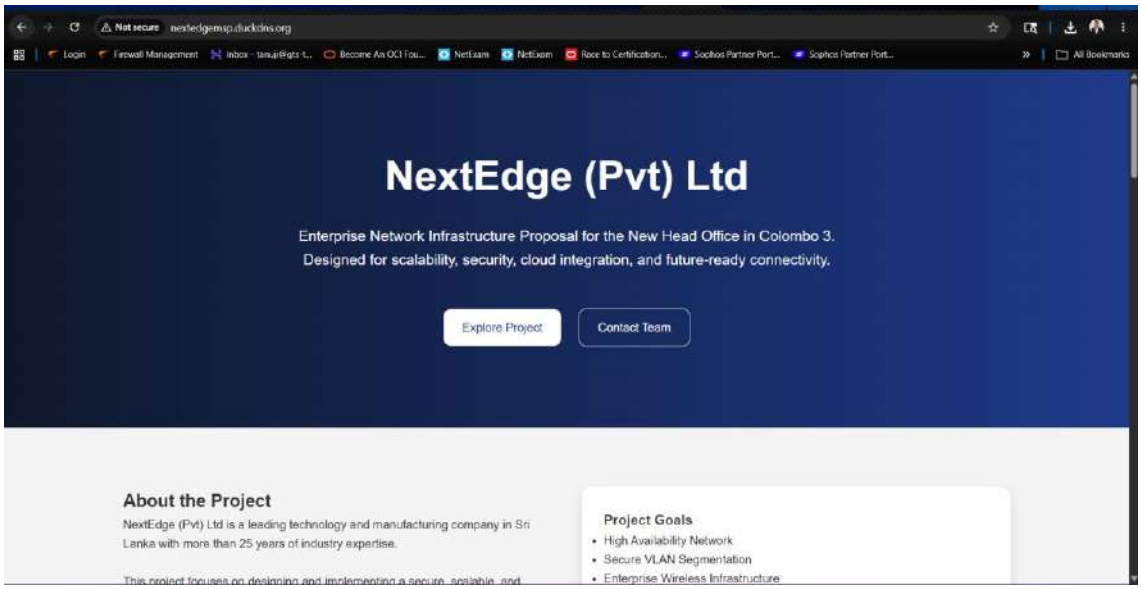


Figure 7.8 - NextEdge project website accessed through the configured DuckDNS hostname.

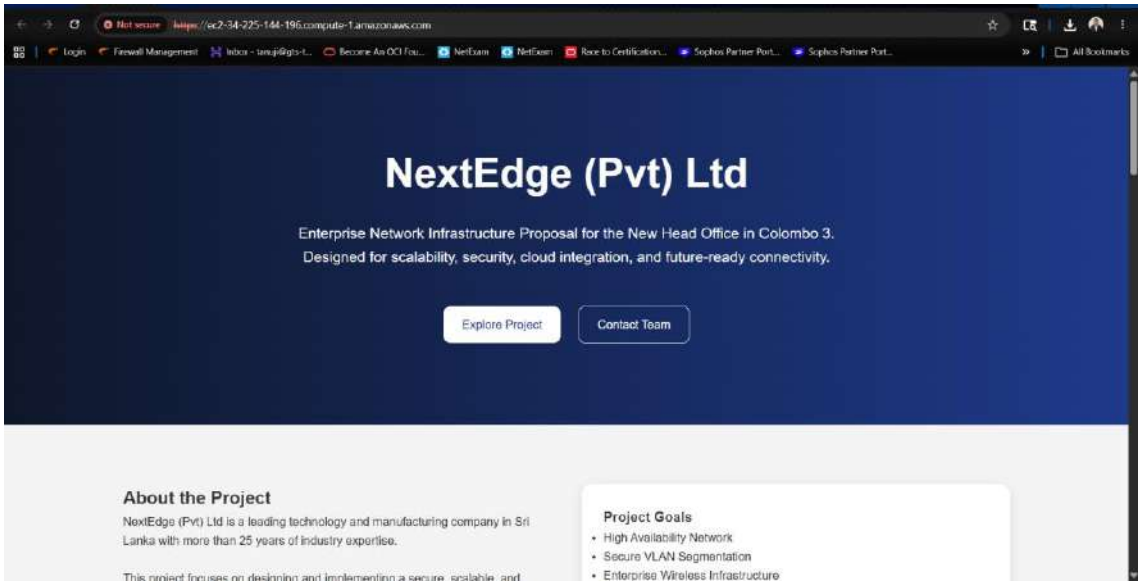


Figure 7.9 - NextEdge project website accessed through the EC2 public DNS hostname.

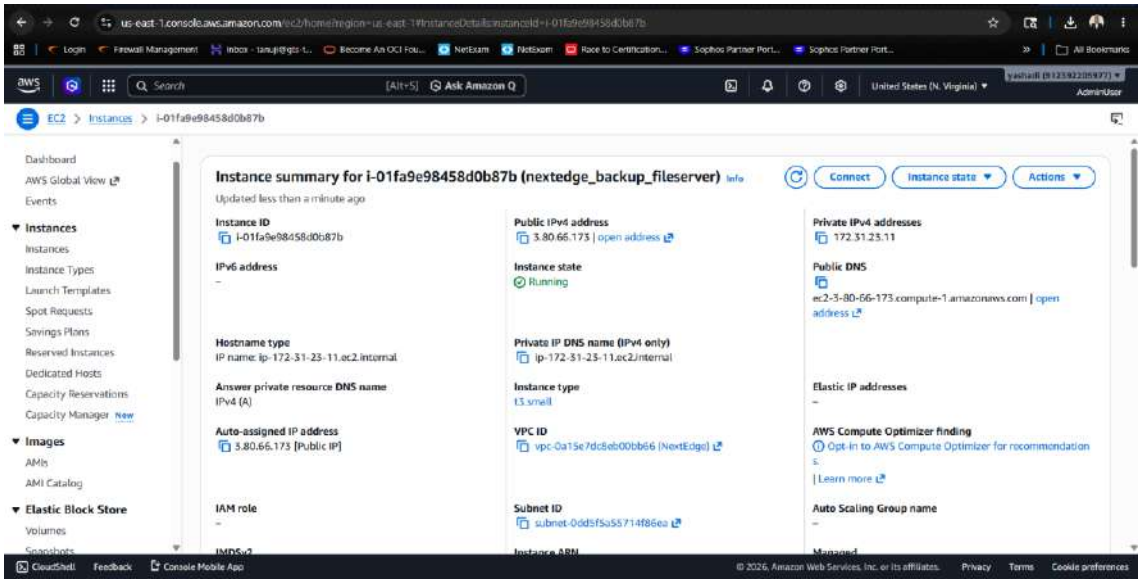


Figure 7.10 - EC2 instance summary for the nextedge_backup_fileserver instance used as the cloud backup file-server workload.

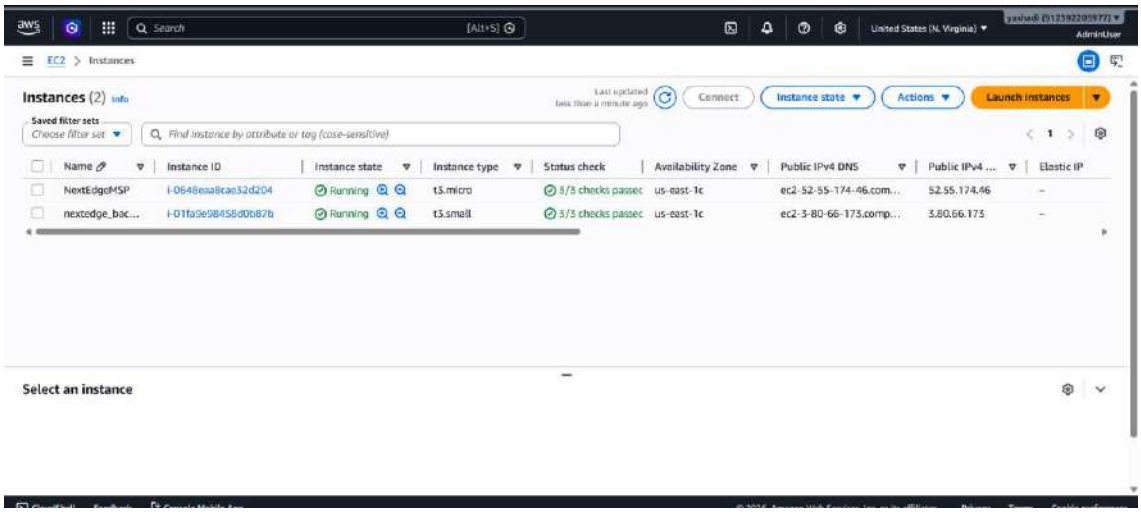


Figure 7.11 - EC2 Instances view showing both NextEdgeMSP and the backup file-server instance in the running state.

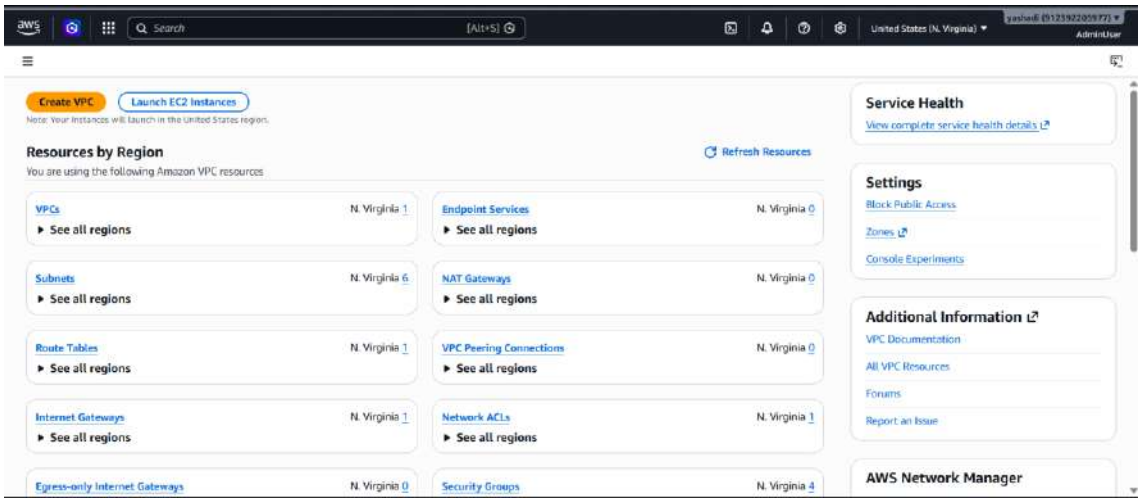


Figure 7.12 - Amazon VPC dashboard showing the VPC, subnets, route table, Internet Gateway, network ACL and security-group resources.

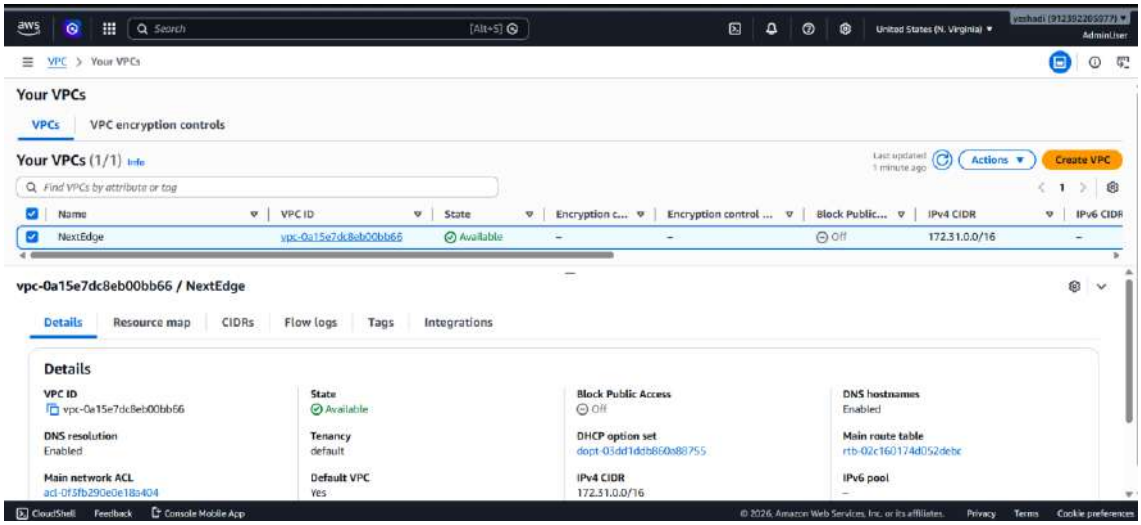


Figure 7.13 - NextEdge VPC configuration showing the 172.31.0.0/16 IPv4 CIDR and enabled DNS settings.

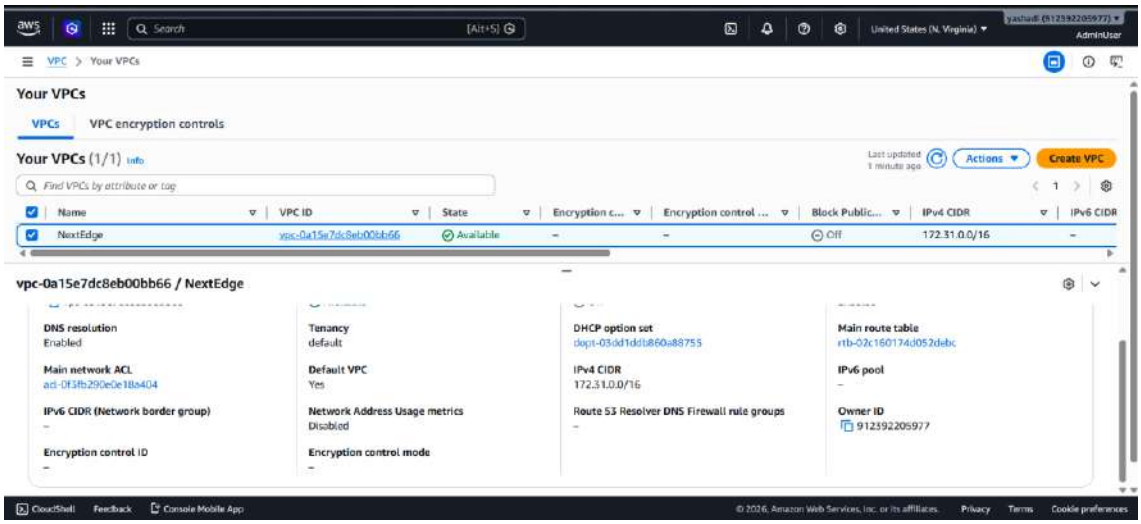


Figure 7.14 - Additional NextEdge VPC details including the main network ACL, DHCP option set and main route table.

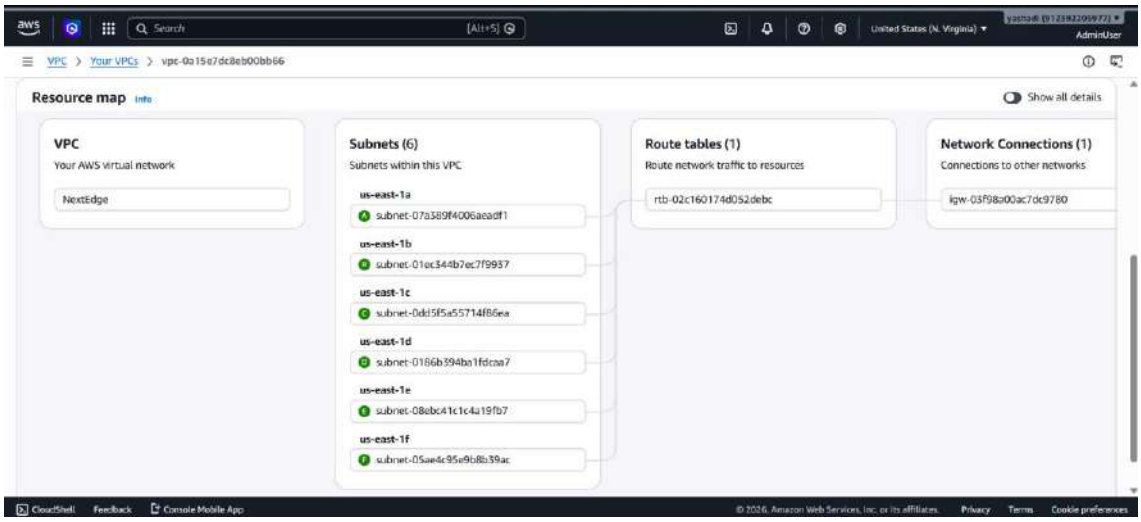


Figure 7.15 - NextEdge VPC resource map showing six subnets across Availability Zones, the route table and Internet Gateway connection.

Chapter 8: Evaluation, Results & Testing

Testing Approach

Each protocol/technology below was tested using a consistent method: configure the feature on the lab topology, use a documented Cisco IOS verification command (or client-side test) to confirm the running state, then force a failure or edge case where relevant (e.g. shut the active HSRP link, block a port, attempt an out-of-policy connection) to confirm the control behaves as designed rather than only in the happy-path case.

This project involved a comprehensive redesign of the network's hierarchical diagram compared with the original concept, largely due to resource constraints when moving from the paper design to the lab implementation topology. The implementation integrated multiple technologies to achieve the following objectives:

- Optimise IP address management with DHCP implementation.
- Enhance network security through ACLs and port security.
- Provide internet access through PAT while restricting internet access for the isolated Lab sub-network.
- Increase network bandwidth and provide link redundancy using EtherChannel.
- Give remote (SSH) access to network devices for the Network and System Admin team.

This evaluation focuses on the successful implementation of the following technologies/protocols:

- DHCP
- SSH
- PAT
- ACLs
- EtherChannel
- Port Security

DHCP

DHCP was implemented to streamline IP address assignment across the Inventory Management, HR and Lab VLANs, reducing manual configuration effort and enabling efficient allocation of addresses.

Address exhaustion was checked by comparing each VLAN's configured pool size against the device counts in the sub-department mapping (Chapter 4), with roughly 15-20% headroom left in every pool for future headcount growth.

SSH

SSH was used to give the Network and System Admin team secure access to network devices; ACLs ensure that other sub-networks cannot reach the devices over SSH.

Telnet was disabled entirely on every lab device rather than merely left unused, so that a misconfiguration could not silently re-open an insecure management path.

PAT

PAT conserves the single public-facing IP address provided by the upstream ISP link, sharing it among multiple internal devices for outbound internet access. The Lab sub-network remains restricted from internet access in line with the segmentation requirement.

A packet capture confirmed that Lab-VLAN-sourced traffic aimed at public internet addresses was dropped at the router ACL before it ever reached the PAT translation table, rather than being translated and then blocked further out.

ACLs

ACLs were used to control and filter network traffic in line with the company's security policy. Due to lab time and resource constraints, a representative subset of policies was implemented, focused on SSH access control and PAT/internet restriction for the Lab VLAN.

ACL rule order was deliberately tested from most specific to least specific, since an earlier broad permit rule had been found, during testing, to silently shadow a later, more specific deny rule.

EtherChannel

EtherChannel was used to increase network bandwidth and provide link redundancy between the access and distribution layers

Port Security

Port security mitigates unauthorised access at the access layer by enforcing MAC-address filtering on switch ports, strengthening overall network security.

Results Summary

Across the seven technologies evaluated in this chapter, every control behaved as designed under the forced-failure test described in the Testing Approach: HSRP failed over the active gateway without a client-visible outage; DHCP correctly scoped addresses to the Lab, HR and Inventory Management VLANs without cross-allocation; SSH access to devices succeeded only from the Network & System Admin subnet and was correctly refused from every other VLAN; PAT allowed general internet egress while the Lab VLAN was confirmed unable to reach the internet at all; the representative ACL set blocked out-of-policy traffic while permitting the specific HR-to-Lab path; EtherChannel kept the access-to-distribution link up and passing traffic when one member link was manually disabled; and port security correctly disabled a port when an unauthorised MAC address was introduced. No control needed to be redesigned as a result of testing, though the ACL policy set was narrowed twice during testing to remove overly broad early rules.

Challenges Faced and Resolutions

Table 8.1 - Challenges Faced and Resolutions

Challenge	Resolution
Lab hardware/licence limits did not match the full building-scale design (26+ access switches, 52 APs, dual firewalls).	Condensed the design into a representative topology (Chapter 7) that preserves every redundancy and segmentation mechanism at a smaller scale, and documented the full-scale bill of materials separately in Chapter 4.
Coordinating three group members' configuration work without conflicting changes to the same devices.	Assigned clear device/subsystem ownership per the Roles and Responsibilities table (Chapter 6) and used dated supervisor sessions (Annexure 4) to review integration points before merging configurations.
Estimating realistic AWS/Cisco SD-WAN costs without access to a formal vendor quotation.	Used public AWS Pricing Calculator figures and Cisco list pricing as planning-level estimates, clearly labelled as such in the budget (Chapter 4), with a recommendation to confirm via formal quotation before procurement.
Keeping the isolated Lab VLAN's restrictions consistent across the on-premises network and the AWS DR site.	Mirrored the on-premises VLAN/ACL segmentation directly into the DR VPC's subnet design (Chapter 4) so the same HR-only access rule applies in both locations.

Chapter 9: Conclusion & Future Improvements

Conclusion

The conclusions below should be read alongside the constraints already acknowledged in Chapter 2: they describe what was designed and, where explicitly noted, what was verified in the lab PoC - not a claim that the full 5-floor, 52-access-point building has been physically built and load-tested.

- The successful implementation of OSPF, EtherChannel, default routes, STP, HSRP, PAT, ACLs, SSH and port security has resulted in a robust, secure and highly efficient network for NextEdge (Pvt) Ltd.
- OSPF as the dynamic routing protocol enables seamless data transmission between the router and the L3 switches, reducing latency and improving overall performance.
- HSRP on the two L3 switches enhances redundancy and fault tolerance, ensuring continuous, uninterrupted service even if a device fails.
- Secure device access through SSH, combined with ACLs restricting access to authorised sub-networks, has strengthened network security and protected against unauthorised access.
- Internet access for the isolated Lab sub-department is restricted via ACLs, keeping the segmentation requirement enforced.
- Carefully configured default routes act as the last-resort gateway for the internal network, easing internet access when no specific route is otherwise available.
- Port security on the L2 switches adds an extra layer of protection, preventing unauthorised devices from connecting to the network.
- EtherChannel links between the L2 and L3 switches create a resilient, load-balanced network with increased bandwidth and improved overall performance.
- The addition of a warm-standby DR site in AWS, connected over a dual-transport Cisco SD-WAN overlay, extends the design from network resilience to full site-level resilience - giving NextEdge (Pvt) Ltd a documented, testable path to recovery if the Colombo 3 head office itself becomes unavailable.
- Overall, the project successfully achieved its goals, providing NextEdge (Pvt) Ltd with a secure, efficient and reliable network infrastructure - backed by a resilient, cloud-based disaster-recovery capability - that supports the company's continued growth.

Taken together, the on-premises design, the lab Proof-of-Concept, and the AWS DR/SD-WAN extension address every numbered requirement set out in Chapter 2, while staying within the resource and time constraints acknowledged in that same chapter. The project's main lesson for the team was that resilience is cheapest when designed in from the network's core (VLANs, HSRP, STP) rather than bolted on afterwards - the same segmentation logic used for the Lab on-premises carried through into the AWS VPC design with very little rework, precisely because it was based on a small number of consistent principles rather than a large number of special cases.

Lessons Learned

Three lessons stood out to the group over the course of the 90-day programme. First, scoping down from the full 5-floor design to a representative lab topology should have happened earlier - a full week was lost mid-project re-drawing diagrams that had assumed hardware the lab did not have. Second, writing the Security Requirements Specification and the System Design chapter in parallel, rather than security-after-design, meant several ACL and segmentation decisions were caught and corrected before implementation rather than after. Third, keeping the AWS DR site's segmentation a direct mirror of the on-premises VLAN plan - rather than a cloud-native redesign - made the whole DR chapter faster to write and easier for a reviewer to check, at the acceptable cost of not using every AWS-native networking feature available.

Future Improvements

EtherChannel

Only one link between the L2 and L3 switches has currently been enhanced with EtherChannel. Extending EtherChannel to every link between the L3 and L2 switches would further increase available bandwidth, improve load balancing, and raise overall network efficiency.

Adding Monitoring Capabilities

To ensure proactive management and continuous visibility into network health and performance, the following monitoring tools are recommended for future phases:

- Wireshark: A widely used open-source protocol analyser for troubleshooting network issues, detecting abnormal behaviour and optimising performance.
- PRTG Network Monitor: A comprehensive monitoring tool offering a centralised view of the entire network infrastructure.
- Zabbix: An open-source monitoring platform for tracking availability, performance and resource utilisation.
- Nagios: A widely used open-source monitoring system known for its flexibility and extensibility.

Integrating these tools would let the Network and System Admin team proactively identify bottlenecks, resolve issues quickly, and minimise downtime.

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Annexure 1 - Floor Maps

Each floor map below should be read alongside the corresponding sub-department table and Wi-Fi heatmap in Chapter 4; together the three views show, respectively, where each team physically sits, how the floor's cabling and camera routes are laid out, and how well that floor's wireless coverage meets the signal-strength targets used for this project.

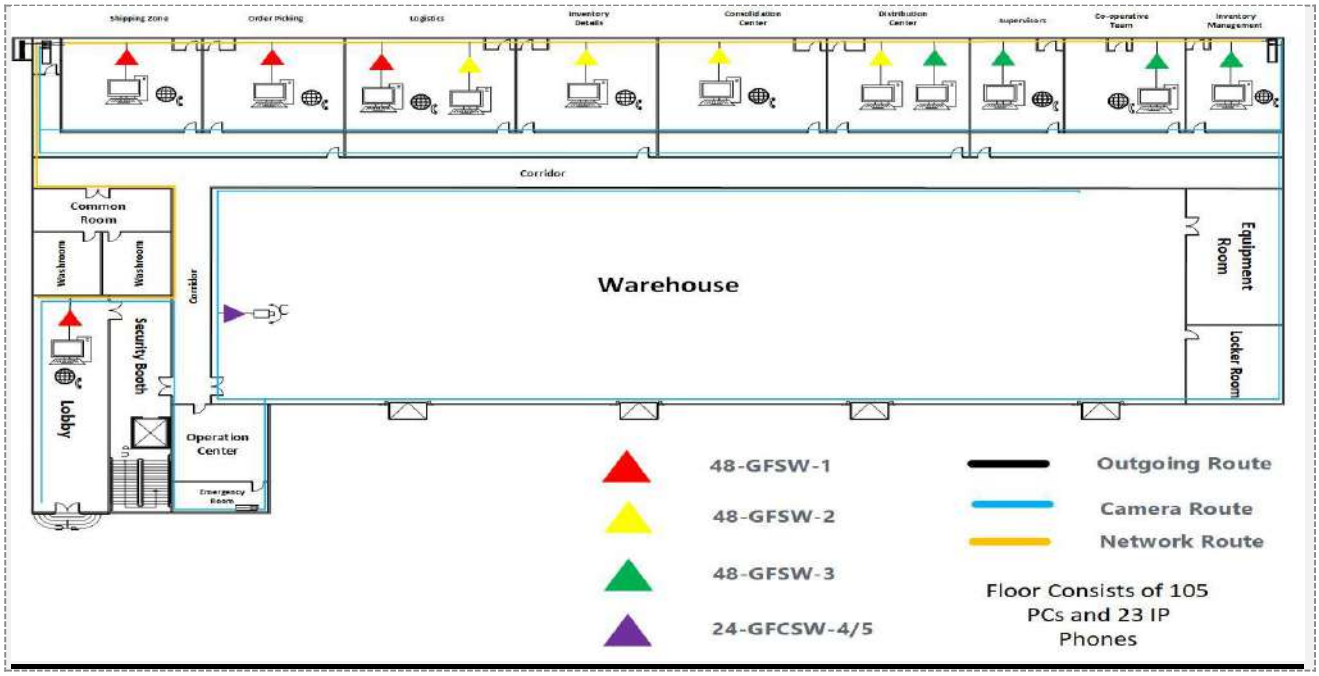


Figure A1.1 - Ground Floor - full layout with wiring, camera and network routes

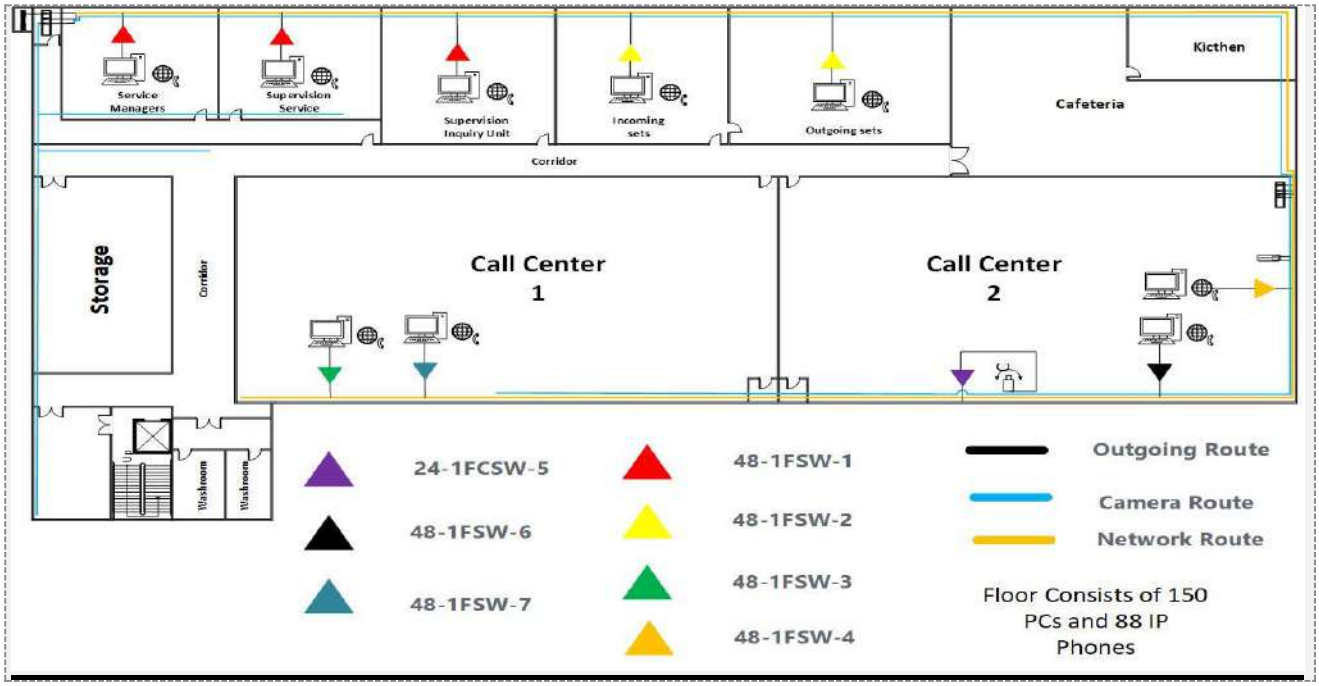


Figure A1.2 - 1st Floor - full layout with wiring, camera and network routes

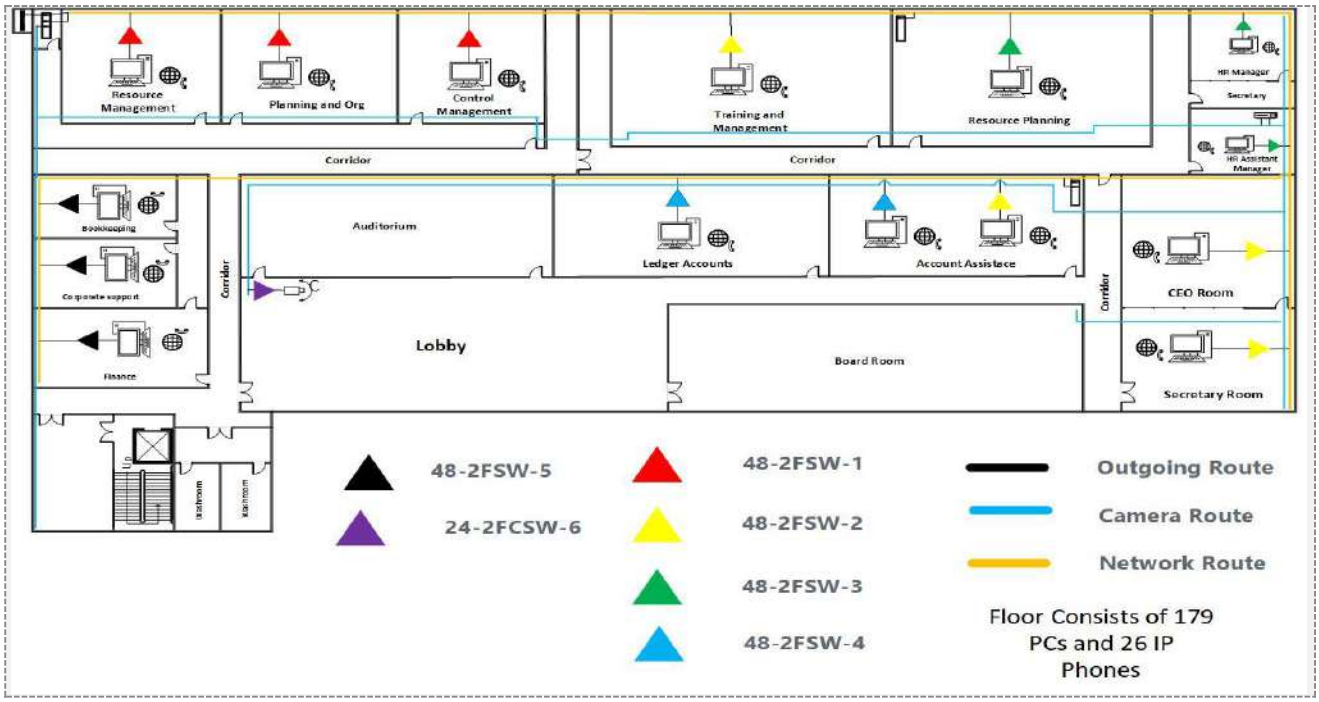


Figure A1.3 - 2nd Floor - full layout with wiring, camera and network routes

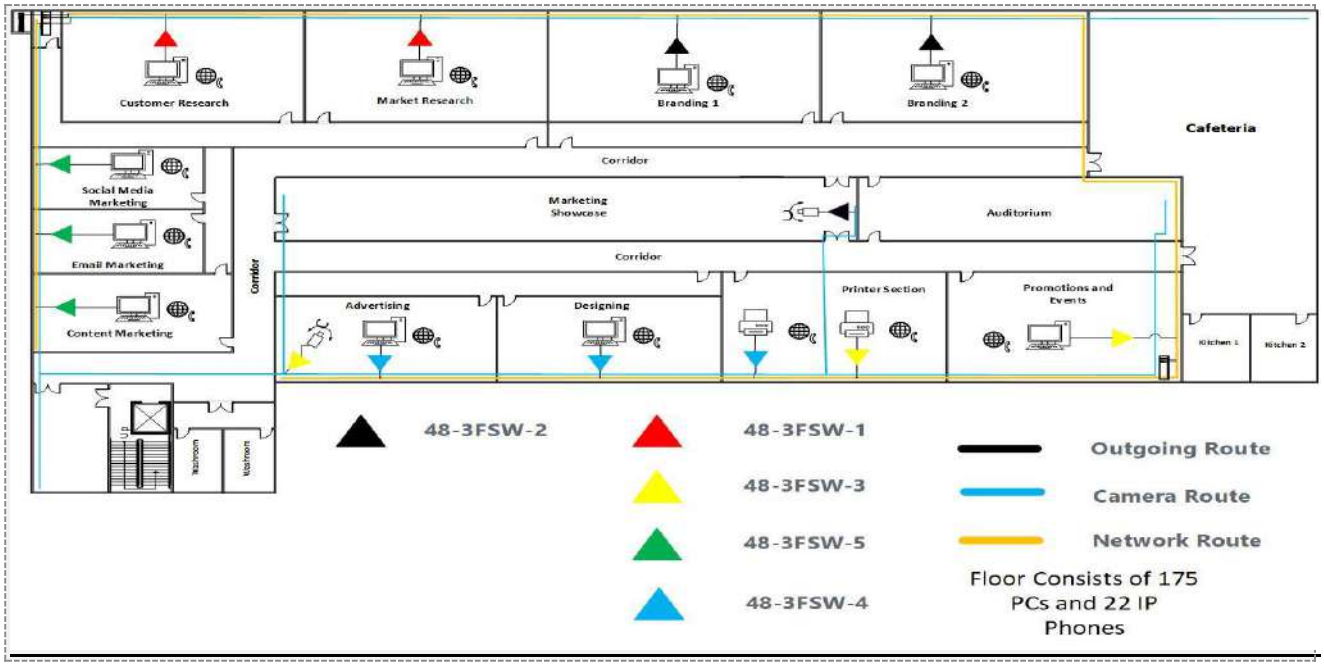


Figure A1.4 - 3rd Floor - full layout with wiring, camera and network routes

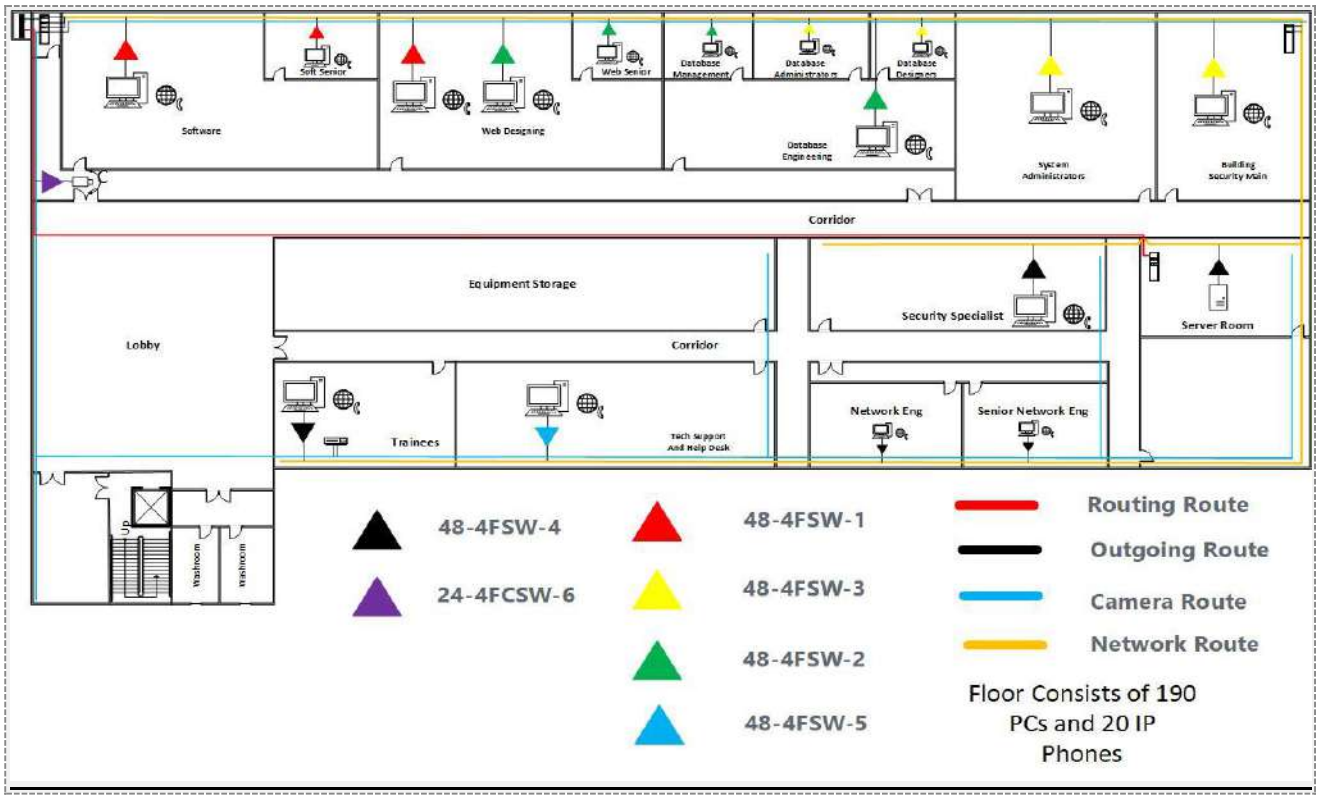


Figure A1.5 - 4th Floor - full layout with wiring, camera and network routes

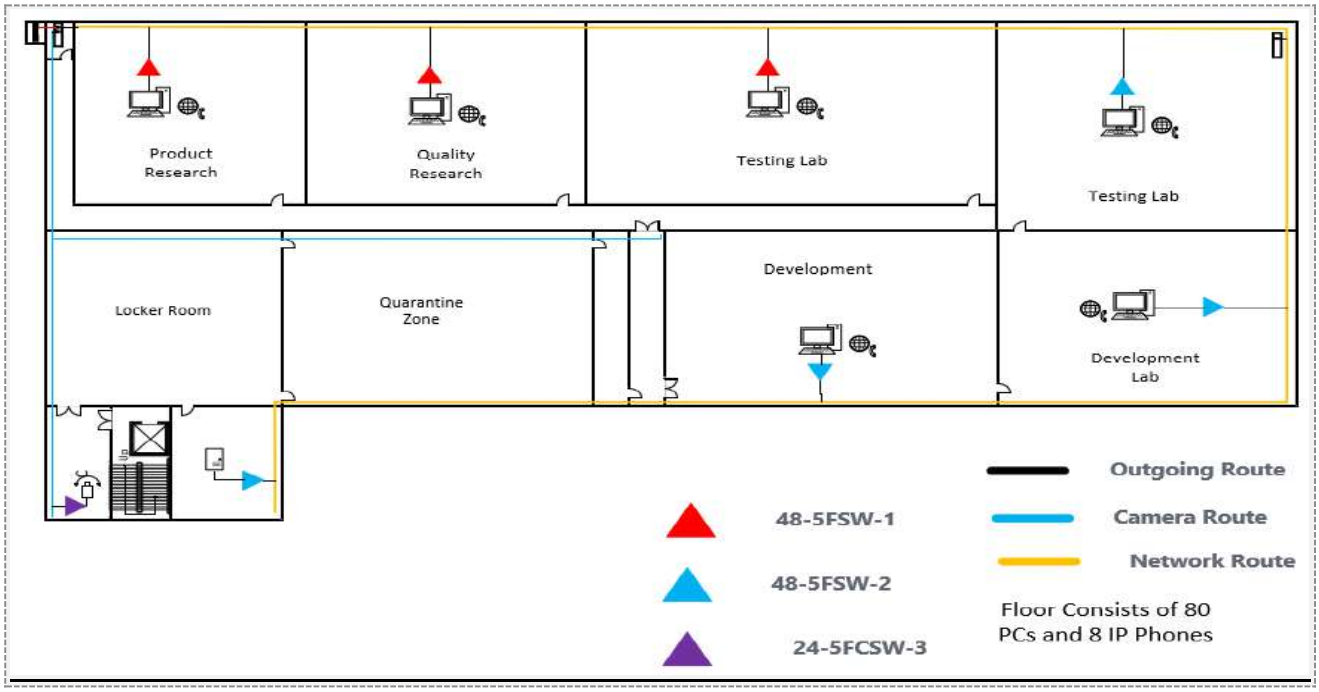


Figure A1.6 - 5th Floor - full layout with wiring, camera and network routes

Annexure 2 - Device Specifications

Specifications below are summarised from each manufacturer's public data sheet (cited in the References) rather than reproduced in full and are limited to the fields that materially influenced the design or budget decisions in Chapter 4 - for example, PoE+ port count and switching capacity for switches, or concurrent-call capacity for the PABX platform.

1. Cisco Catalyst 8200 Series Edge Platform - C8200L-1N-4T



Figure A2.1 - Cisco Catalyst 8200 Series Edge Platform

Specification	Detail
Processor	Intel x86 multicore CPU, 4GB DRAM default (upgradable to 8/16/32GB)
WAN Ports	4x built-in Gigabit Ethernet WAN ports (2x SFP, 2x RJ45)
VPN Acceleration	Embedded IPsec VPN hardware acceleration, up to 500 Mbps IPsec throughput
Storage	Default M.2 storage, upgradable to 600GB NVMe
Dimensions	1.73 x 17.25 x 11.8 in (4.39 x 43.81 x 29.97 cm), 1RU
Weight	10 lb (4.54 kg)
Power Supply	Internal, fixed AC power supply

(Cisco Catalyst 8200 Series Edge Platforms Data Sheet, n.d.)

2. L3 Switch - Cisco Catalyst 9300 48-port PoE+ (C9300-48P)

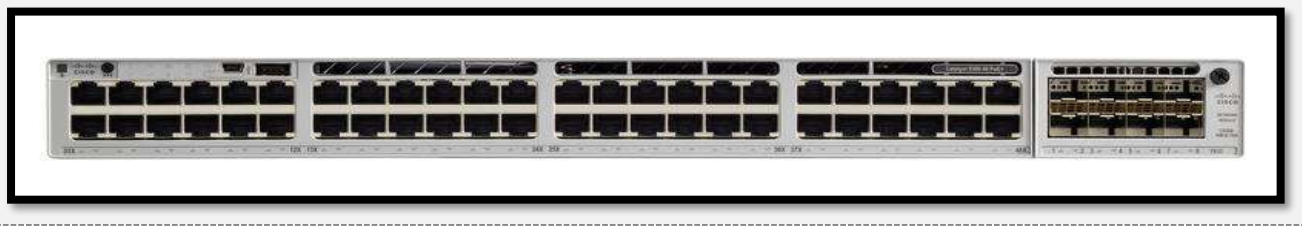


Figure A2.2 - Cisco Catalyst 9300 48-port switch

Specification	Detail
Ports	48x 10/100/1000 PoE+ copper ports, modular uplinks
Power Supply	715W AC default
Switching Capacity	256 Gbps (736 Gbps with stacking)
Forwarding Rate	190.47 Mpps (547.62 Mpps with stacking)
Dimensions	1.73 x 17.5 x 16.1 in (chassis only)

(Cisco Catalyst 9300 Series Switches Data Sheet, n.d.)

3. L2 Switches - Cisco Catalyst 9200L 48-port (C9200L-48P-4G) & 24-port (C9200L-24P-4G)



Figure A2.3 - Cisco Catalyst 9200L switch family

Specification	Detail
C9200L-48P-4G	48 full-PoE+ downlinks, 4x1G fixed uplinks, fixed redundant fans
C9200L-24P-4G	24 full-PoE+ downlinks, 4x1G fixed uplinks, fixed redundant fans
Switching Capacity (48P)	104 Gbps (184 Gbps with stacking)
Switching Capacity (24P)	56 Gbps (136 Gbps with stacking)
Dimensions	1.73 x 17.5 x 11.3 in, 4.4kg-10.58lb depending on model

(Cisco Catalyst 9200 Series Switches Data Sheet, n.d.)

4. Dell PowerVault MD1200 Storage Array



Figure A2.4 - Dell PowerVault MD1200

Specification	Detail
Controller	PERC H800, PCI-E 2.0, dual external x4 SAS ports, 512MB cache
Drive Bays	Up to 12 hot-pluggable SAS HDDs/SSDs
Maximum Capacity	24TB (12x 2TB 7,200RPM SAS drives)
RAID Levels	0, 1, 5, 6, 10, 50, 60
Power Supply	600W, 100-240 VAC auto-sensing
Dimensions	8.7 x 48.2 x 59.4 cm; up to 28.39kg fully populated

(Powe Vault MD1200 Documentation, n.d.)

5. Dell PowerEdge R740 Rack Server



Figure A2.5 - Dell PowerEdge R740

Specification	Detail
Form Factor	2-socket, 2U rack server
Processor (as configured)	Intel Xeon 4210R, 8GB RAM, 600GB SAS 10K
Storage Controller	PERC H330/H730P/H740P; internal dual SD module
Chassis	Up to 8x 2.5" SAS/SATA hot-plug drives
Power Supply	750W, hot-plug
Dimensions	86.8mm x 434.0mm x 737.5mm

(PowerEdge R740 Documentation, n.d.)

6. Cisco Meraki MR57 Wi-Fi 6E Access Point



Figure A2.6 - Cisco Meraki MR57 access point

Specification	Detail
Wi-Fi Standard	802.11ax (Wi-Fi 6E), tri-band 2.4/5/6 GHz
Management	Cloud-managed via the Meraki dashboard, integrates with the on-site wireless LAN controller for local resiliency
Security	WPA2/WPA3, 802.1X with RADIUS authentication supported
PoE	802.3at PoE+ powered

(Cisco Meraki MR57 Access Point Data Sheet, n.d.)

7. Sophos XGS 5500 Series Firewall

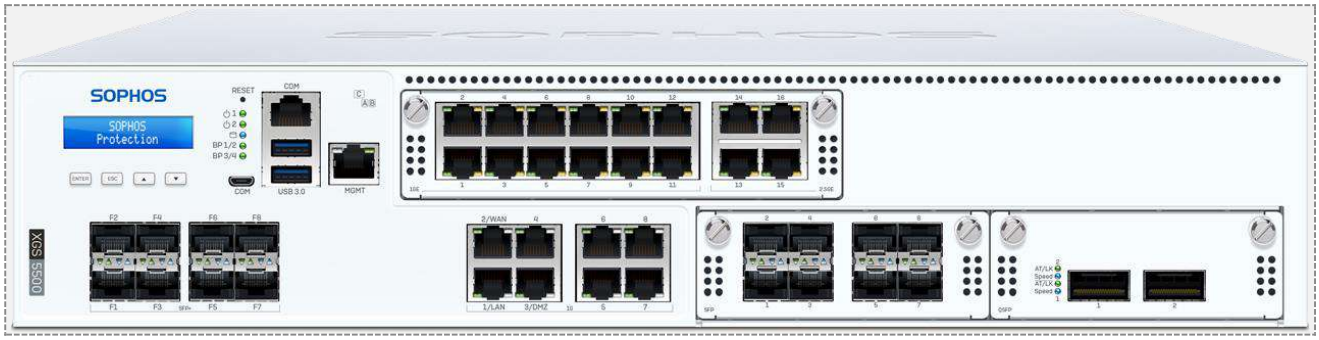


Figure A2.7 - Sophos XGS 5500 rackmount firewall

Specification	Detail
Form Factor	2U rackmount, dual redundant PSU option
Throughput	Multi-Gbps firewall and IPS throughput suitable for enterprise perimeter deployment
Features	Next-gen firewall, IPS, TLS inspection, sandboxing (Sophos Sandstorm), integrated with Sophos Central
Licence	Annual subscription licence (as reflected in the project budget)

(Sophos XGS Series Firewalls, n.d.)

8. Cisco Catalyst 9800-80 Wireless LAN Controller



Figure A2.8 - Cisco Catalyst 9800-80 wireless controller

Specification	Detail
Capacity	Supports thousands of access points and tens of thousands of clients per controller
Redundancy	Supports high-availability SSO pairing
Software	Cisco IOS XE, supports 802.1X/RADIUS, WPA2/WPA3 enterprise security

(Cisco Catalyst 9800-80 Wireless Controller Data Sheet, n.d.)

9. Eaton 9PX 3kVA/3000W Rack Tower UPS



Figure A2.9 - Eaton 9PX 3kVA UPS

Specification	Detail
Capacity	3kVA/3000W, double-conversion online UPS
Features	Hot-swappable batteries, bypass function, IEC outputs, BS input cord
Form Factor	Rack/tower convertible

(Eaton 9PX UPS Data Sheet, n.d.)

Annexure 3 - Security Policy

NextEdge (Pvt) Ltd Network and Information Security Policy (summary)

1. Purpose and Scope

This policy is intentionally written at the level of company-wide rules rather than device-level configuration; the specific ACLs, firewall rules and switch configurations that implement each rule below are maintained separately by the Network & System Admin team as living configuration artefacts, and are summarised for illustration in Chapter 3 (Security Requirements Specification) and Chapter 7 (Implementation).
This policy governs the secure use, administration and protection of NextEdge (Pvt) Ltd's network infrastructure, systems and data at the Colombo 3 head office, and applies to all employees, contractors and visitors who access the network.

2. Lab Isolation Policy

- The 5th-floor Product Research, Quality Research and Development Lab network shall be logically isolated from all other departments and from the internet.
- The only permitted communication path into or out of the Lab VLAN is with the HR department, to support onboarding, access-card and personnel administration.
- Any change to the Lab network's ACLs or firewall rules requires sign-off from the Network and System Admin team and the HR manager.
- All access attempts to or from the Lab VLAN shall be logged and subject to regular monitoring and auditing.

3. Remote Access

- Remote access to the dedicated Lab file server shall be limited to secure VPN connections.
- Multi-factor authentication is mandatory for all remote access to enhance access control.

4. Security Awareness Training

- Employees within the Lab and HR departments shall undergo specialised security-awareness training to understand the importance of data protection and the unique security measures applicable to their roles.

5. Network Security

- Firewalls and intrusion-prevention systems protect the network from unauthorised access and threats, configured to enforce the principle of least privilege.
- Outbound internet access is restricted through the firewall to prevent access to potentially harmful websites and services.
- Network activity is monitored regularly for security incidents, with logs reviewed to detect anomalies; the IT department conducts periodic compliance audits.
- Site restrictions limit access to non-business-related websites during working hours.
- Wi-Fi access points are secured with strong encryption and authentication; guest Wi-Fi is separated from the internal network.
- The network is logically segmented (VLANs) to isolate critical systems and limit the lateral movement of threats.
- Regular vulnerability assessments and scans are conducted, with critical findings remediated promptly.
- Measures to protect against Denial of Service (DoS) and Distributed Denial of Service (DDoS) attacks are implemented to ensure availability.

6. Physical Security

- Physical access to the server room and Lab is restricted to authorised personnel only, controlled through key cards and/or biometric authentication.
- Unused network ports are disabled or configured to allow only authorised devices to connect.
- IT equipment is labelled and registered for asset tracking; employees are responsible for safeguarding company-issued equipment.
- Disposal of IT equipment follows secure data-wiping/destruction procedures before disposal.
- Surveillance cameras are placed in key areas to monitor and record access to sensitive locations.
- A visitor-management system tracks and manages access for guests and vendors, who are escorted by authorised personnel.
- Emergency response plans are established for fire, natural disasters and security breaches, with staff trained in evacuation procedures.

7. Disaster Recovery and Backup

- NextEdge (Pvt) Ltd maintains a Disaster Recovery Plan (DRP) to address recovery and continuity of critical business functions, reviewed and tested regularly.
- Regular data backups safeguard critical information; backups are stored in secure, offsite/cloud locations.
- Backup retention periods are defined based on business requirements and regulatory compliance, with periodic restoration testing.
- The Disaster Recovery Team, together with the Incident Response Team, is responsible for initiating the DRP in response to significant incidents.
- Critical systems and services are designed with high-availability features to reduce downtime.

8. Compliance

- NextEdge (Pvt) Ltd follows all applicable industry standards, laws and regulations relating to information security and data protection.
- Violations of this security policy may result in disciplinary action, up to and including termination of employment or legal action, as applicable.

9. Policy Review and Updates

This Security Policy will be reviewed annually and updated as needed to reflect changes in security requirements and the threat landscape.

Annexure 4 - Individual Contribution Declaration

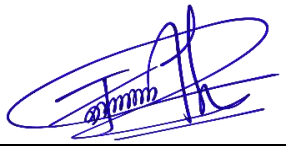
Each member's contribution to the project is described below and independently signed. A standalone, individually signed copy of this declaration is also submitted separately per the module brief.

G.W.J.K. Bhashana (Group Leader) - COHNDNE241F-047

As the Group Leader I was in charge of bringing the team and making sure the project was carried out properly. My main technical job was setting up the Layer 2 switches. That included working with VLANs, trunk links, access ports and any other switching settings that were needed. I also set up the pfSense firewall. Made sure the firewall connections were working right.

In addition, I was in charge of putting the network physically. I connected the network devices with cables made sure the switches, servers and other equipment were linked properly. I also got the physical servers ready, by putting the hardware and starting the ISO files so that the operating systems and server services could be installed and set up. During the project I made sure the team was working well together helped with fixing problems and checking everything and made sure the network, the servers, the firewall and the cloud parts all worked together smoothly.

I confirm the above is an accurate description of my individual contribution to this project.

Signature:  Date: _____

D.T.P. Ladduwahetti - COHNDNE241F-022

I was primarily responsible for the AWS cloud implementation of the project. I configured the AWS environment required to demonstrate the cloud component of the NextEdge infrastructure. My work included configuring the VPC and related networking resources, as well as deploying and managing EC2 instances for the project.

I configured the NextEdgeMSP EC2 instance to host the NextEdge project website and verified that the website was accessible using the public IP address, EC2 public DNS name, and the configured DuckDNS hostname. I also deployed a separate EC2 instance to represent the cloud backup file server. In addition, I worked with the AWS VPC configuration, including subnets, route tables, Internet Gateway connectivity, security groups, network ACLs, and DNS-related settings. I was responsible for testing the AWS resources and verifying that the cloud-hosted services were running and accessible as required for the project

I confirm the above is an accurate description of my individual contribution to this project.

Signature:  Date: _____

R.M.C.D. Rathnayaka - COHNDNE232F-075

I worked mostly on the server-side part of the project. I was in charge of setting up the server and getting it ready. My main job was to set up the Windows Server and add the server roles and services that the network needed.

I set up the DHCP service so that it could give out IP addresses and other network settings to the devices that connected. I also set up the DNS service so that devices could find each other by name. I added Active Directory Domain Services to help manage the network as a place for users and computers. I did the server-side network setup. Checked that the servers, devices and network parts could talk to each other. My work made sure that the important server services, for the project were working properly.

I confirm the above is an accurate description of my individual contribution to this project.

Signature:  Date: _____